End to End Project ADF

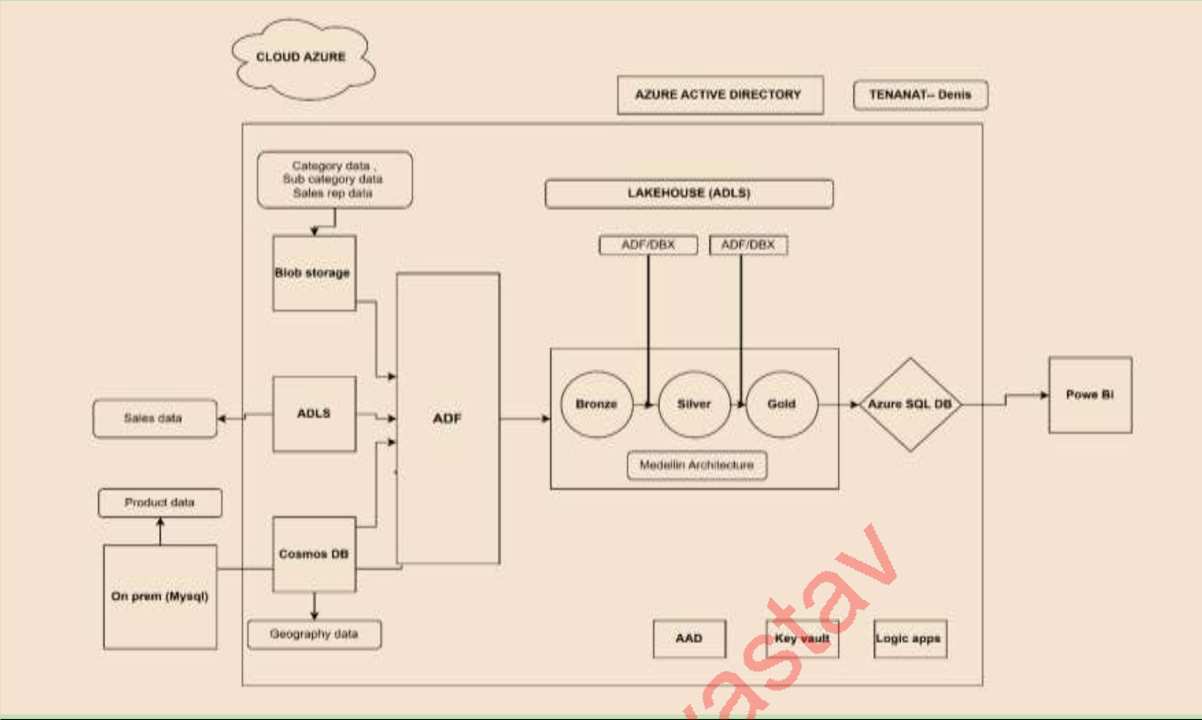
Project process:

This document outlines the implementation of an end-to-end data pipeline using Azure Data Factory (ADF) following the Medellin architecture framework. The project is designed to integrate data from multiple sources, transform it into actionable insights, and deliver it to the reporting layer for visualization.

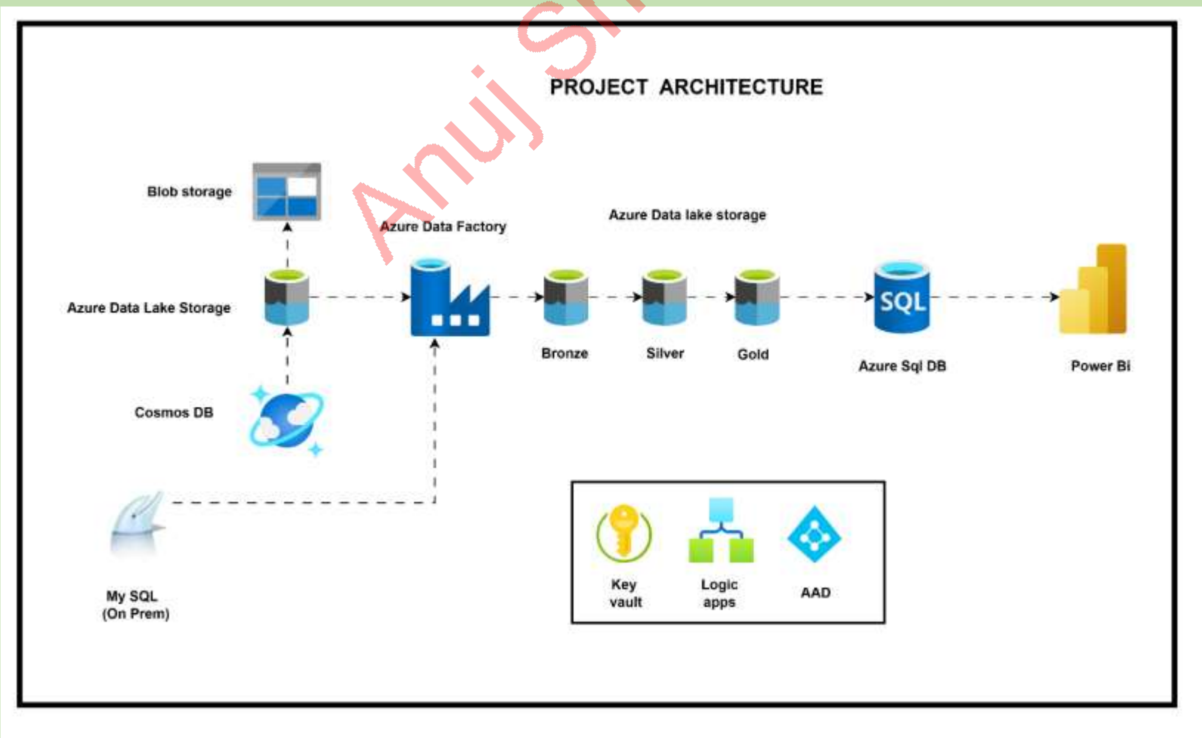
Key Objectives:

1. Efficient Data Integration: Automate the ingestion of data from on-premise and cloud-based sources, such as MySQL, Cosmos DB, and Blob Storage, into Azure Data Lake Storage (ADLS).
2. Data Transformation: Implement the Bronze-Silver-Gold architecture to incrementally process, clean, and optimize data for analytics.
3. Seamless Data Delivery: Load transformed data into Azure SQL Database for Power BI reporting.
4. Security & Scalability: Use Azure Active Directory (AAD), Key Vault, and Logic Apps for secure and scalable workflows.

Architecture Diagram:



Project Architecture:



Medallion Architecture: The Medallion Architecture in Azure Data Factory (ADF) structures data into three layers for better data management and analytics:

Bronze Layer (Raw Data - Landing Zone)

- Stores raw data from MySQL, APIs, CSV, etc.
- Stored in: Azure Data Lake Storage (ADLS) Gen2.
- ADF Activity: Copy Activity (loads raw data as Parquet/CSV).

Silver Layer (Cleansed Data - Curated Zone)

- Data is cleaned & transformed (deduplication, type conversion, joins).
- Processed using: ADF Mapping Data Flows / Azure Databricks.
- Stored in: ADLS Gen2 (Parquet/Delta) or Synapse.

Gold Layer (Aggregated Data - Consumption Zone)

- Final business-ready data for reporting & analytics.
- Stored in: Azure Synapse / Power BI datasets.
- Used in: Dashboards, ML models, analytics reports.

ADF Implementation

- Bronze: Copy Activity loads raw data.
- Silver: Data Flows or Databricks cleanses data.
- Gold: Synapse/Power BI for analytics.

Project Overview:

- Based on the architecture, the first step is to migrate data from **on-premises systems, Blob Storage, and Cosmos DB** to **Azure Data Lake Storage**.
- Within **Azure Data Lake Storage**, the data is first stored in the **bronze layer**, which retains a raw copy of the source data.
- The data is then moved to the **silver layer**, where initial **data transformations** are applied to enhance quality and structure.
- Next, the data is transferred to the **gold layer**, where final **data transformations** and optimizations are performed to prepare it for reporting and analytics.
- The processed data is then loaded into **Azure SQL Database** for structured querying and storage.
- Finally, data from **Azure SQL Database** is connected to **Power BI**, where further **data analysis and visualization** are performed to generate business insights.

Required Azure Services: To implement the project, the following **Azure services** need to be created:

- **Azure Blob Storage** – For storing unstructured data.
- **Azure Data Lake Storage** – For scalable and secure data storage.
- **Azure Data Factory** – For orchestrating data movement and transformation.
- **Azure SQL Database** – For structured data storage and querying.
- **Azure Cosmos DB** – For globally distributed NoSQL data management.
- **Azure Key Vault** – For securely managing secrets, keys, and credentials.
- **Azure Logic Apps** – For automating workflows and integrations.

Business / Functional Requirement Document:

Objective: To create one centralized Dataware housing system

1. Data Gathering / Requirement: The objective is to assemble a **data warehousing system** with various tables to provide **Sales Insights** in a single fact table. The data sources include:

1. Sales (folder by year)
2. Categories (Excel)
3. Geography (Excel)
4. Product (CSV /Database)
5. SalesRep (Excel)
6. SubCategories (Excel)

Task 1.1: Sales Fact Table Mechanism

Create a mechanism to load all the files from the sales folder in a single Sales fact table.The mechanism needs to be resilient as:

- Removing a file from the **Sales** folder does **not** cause errors due to missing files.
- Adding a new yearly **Sales** file is automatically included in the fact query upon refresh.

2. Data Modelling:

Task 2.1:

- Do the respective transformations to the Sales fact table in order to split the Country form the city in field “Location”. Make sure you set up the correct Data Type to allow Geo maps.
- Do the necessary updates in the Date field to make sure you can setup the Date format.

Task 2.2:

- Create unique key (GeoKey) in Sales and Geography table

Task 2.3:

- The Dimensional queries SalesRep and Sub Category need additional treatment. Some ID columns have the following format:

SalesRepID
ID - 6
ID - 7
ID - 5
ID - 3
ID - 1
ID - 2
ID - 4

Create a small function that removes the "ID - " part of these columns that you can invoke and reuse for these two queries to clean the IDs.

Task 2.4:

- Create the Data Model connecting all tables and using the Calendar table already set up in the pbix.

3. Additional column calculations:

Task 3.1:

- Calculate **Total Revenue** in Sales table, using the Product's Retail Price, and multiplying it by the Units.

Task 3.2:

- Calculate **Total Cost** in Sales table, using the Product's Standard Cost, and multiplying it by the Units.

Task 3.3:

- Calculate **Gross Profit** in Sales: Total Revenue – Total Cost

Task 3.4:

- Calculate a measure for **AVG sales per day** – this is the average sum of **Total Revenue** per day based on the Dates of actual Sales.

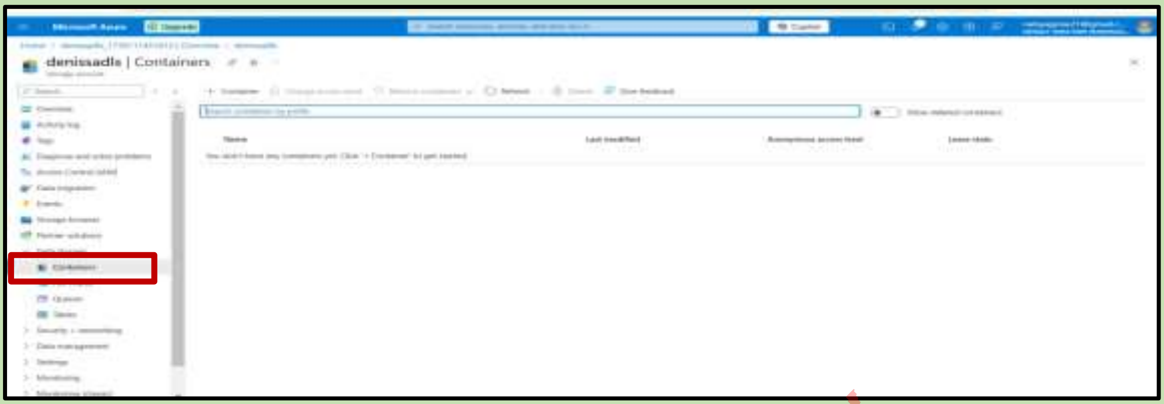
Task 3.5:

- Breakdown Analysis by **Product (drop or increase)**

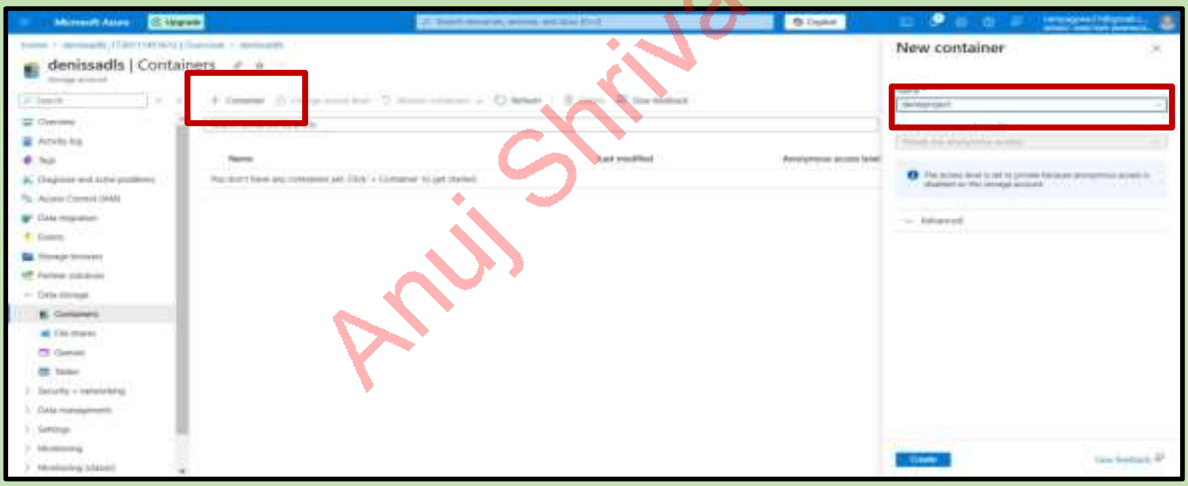
4. Load the target tables into Azure SQL base. The number of analytical tables is developer choice. Feel free to add own ideas and come up with end-to-end process using azure services.

Project Execution: The project execution process is documented with **step-by-step screenshots**, illustrating each stage of implementation in detail.

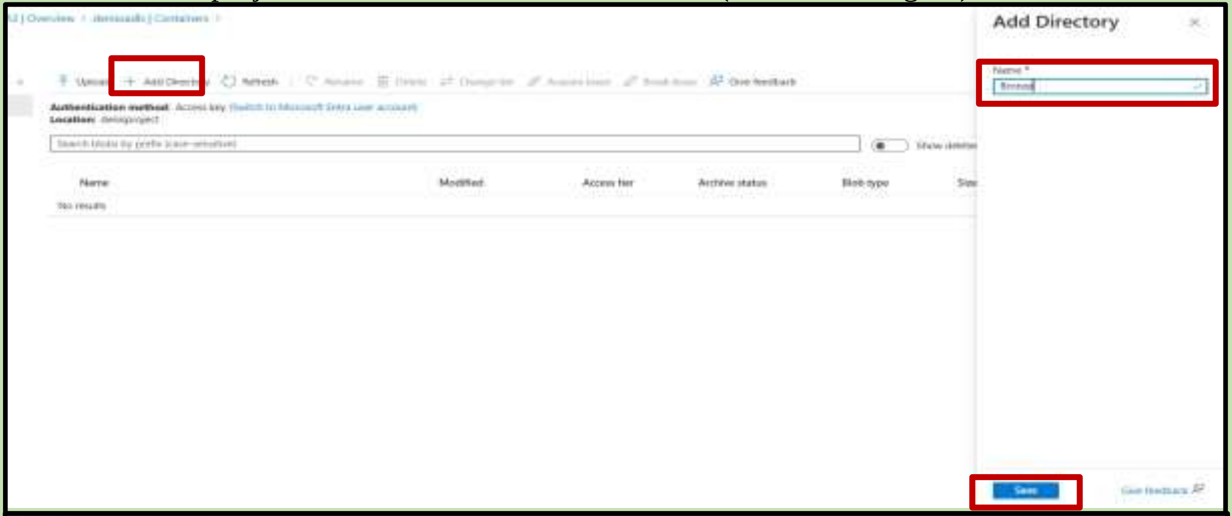
- Firstly, we will open Azure data lake Storage and go to containers.

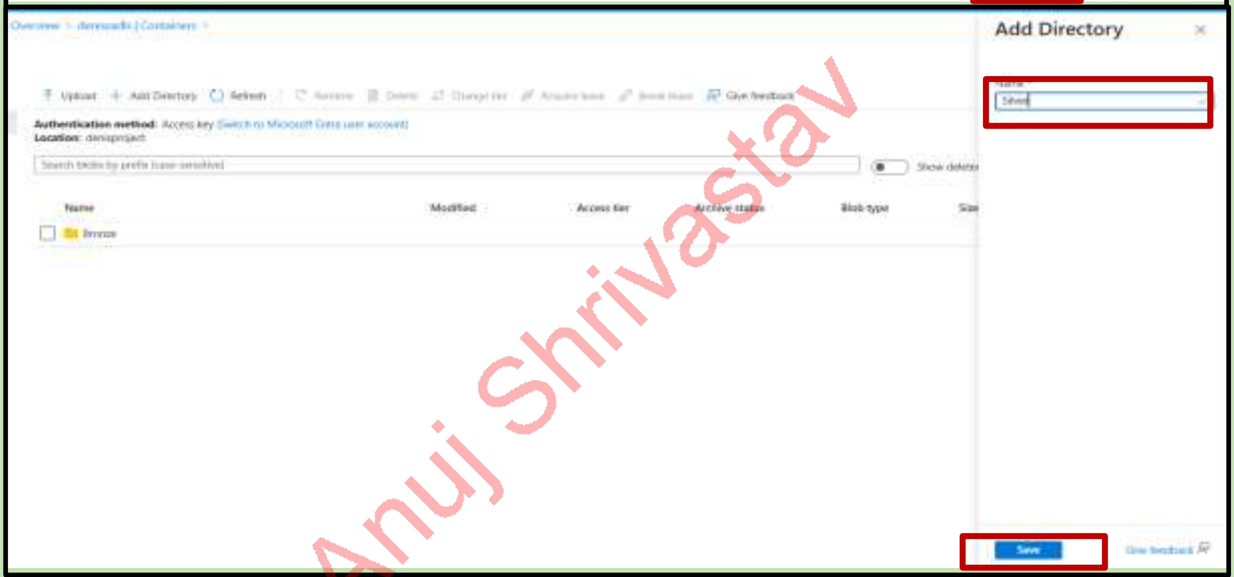


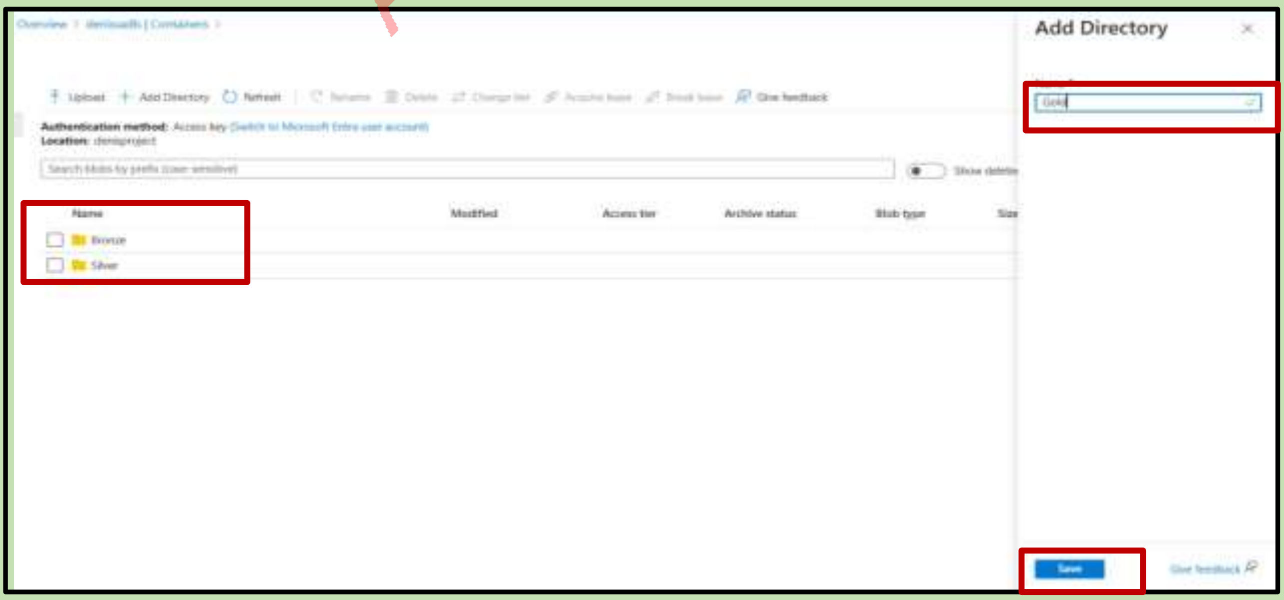
- We need to create new container as denisproject.



- Under denisproject container ---create directories---(bronze,silver,gold)



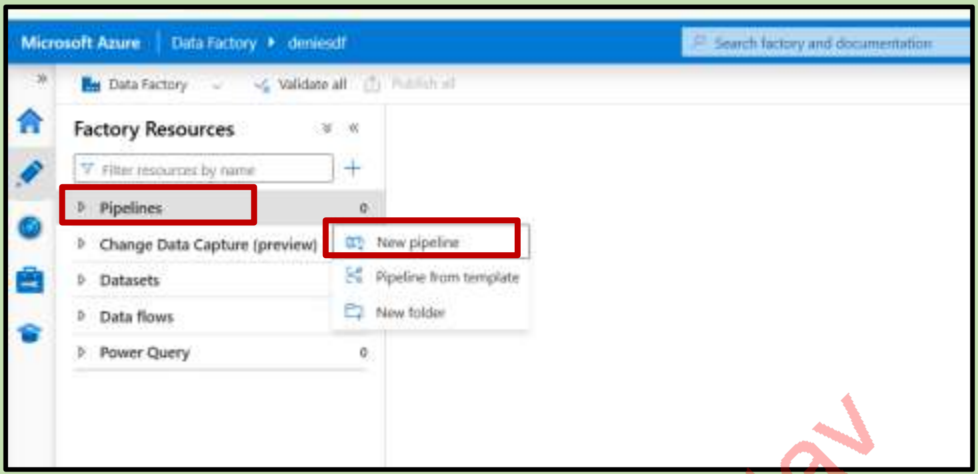




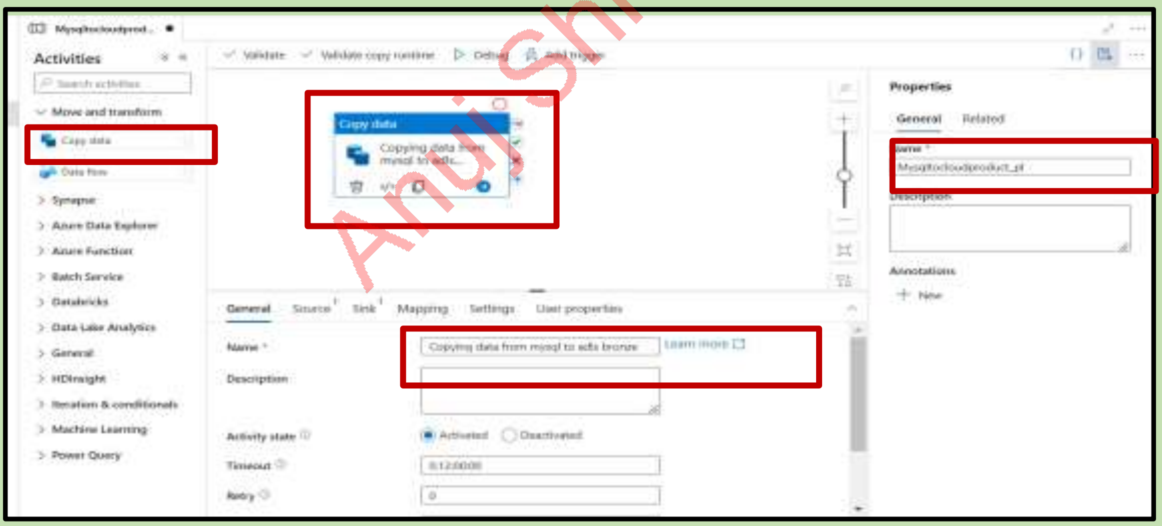
Pipeline 1:

Mysql : Azure data lake storage

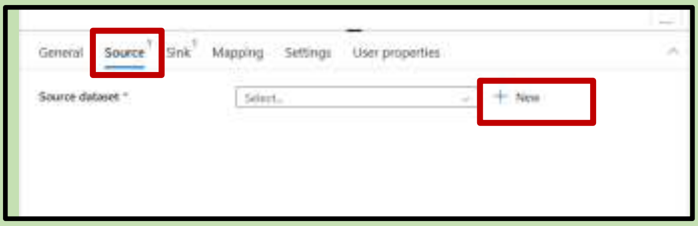
Now open Azure data factory and click on pipeline click on create new



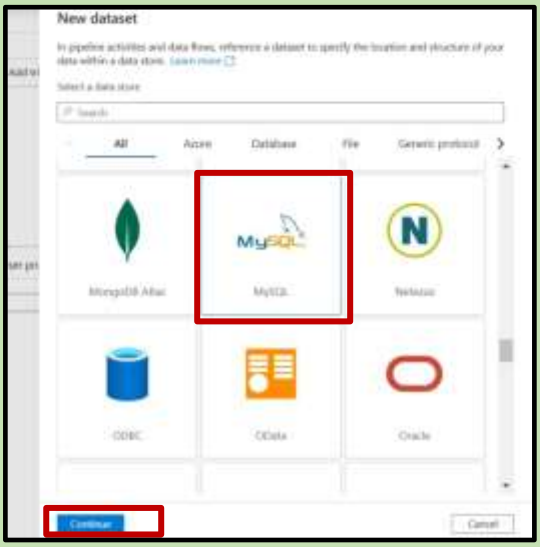
Activities → Copy → give name(mysqltocloudproduct_pl) →(give a name to copy activity)



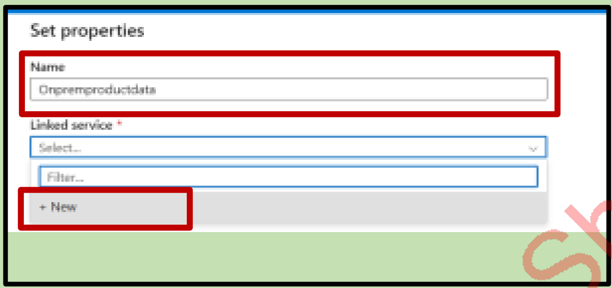
Click on source click on new



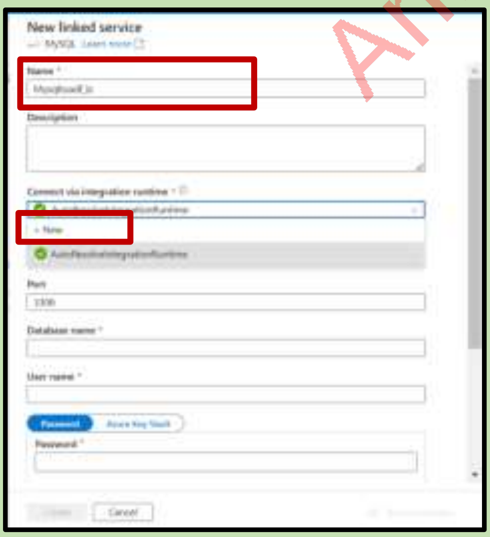
Click on mysql data base and click on continue.



Give the name as onpremproductdata and click new

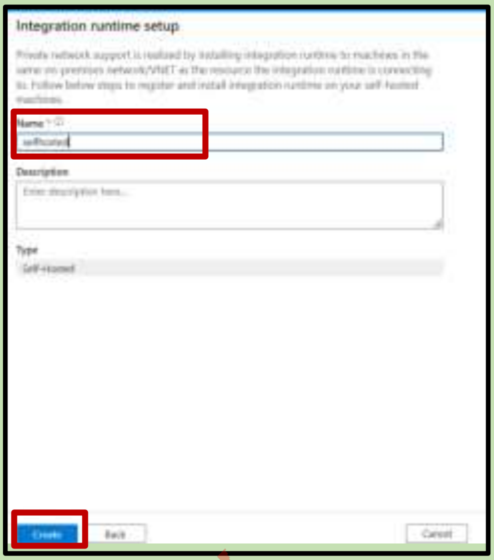


Now give the linked service name and click on new for integration run time

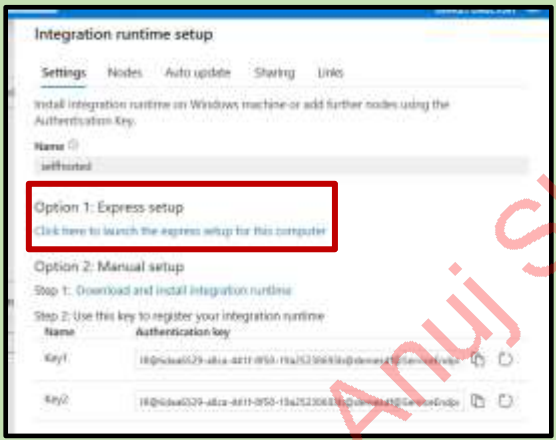


Click on selfhosted

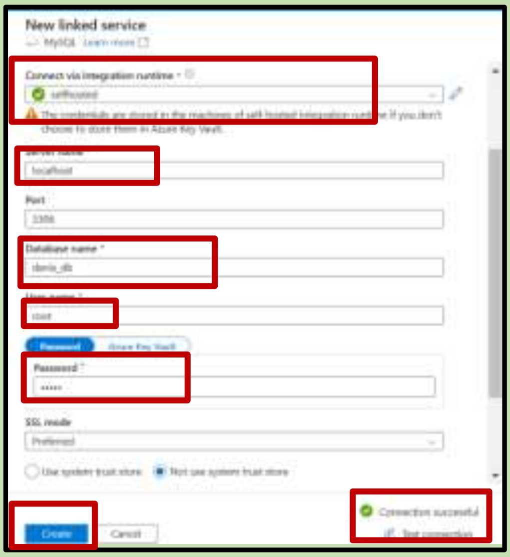
Give the name as selfhosted click on create



Click on express setup and download Integration Runtime

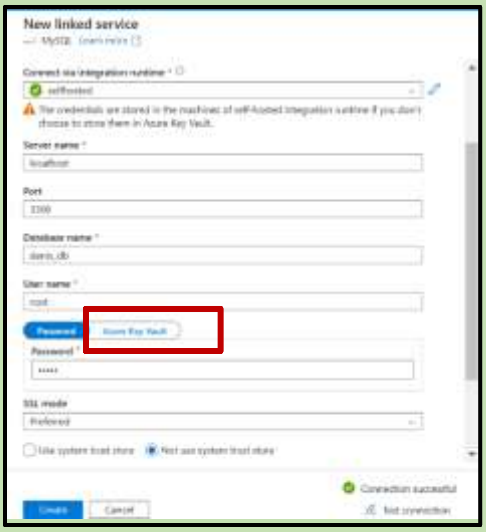


Now selfhosted is selected we need to give the server name database name username and password

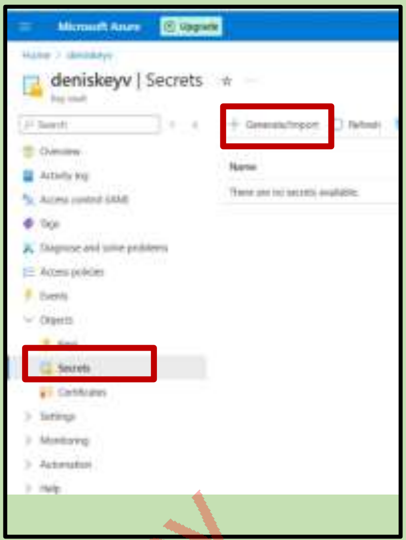


Method 2:

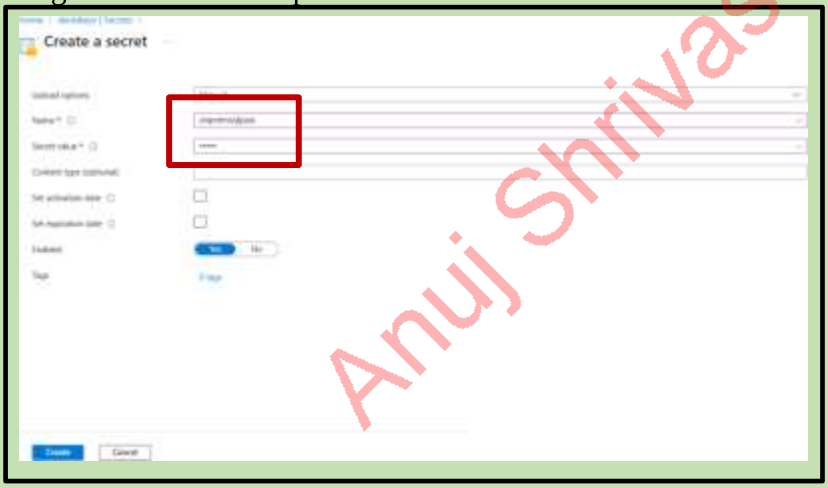
You need to select azure key vault



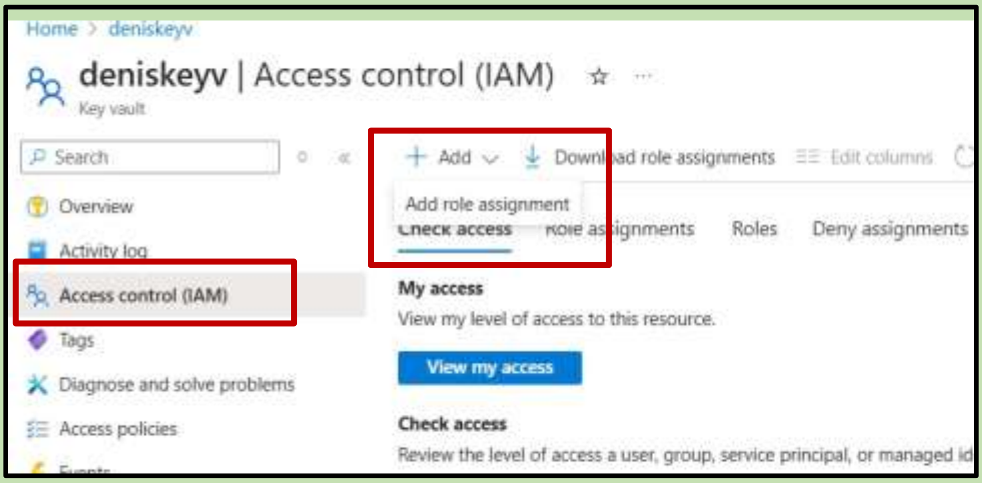
Open the keyvault select secret , generate



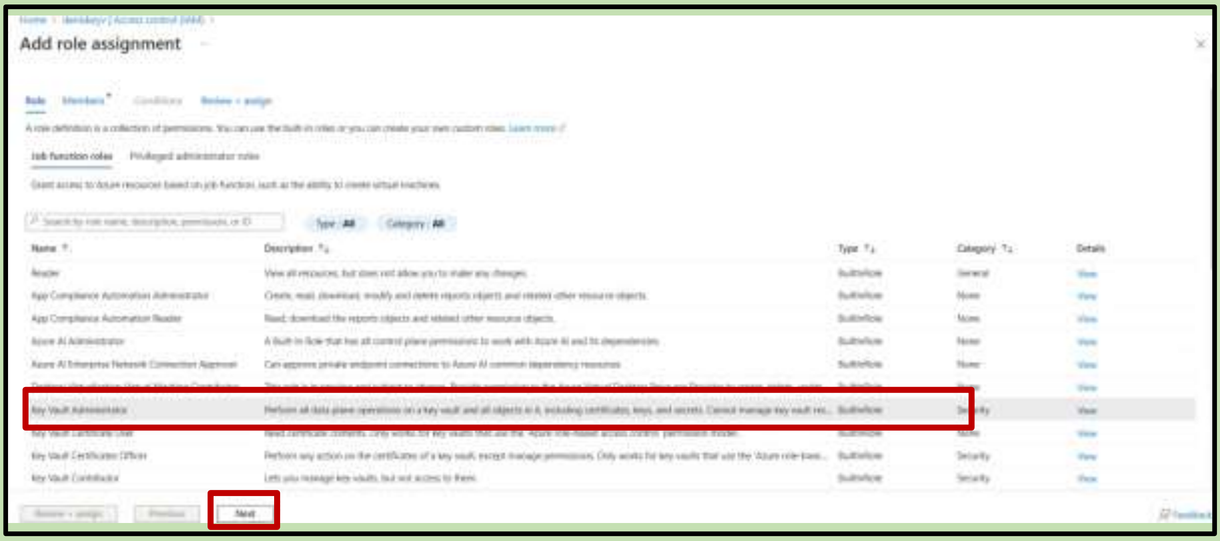
Now give the name and password here



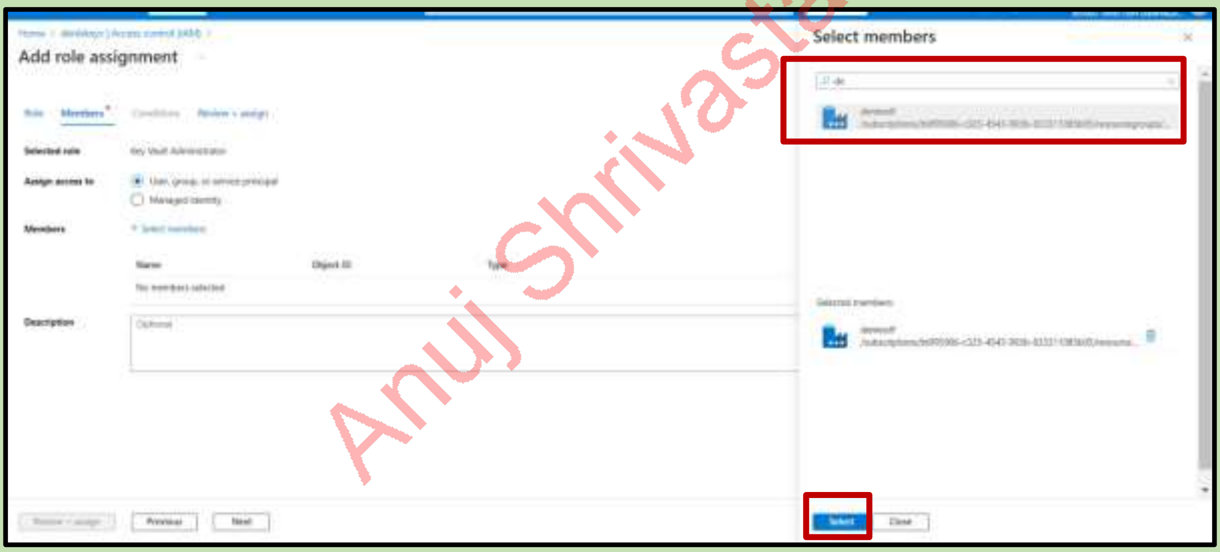
Click on access Control and add role assignment



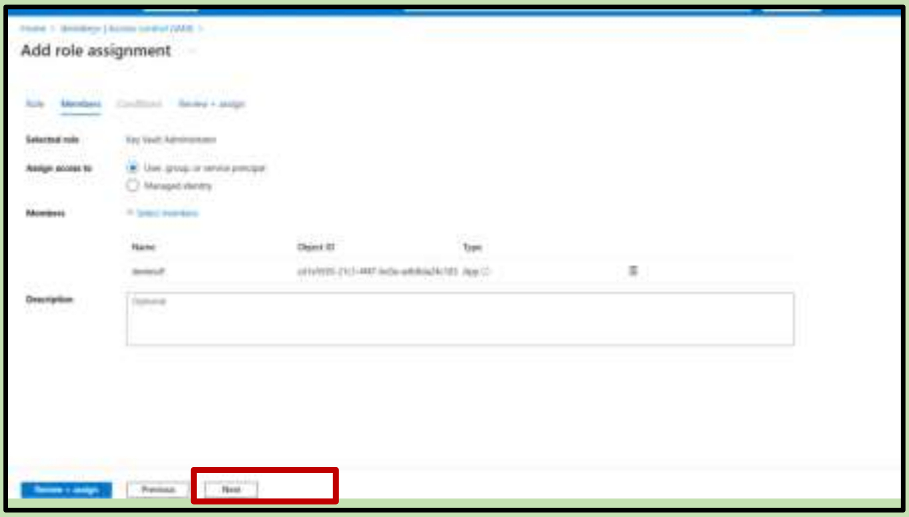
Add key vault administration and click on next



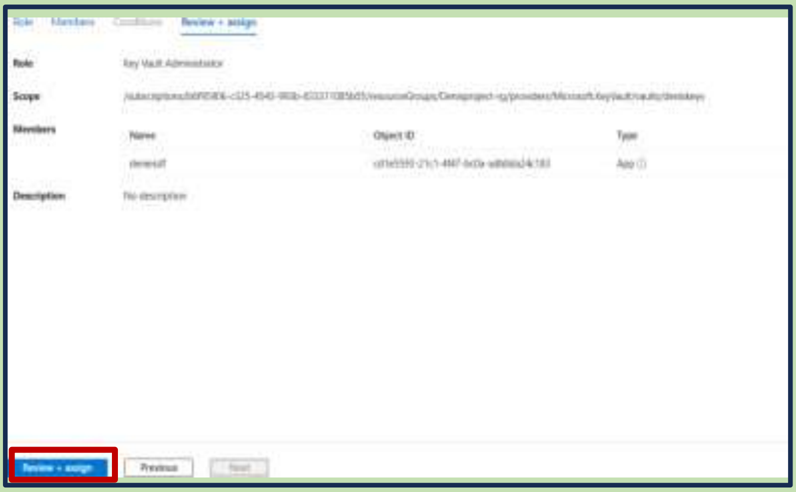
Add azure data factory click on select



Click next

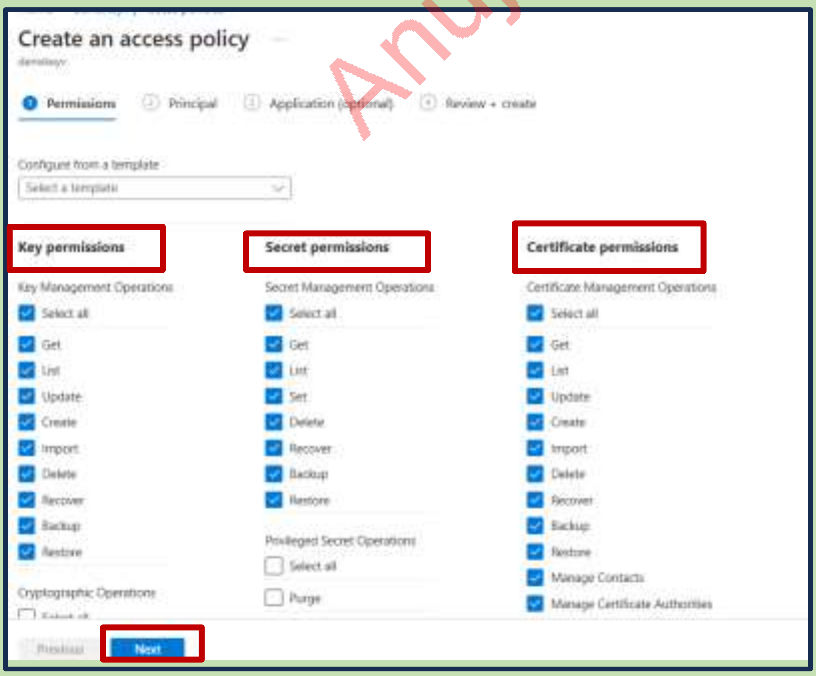


Click on review and assign

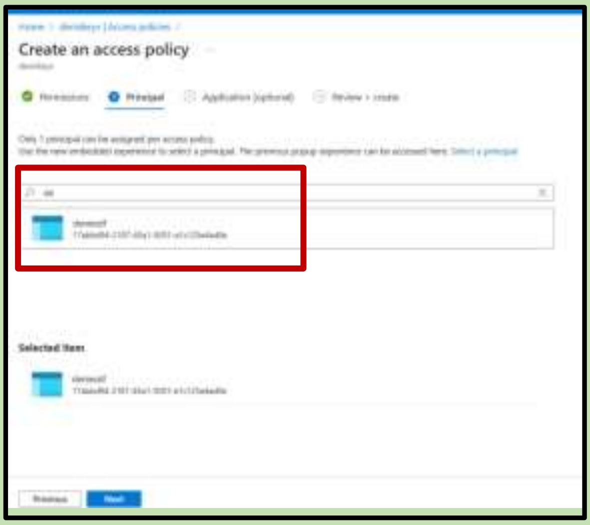


Access policies → create

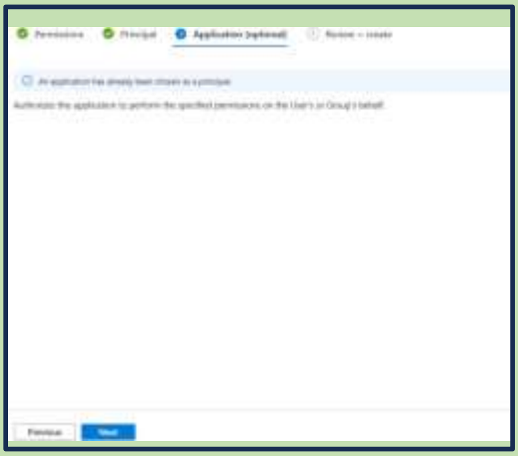
Click on below permissions click next



Give a\Azure data factory click next

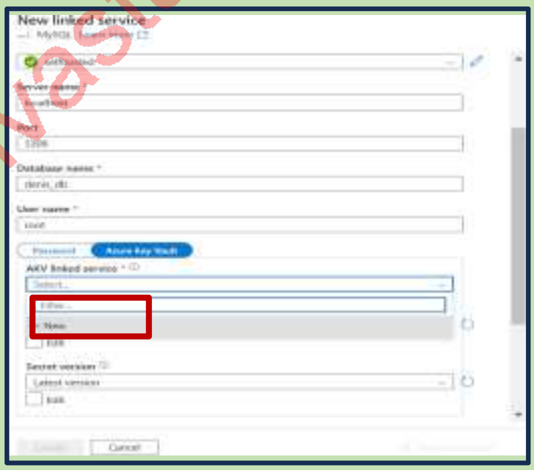


Click next

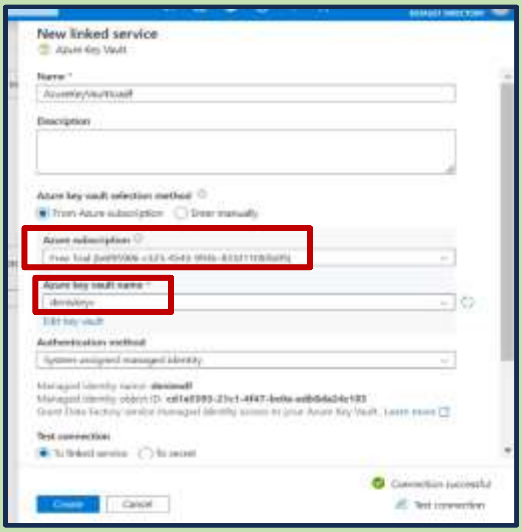


Create→

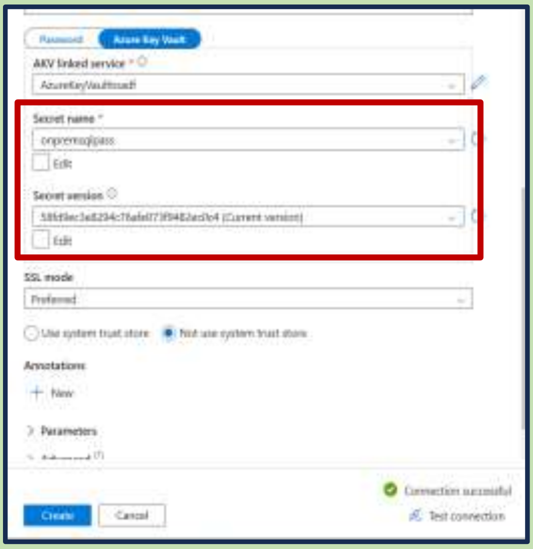
New



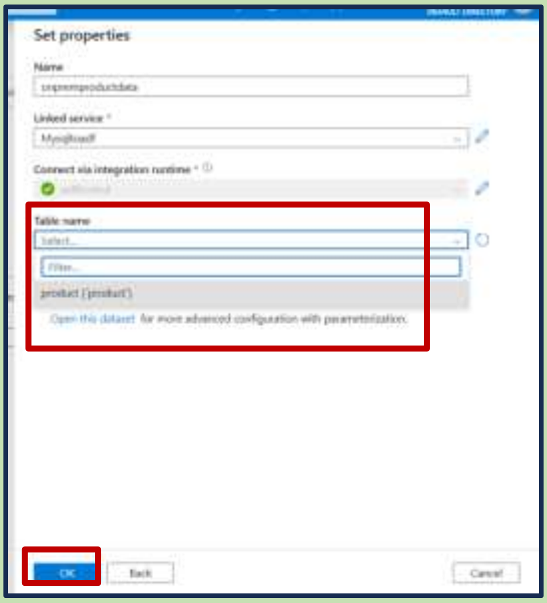
Give the details



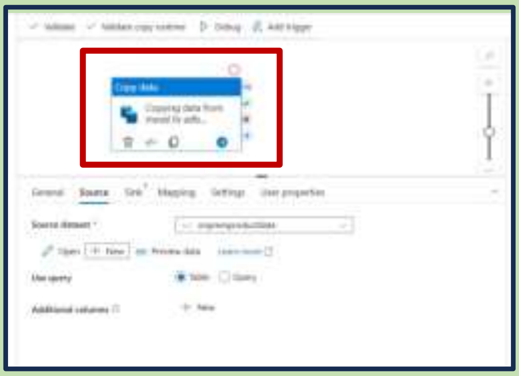
create



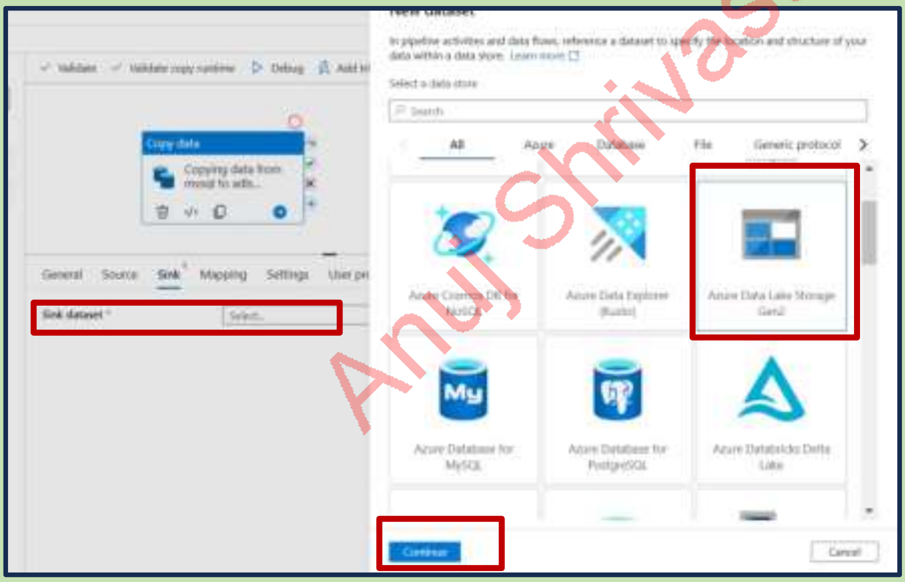
Select → Table



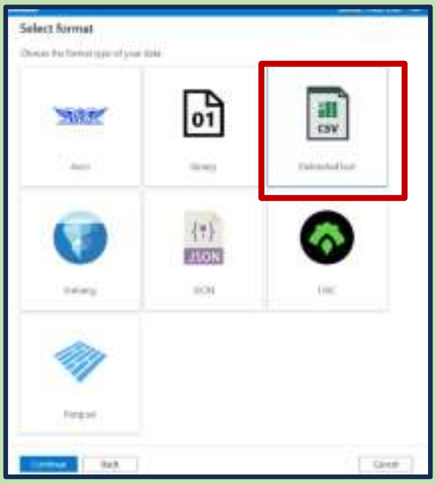
Give the name of source



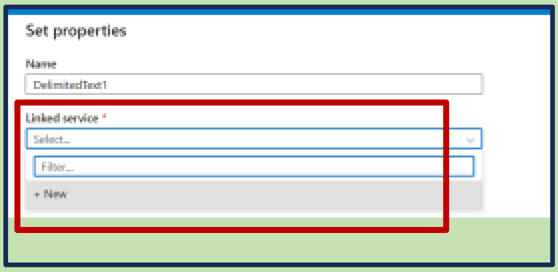
Now select the sink data set



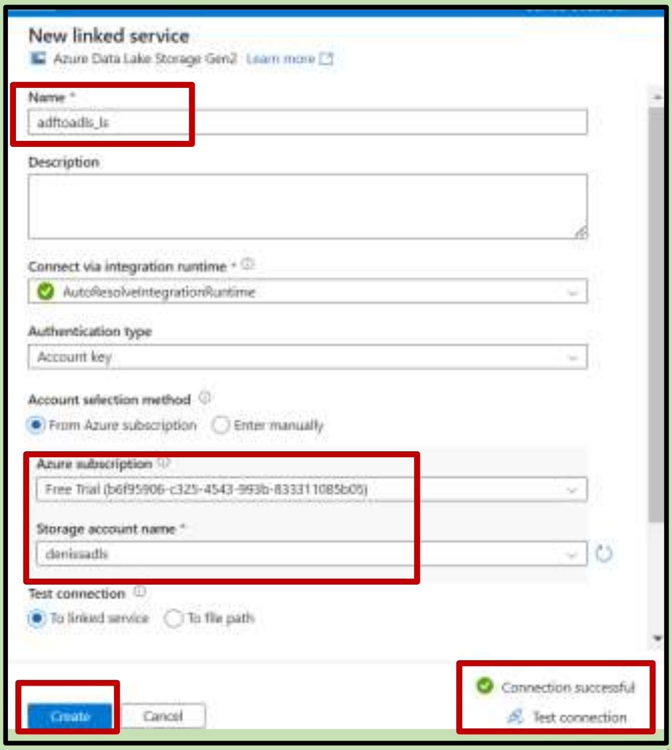
Select the csv format



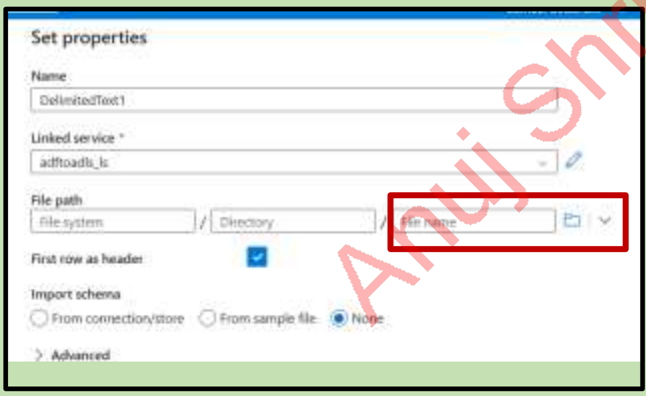
Select the new linked service



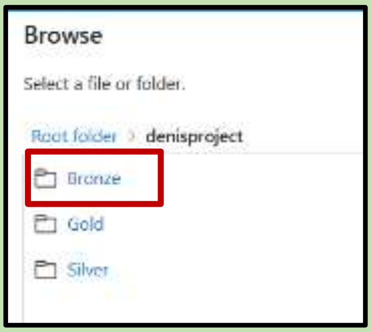
Give details→ create linked service



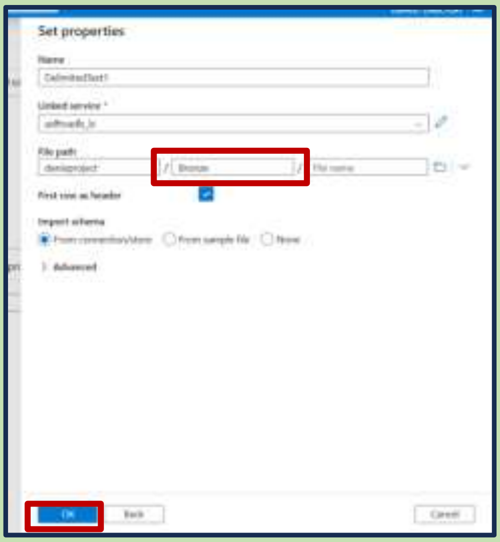
Select the file name



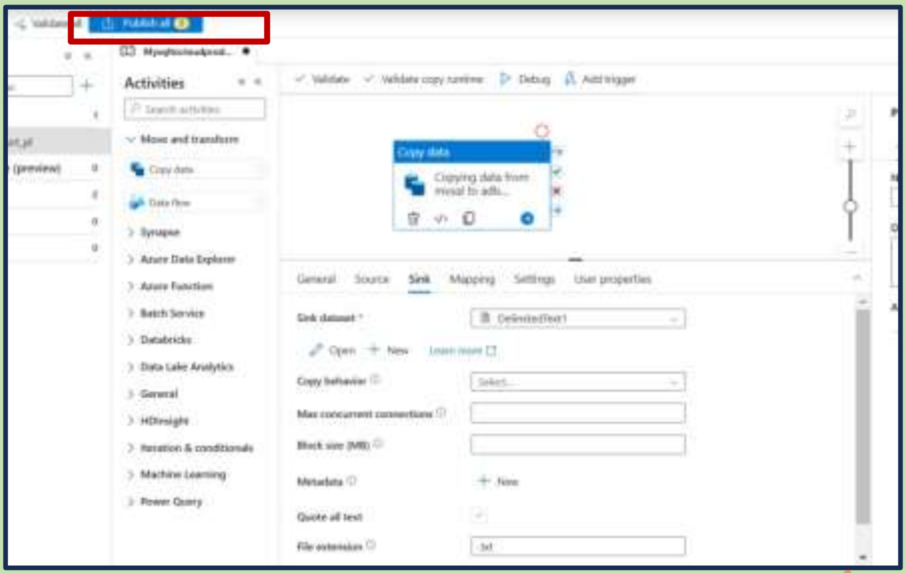
select bronze



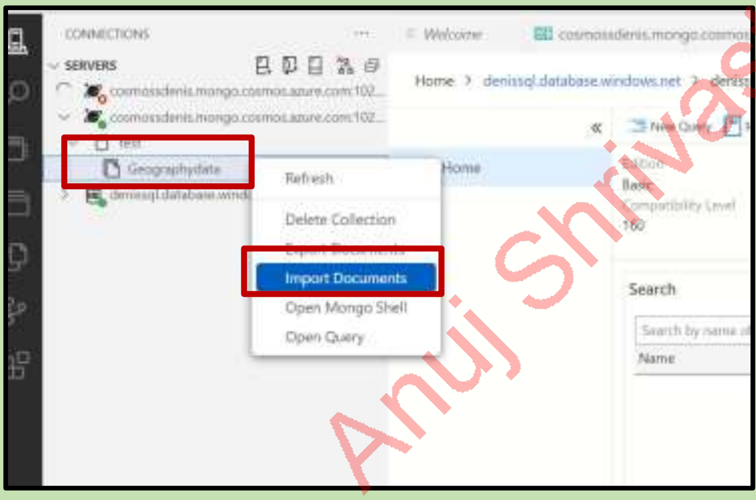
Now click on ok



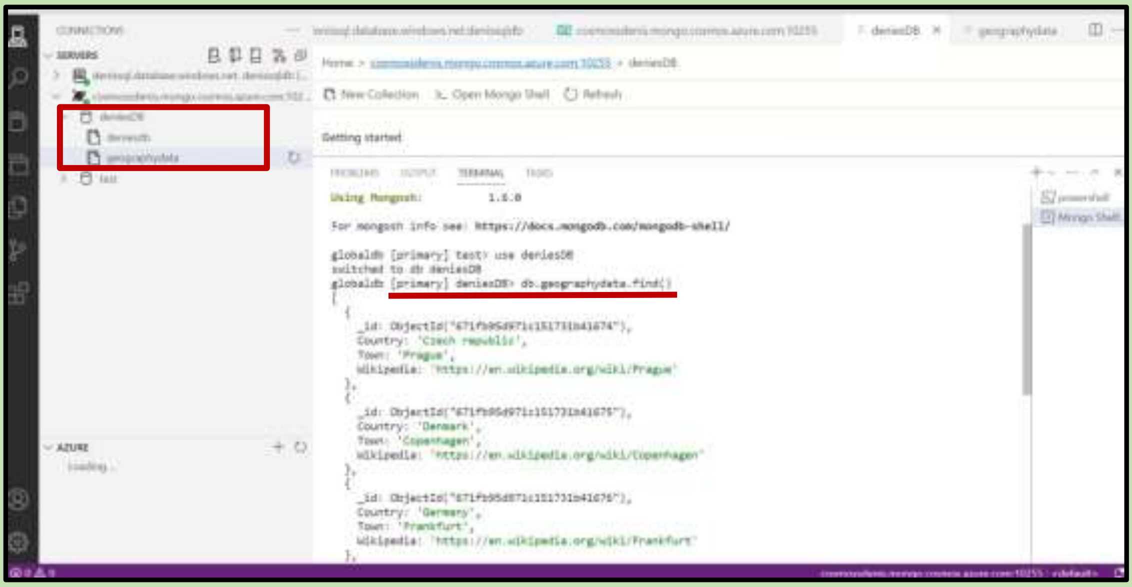
Publish:



Now open cosmos db create geography data and import the Json file



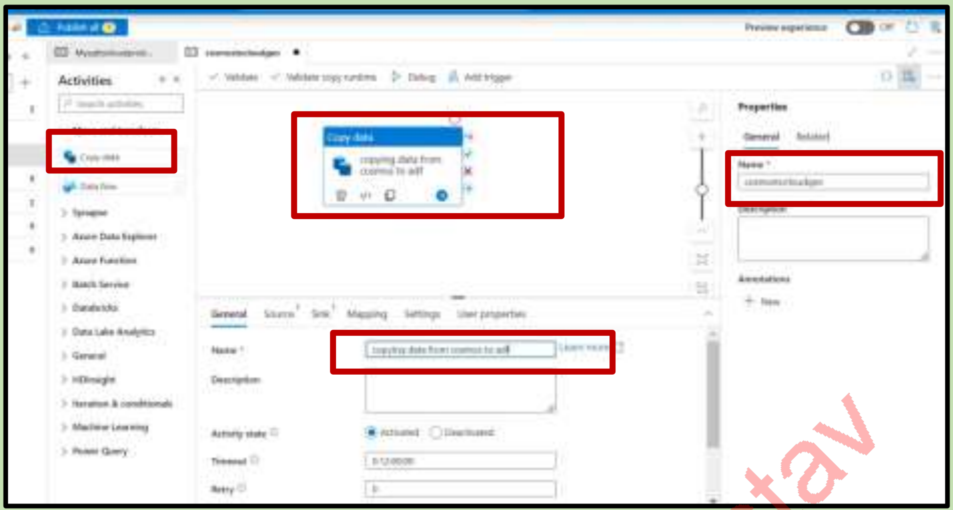
Now we can see the data



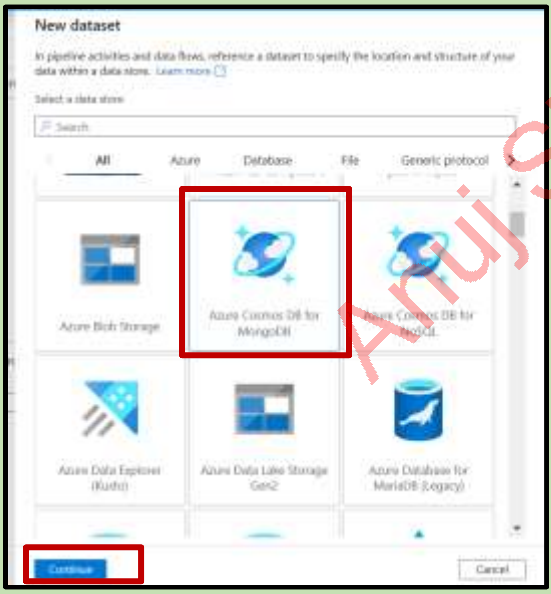
Pipeline 2:

Cosmos: Azure data lake storage

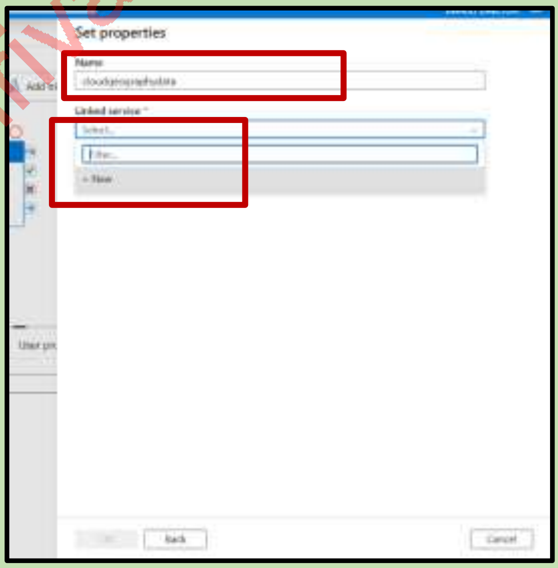
Create one new pipeline → give name



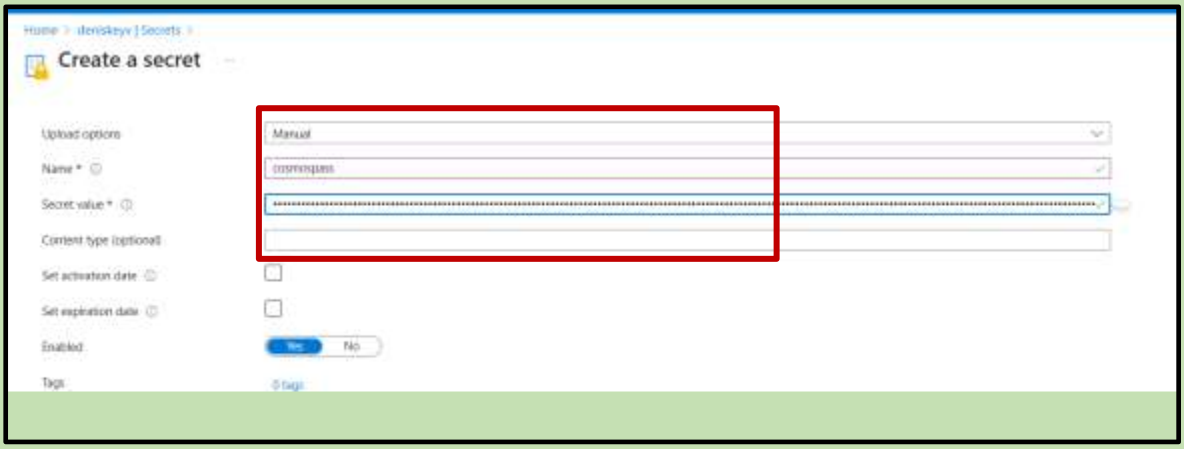
Now select the data set



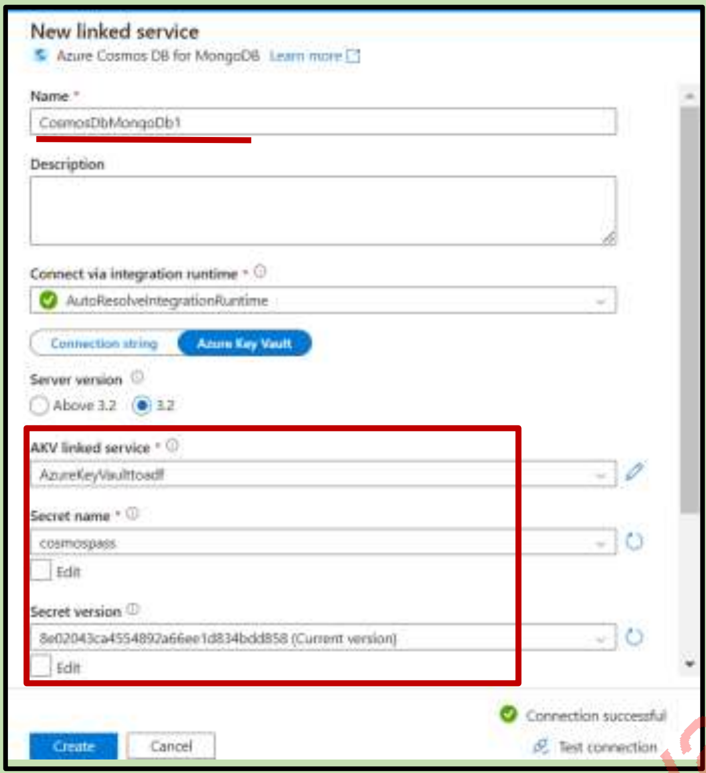
give the name and link service



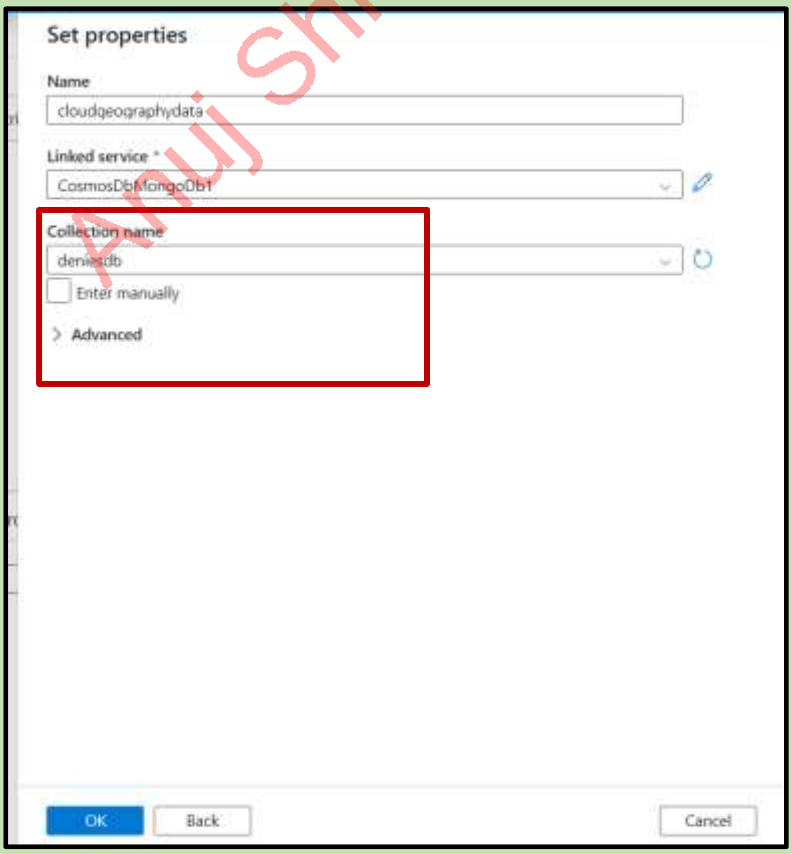
Give the details primary password



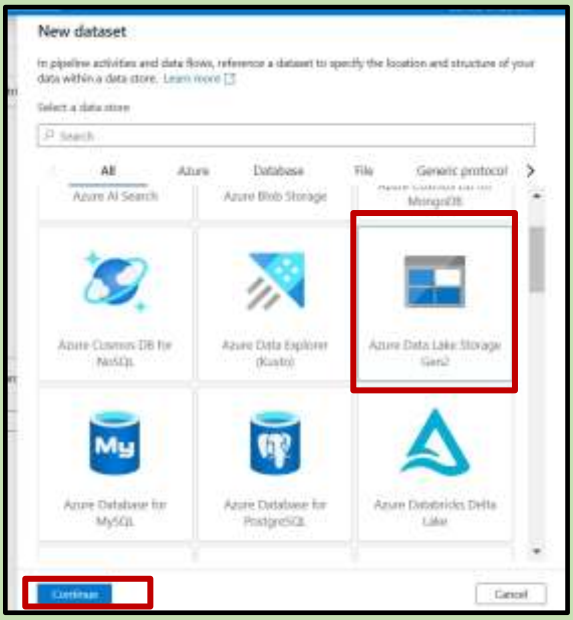
Provide details:



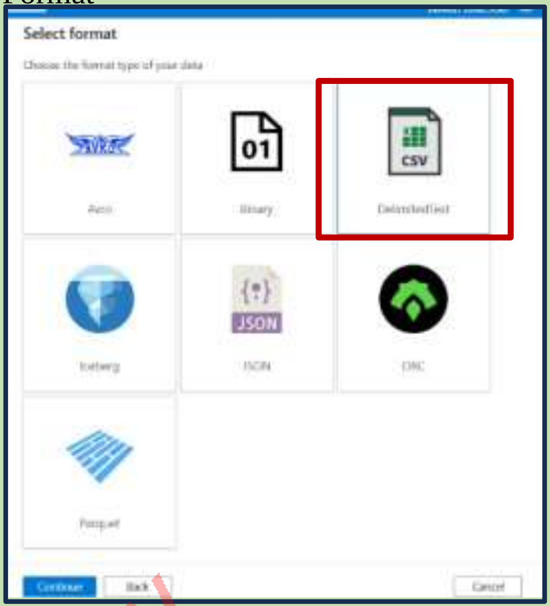
Select the collection name



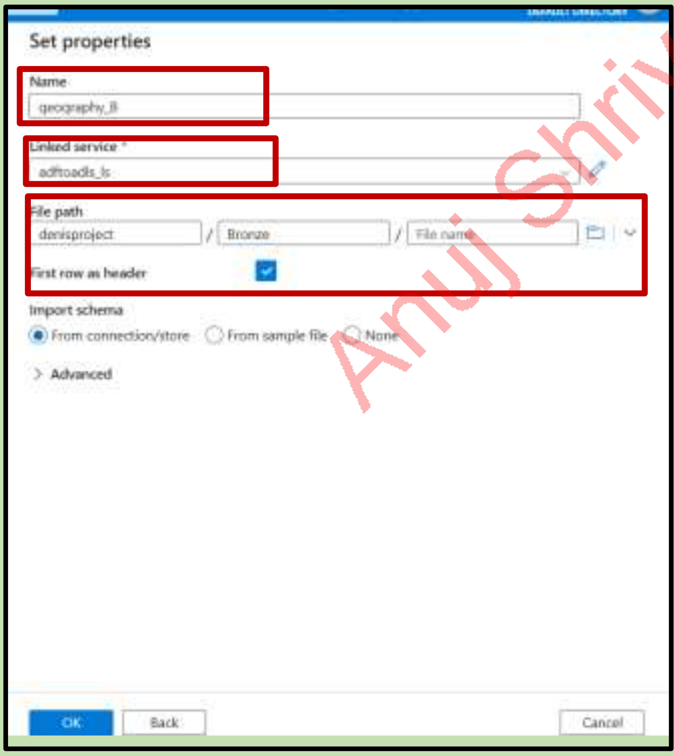
Select ADLS



Format



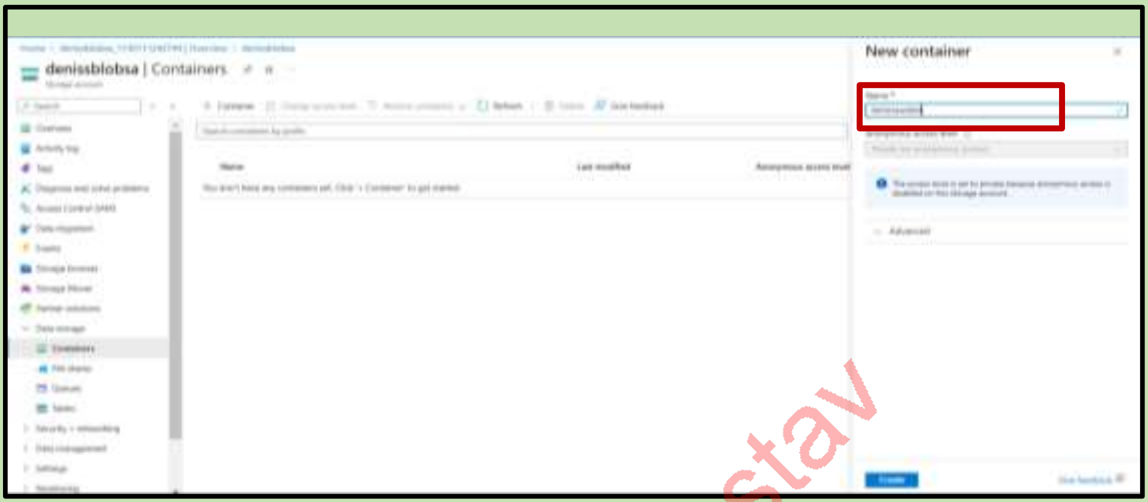
Select the file folder and publish



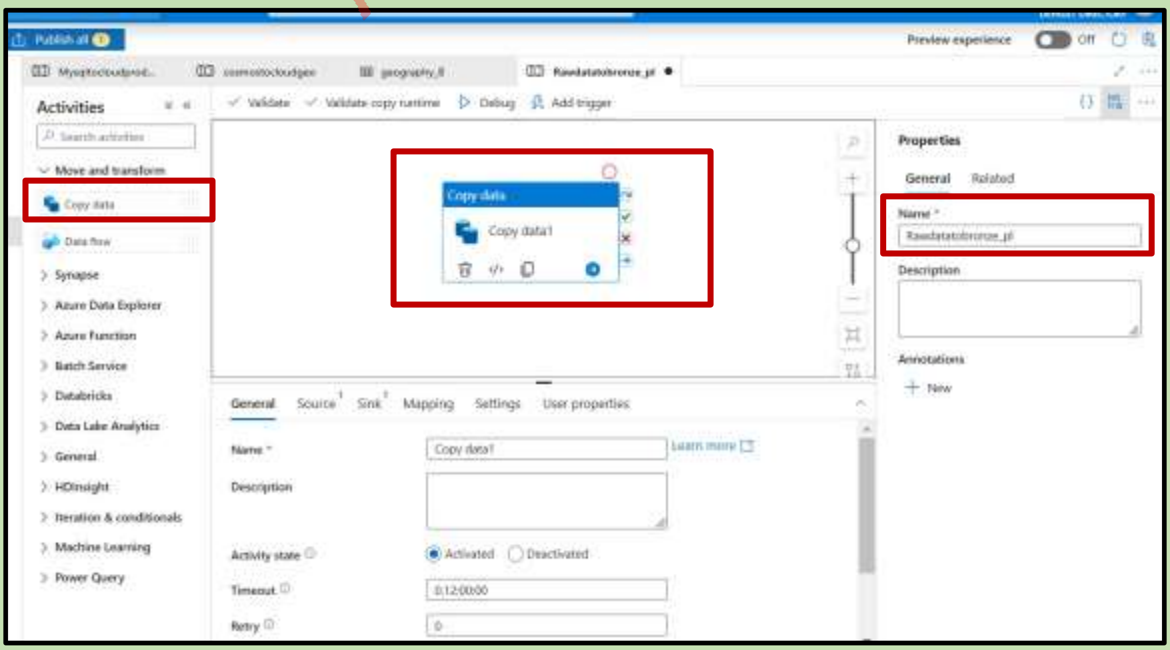
Pipeline 3:

Azure blob storage →Azure data lake storage →Bronze

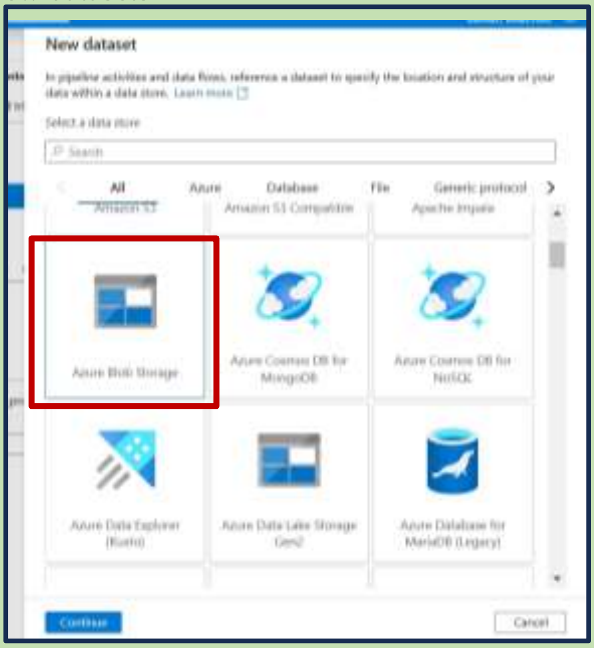
Open Blob storage create container and upload the raw files



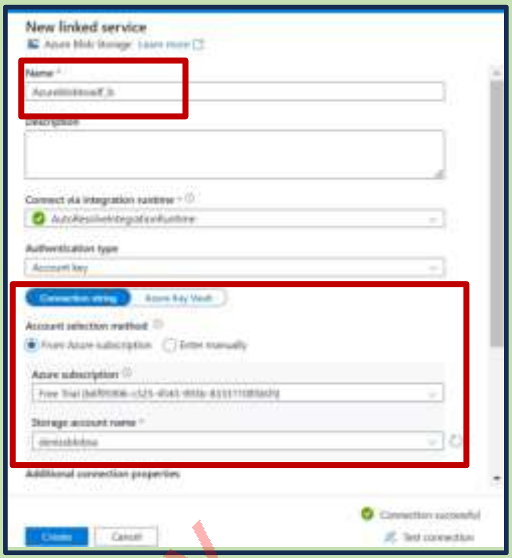
Give the pipeline name



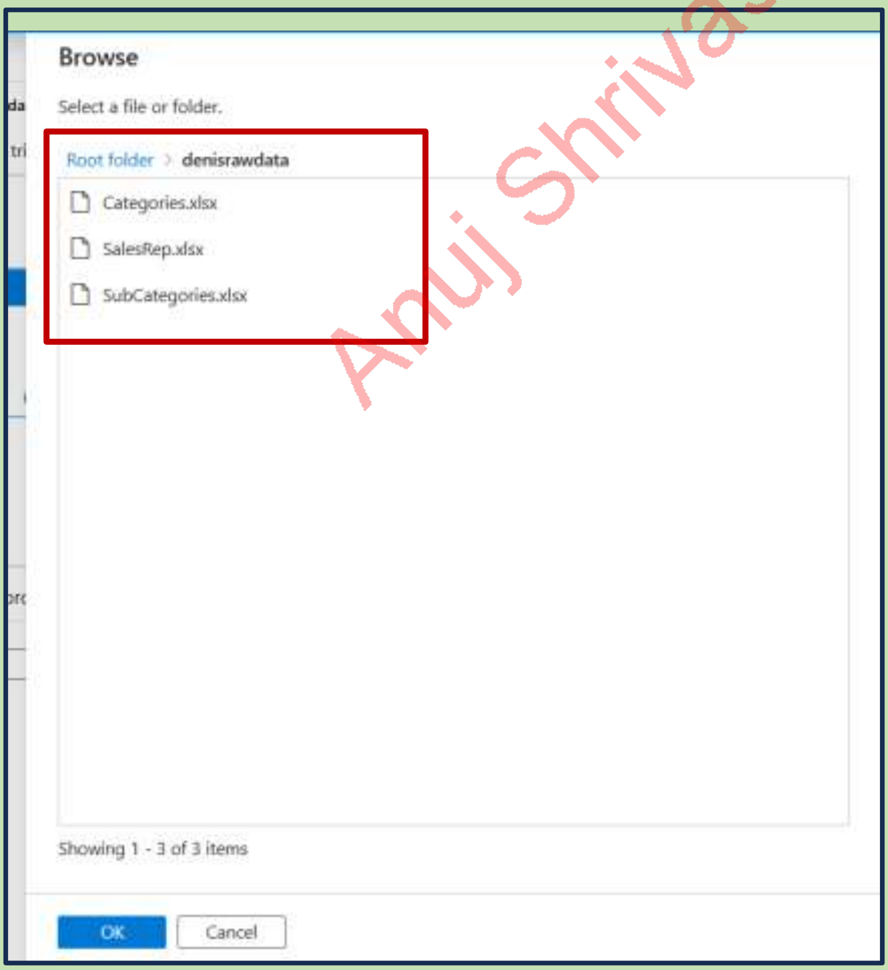
New datasets



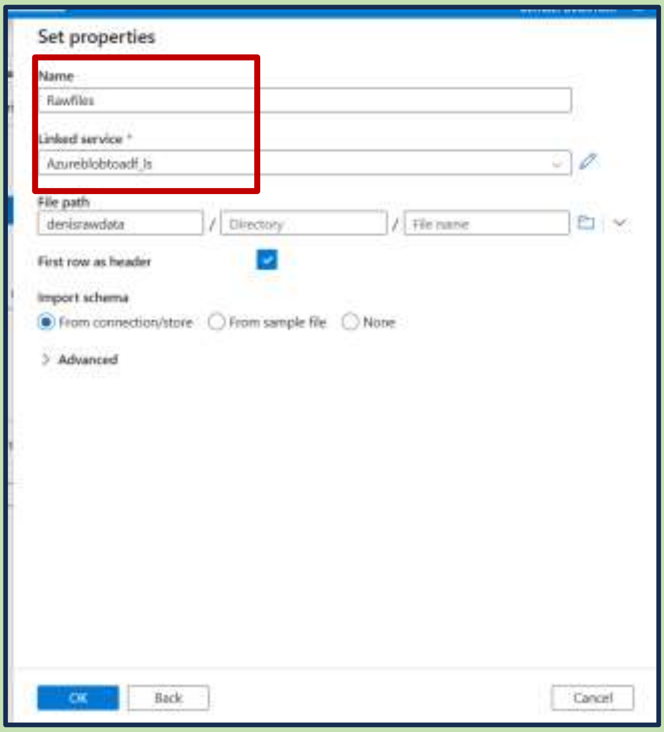
Give details



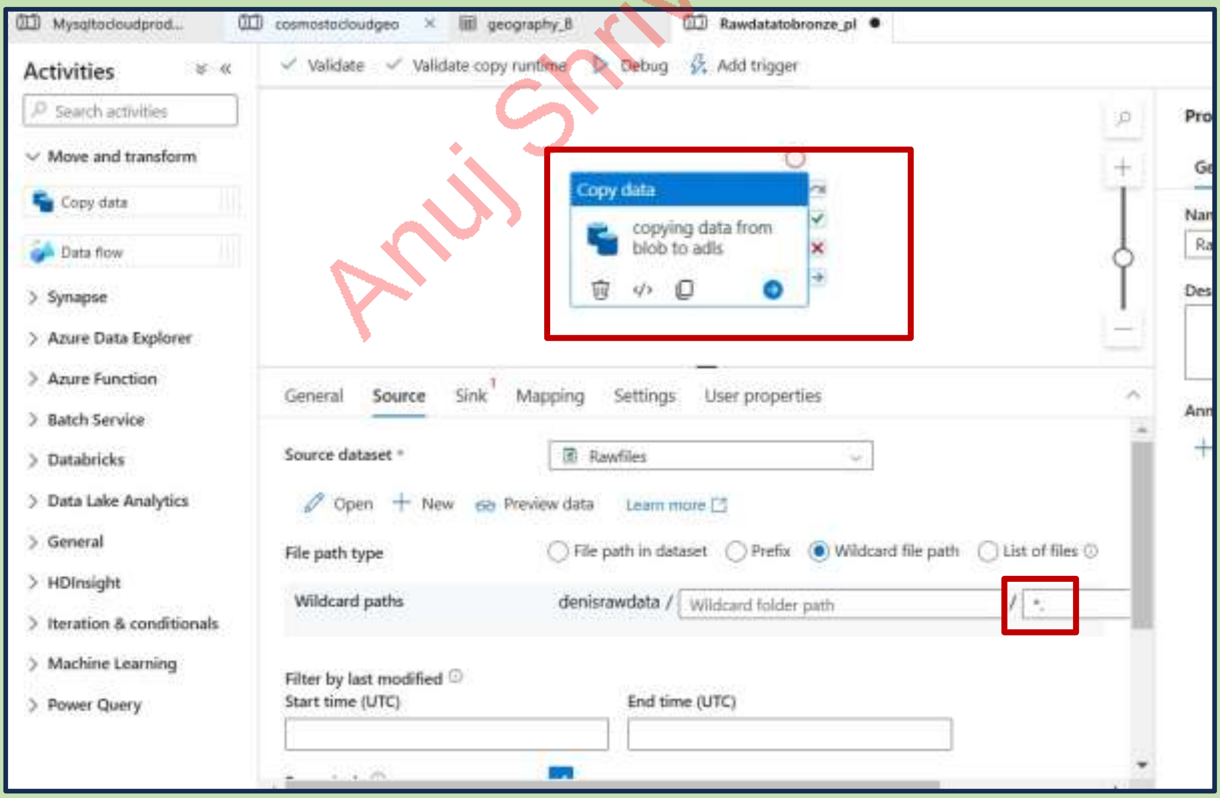
Select the source data

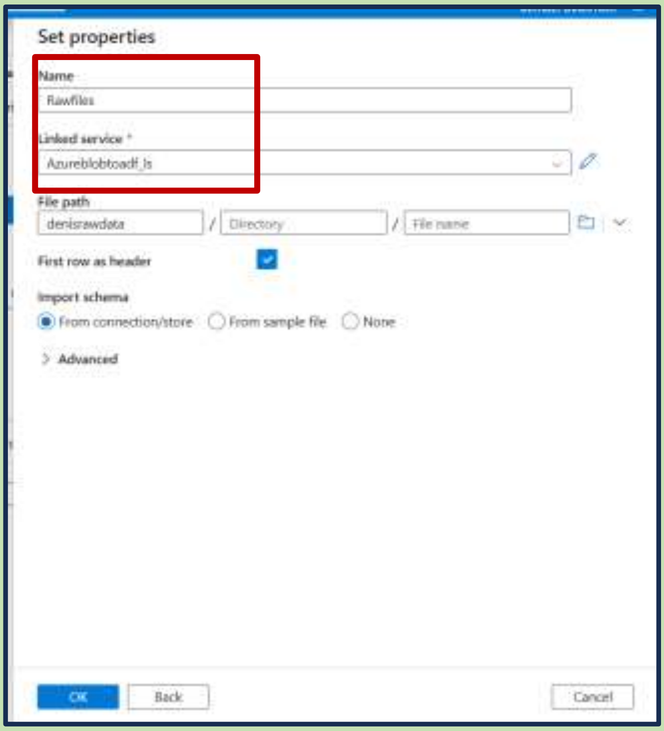


Name → linked servuce

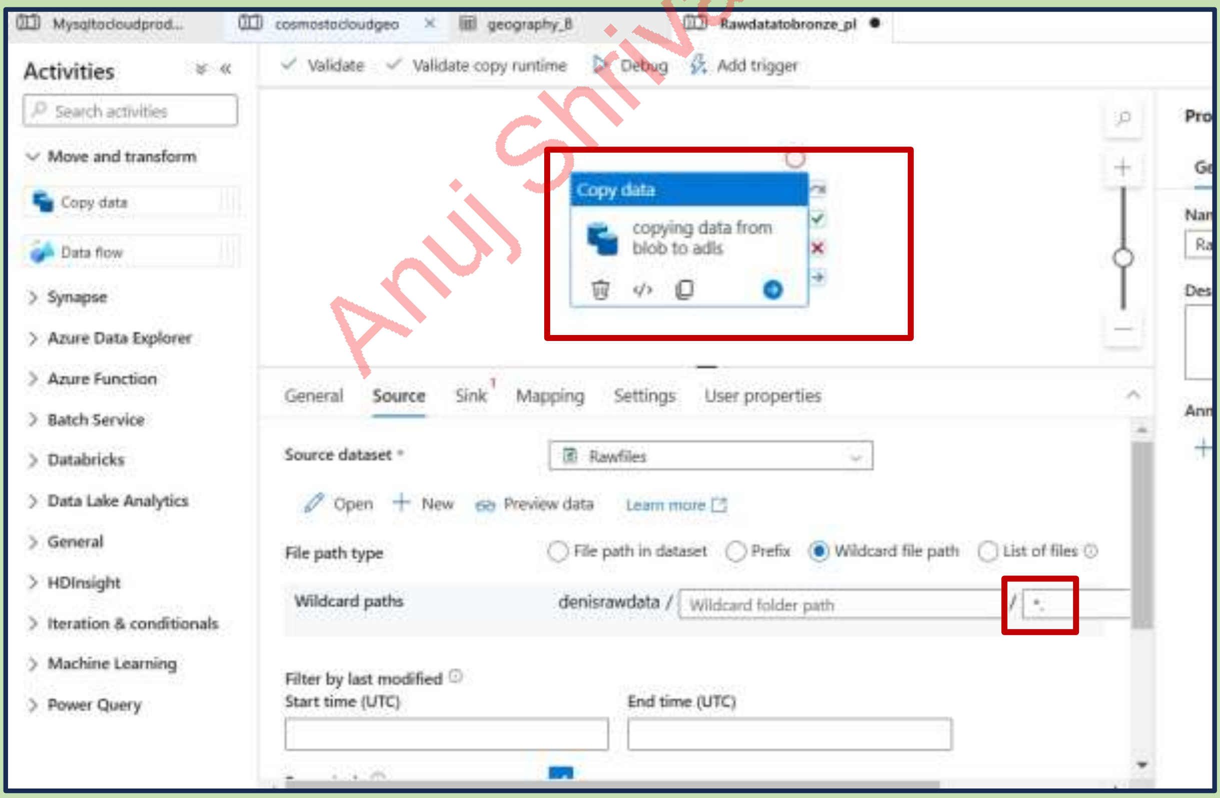


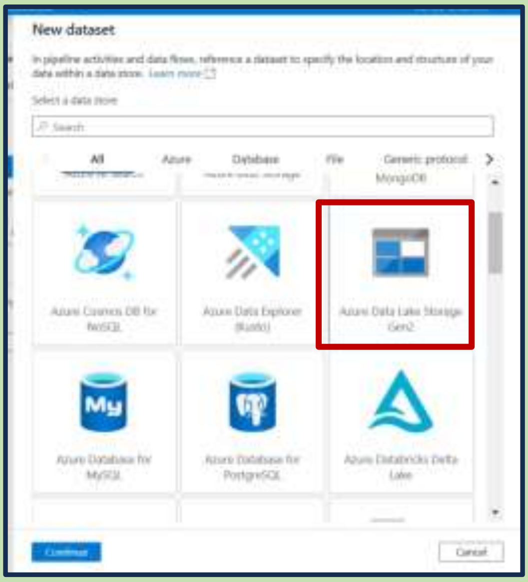
Source is done

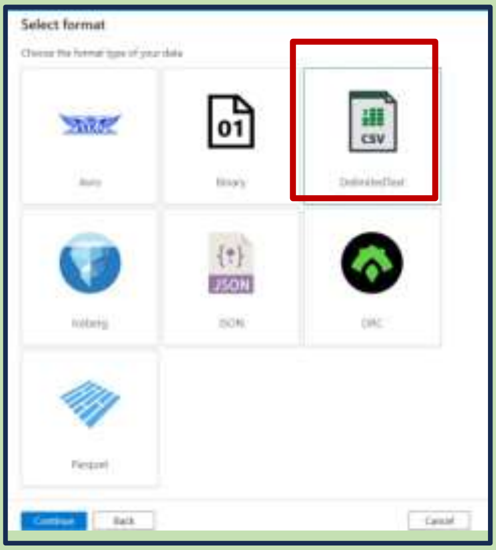




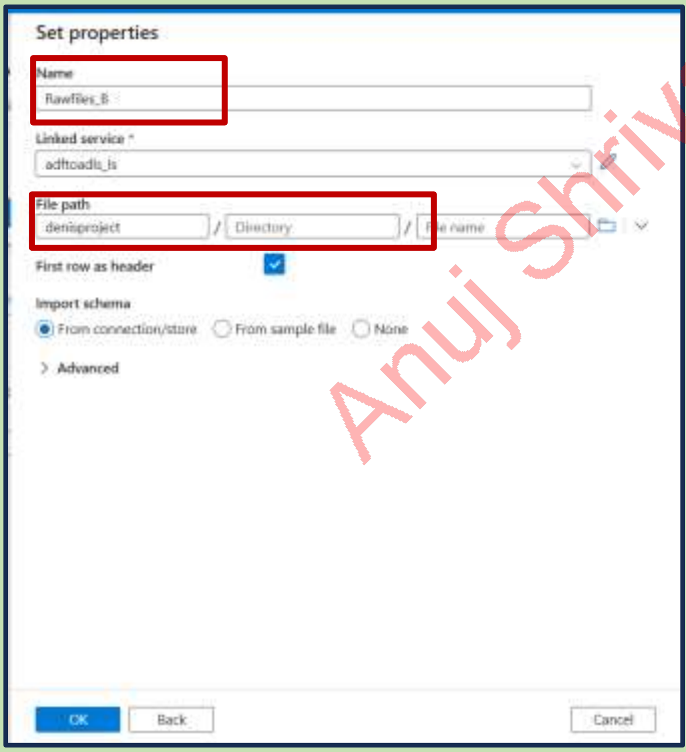
Source is done

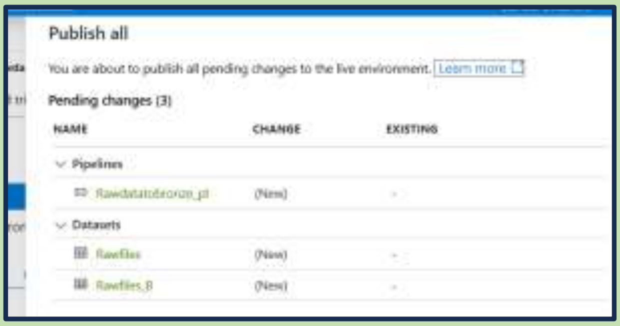






Give the name and select the file path and linked service

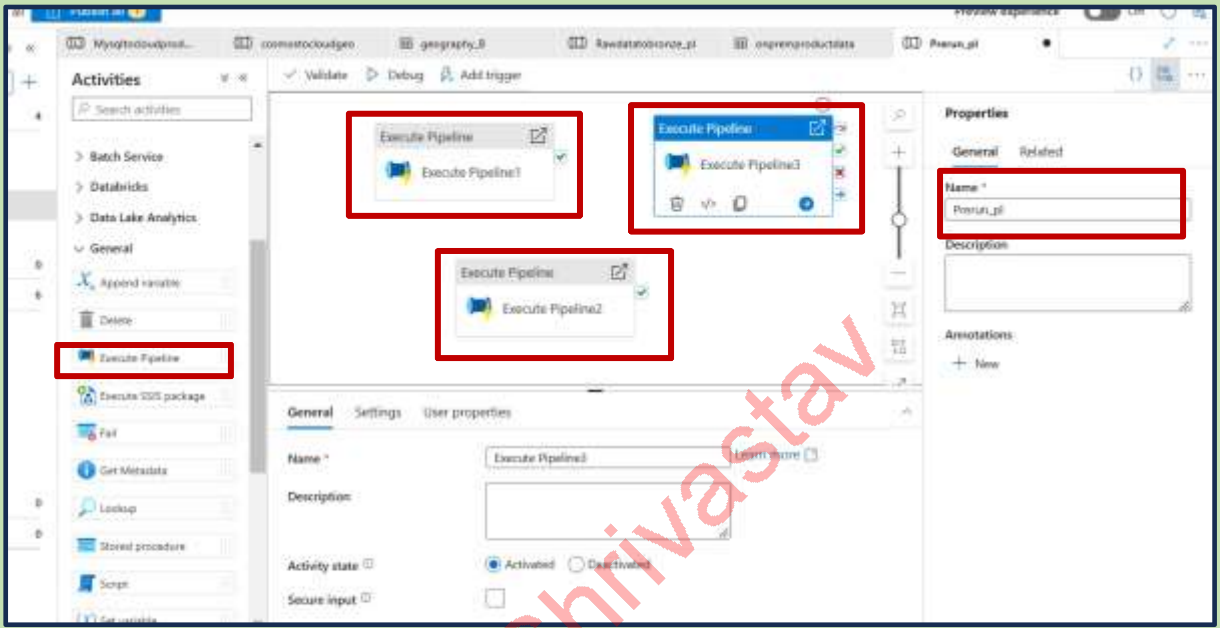




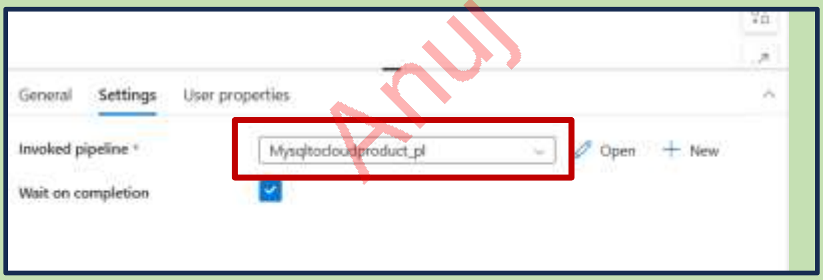
Pipeline 4:

Azure blob storage **Prerunpipeline →Prerunpipeline → Mysqltocloud → cosmostocloud → rawdatatobronze**

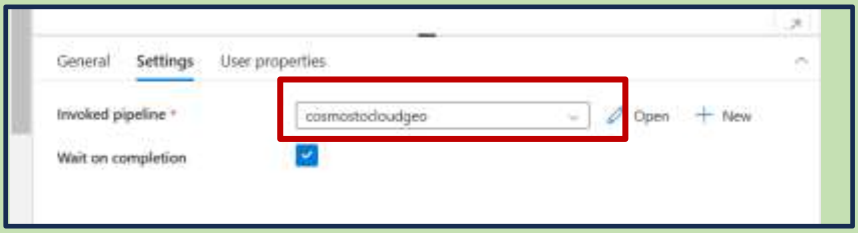
Now we need to run all the 3 pipelines together for that we need to create new pipeline Need to move execute pipeline and give the name of the pipeline



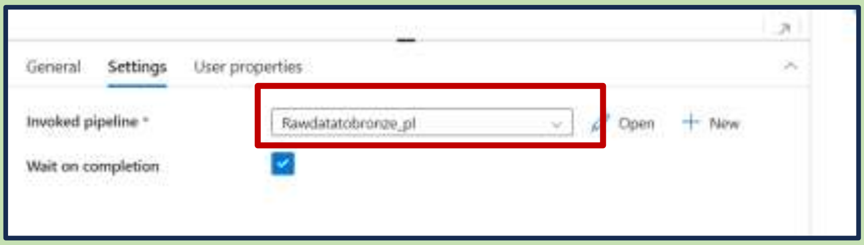
Select the 1st pipeline



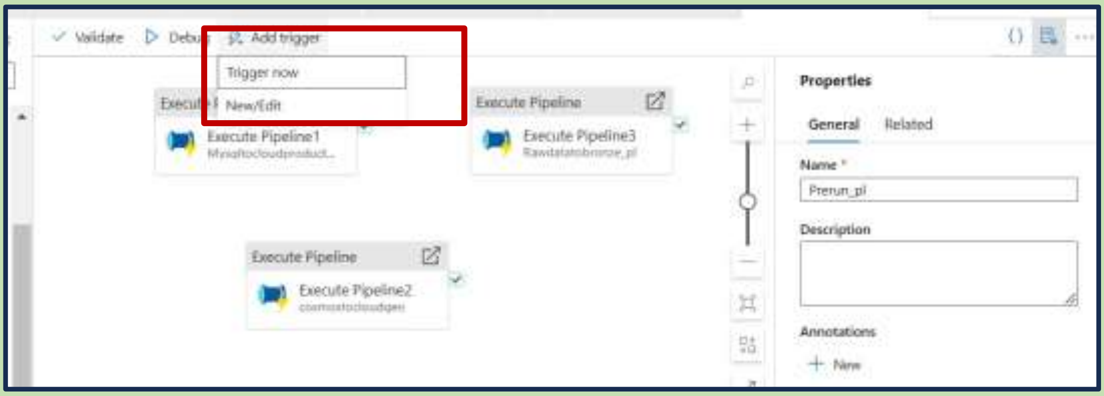
Select the 2nd pipeline



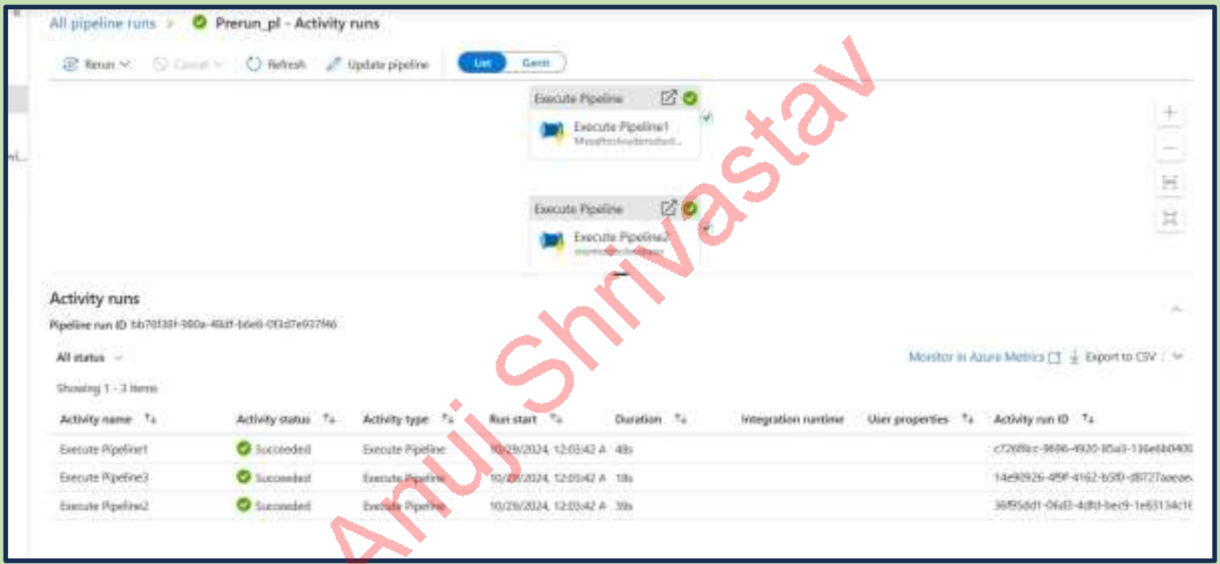
select the 3rd pipeline



Publish → trigger pipelines

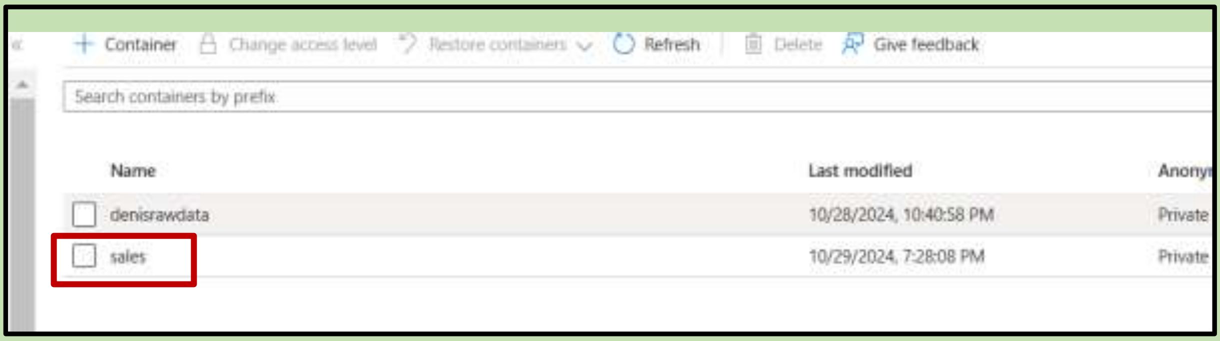


Pipeline runs successfully

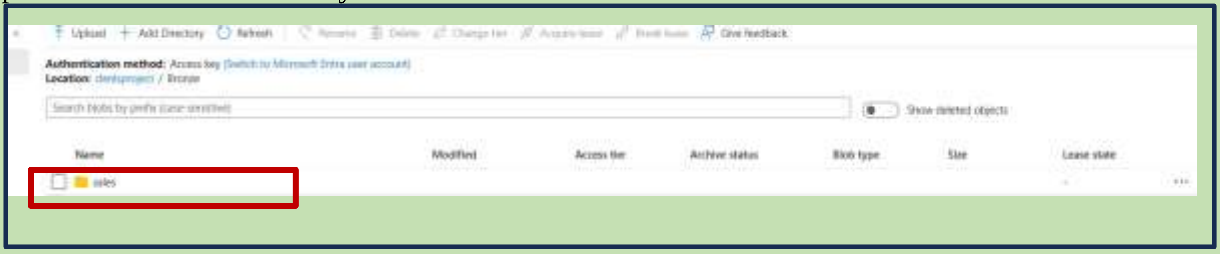


Pipeline 5:

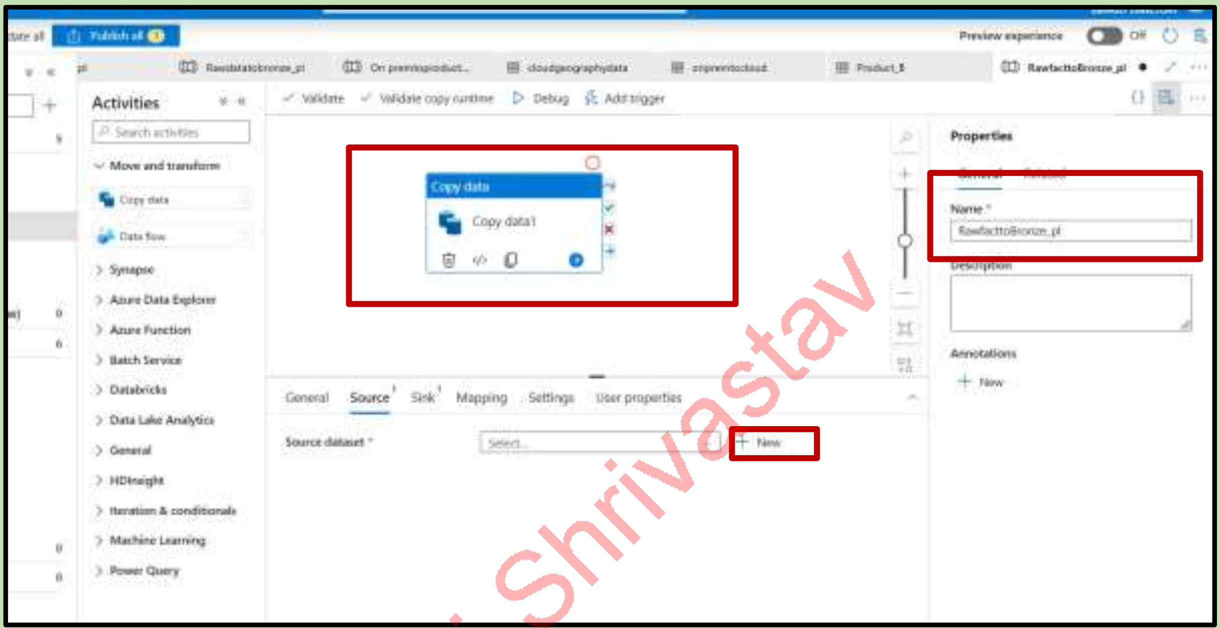
Now we need to dump the sales data in azure blob storage



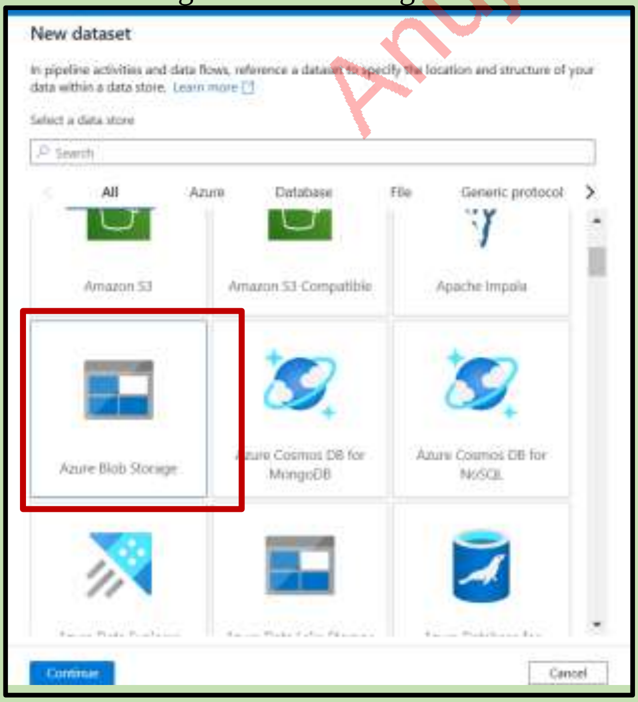
Open adls → create directory → sales



Now we need to create one more pipeline give the name of the pipeline and click on new

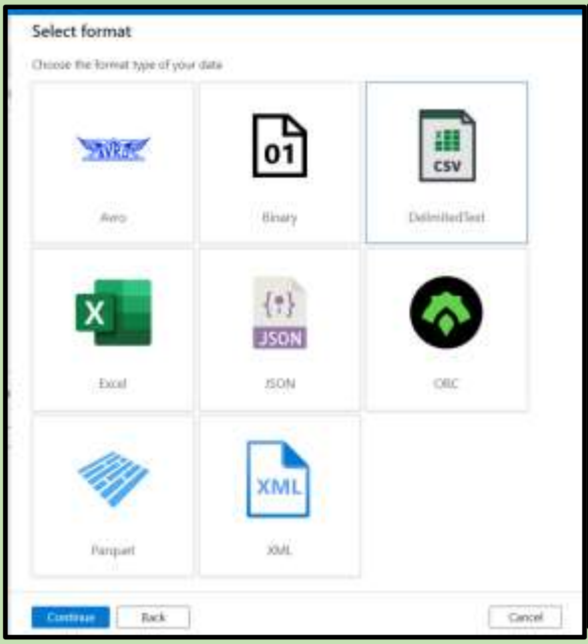


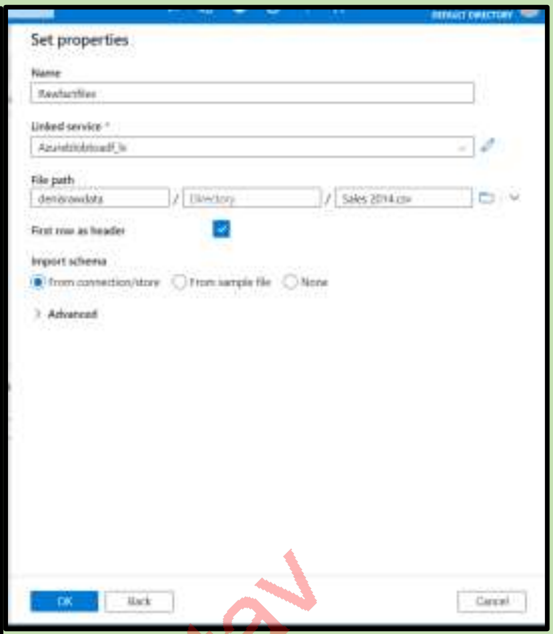
Data is coming from blob storage need to select blob and click on continue



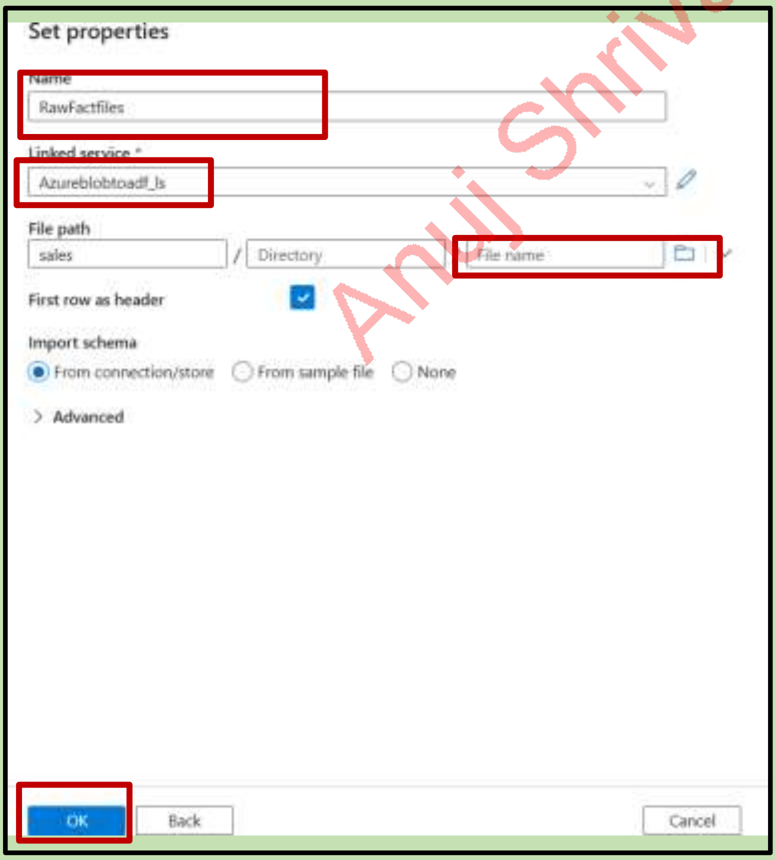
Click→ format

select →ls→one file name

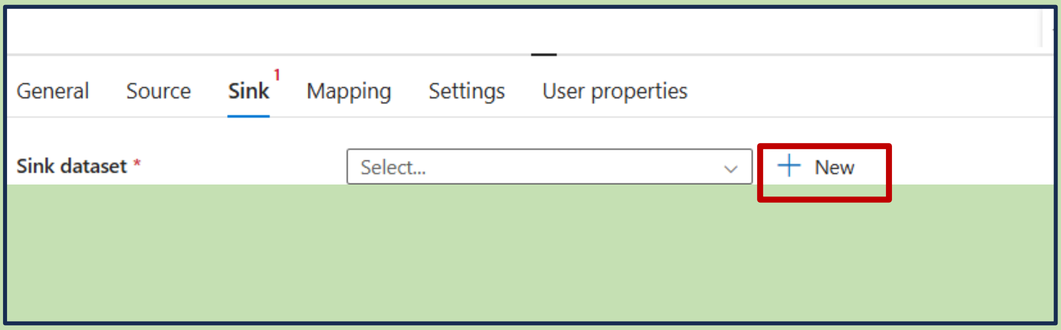




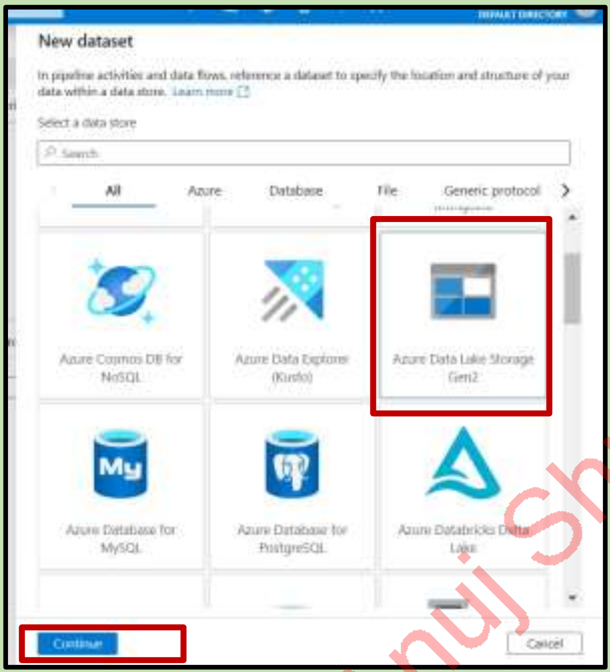
Name ls→ file path



Sink→ new

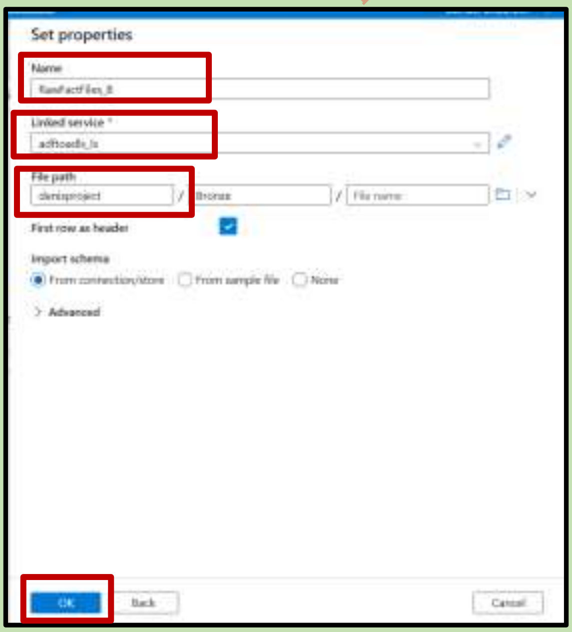


Click on new data set

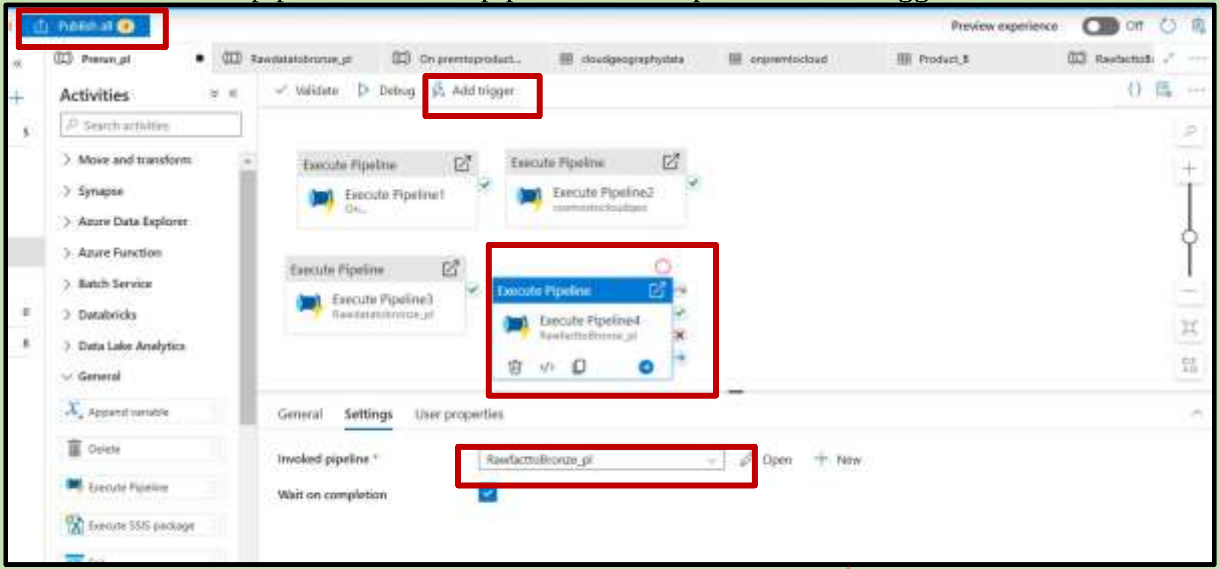


Select the format

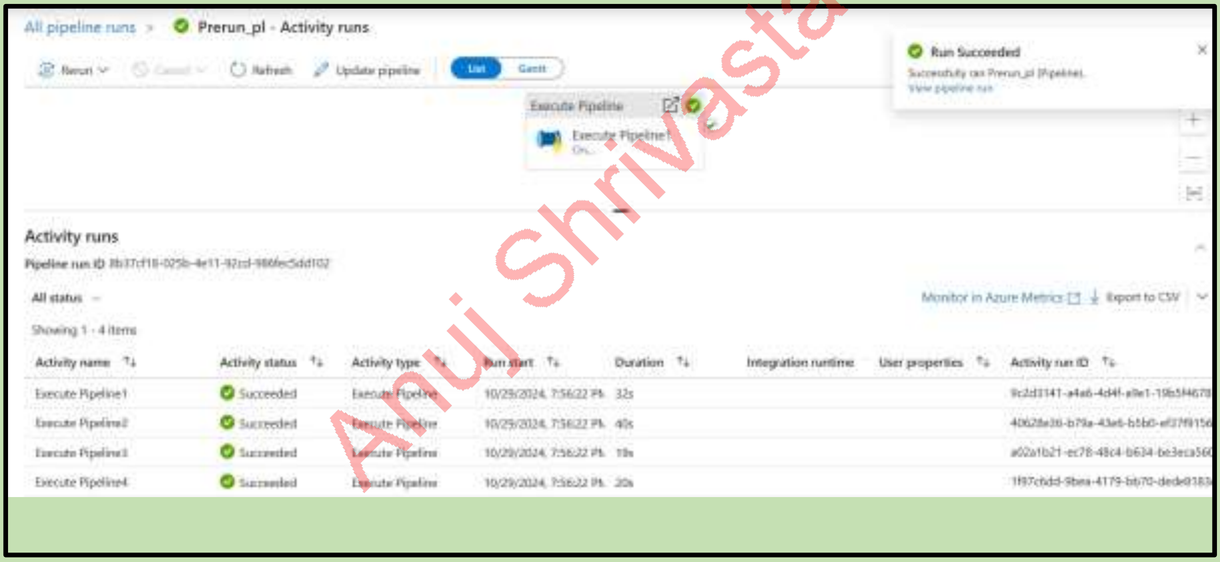
Now we can give the name ls and filepath



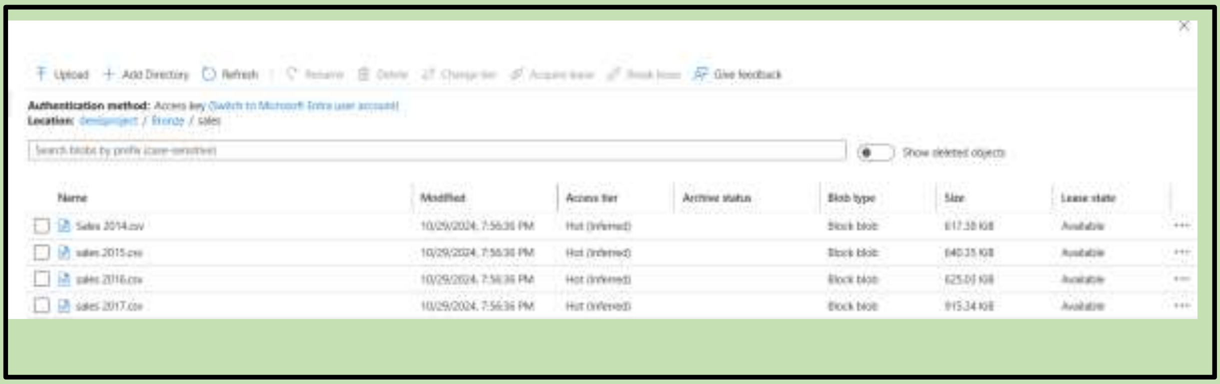
General→ execution pipeline→invoke pipeline name→publish→run trigger



Pipelines got run successfully



As we can see the data moved to adls--->bronze--->Sales



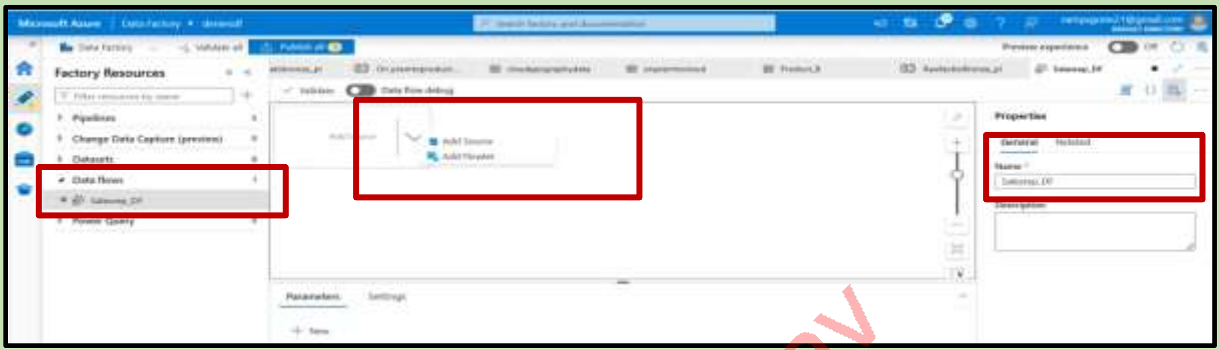
DATA TRANSFORMATION:

Data Flow 1:

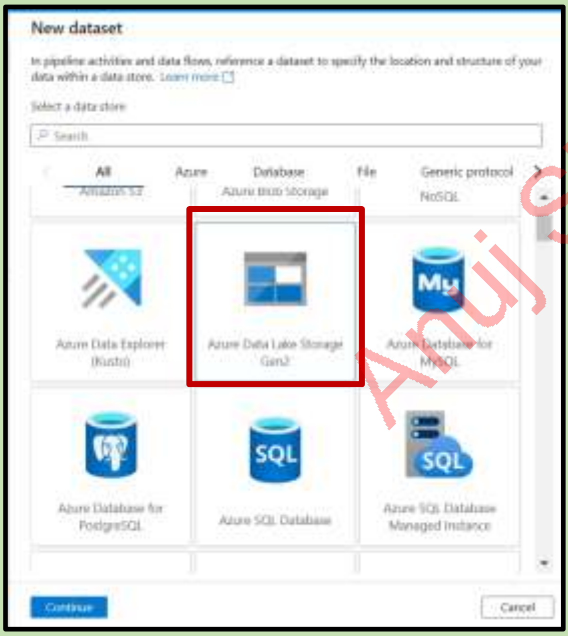
- **Salesrepdataflow**

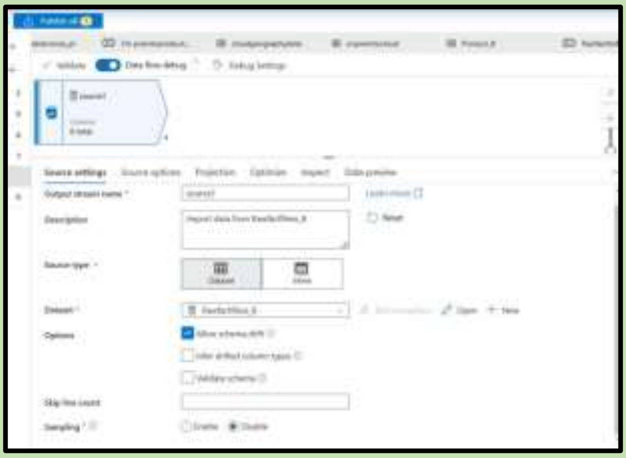
Click on Data flows --->New data flow --->Addsource

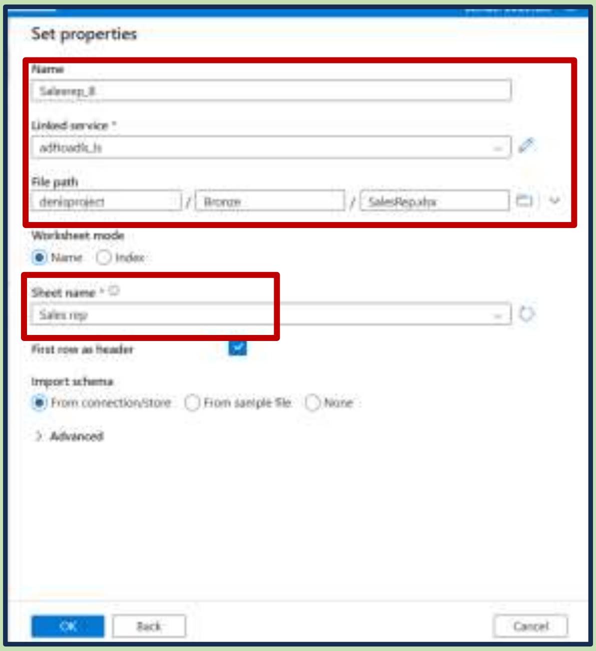
Give name



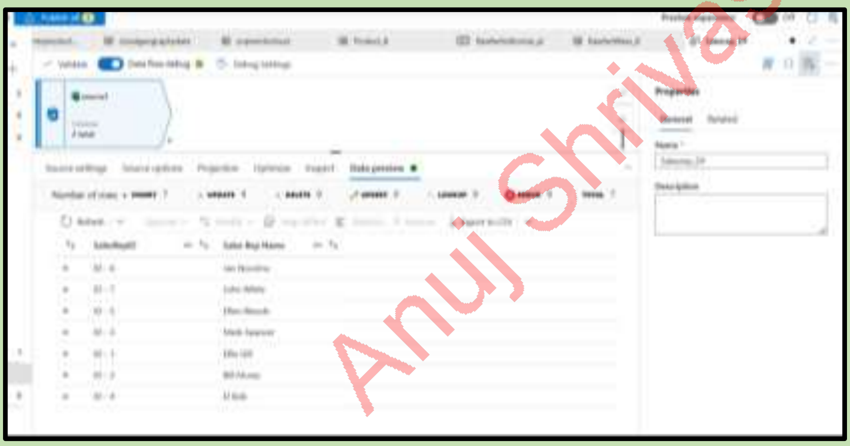
Datasets details





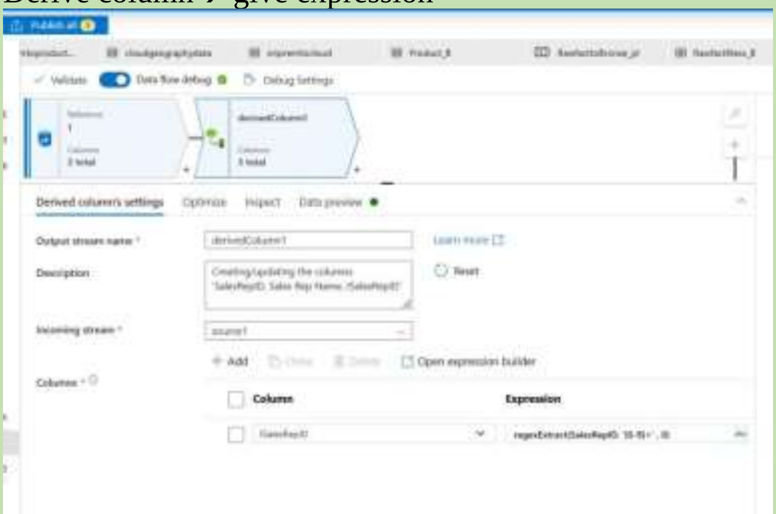


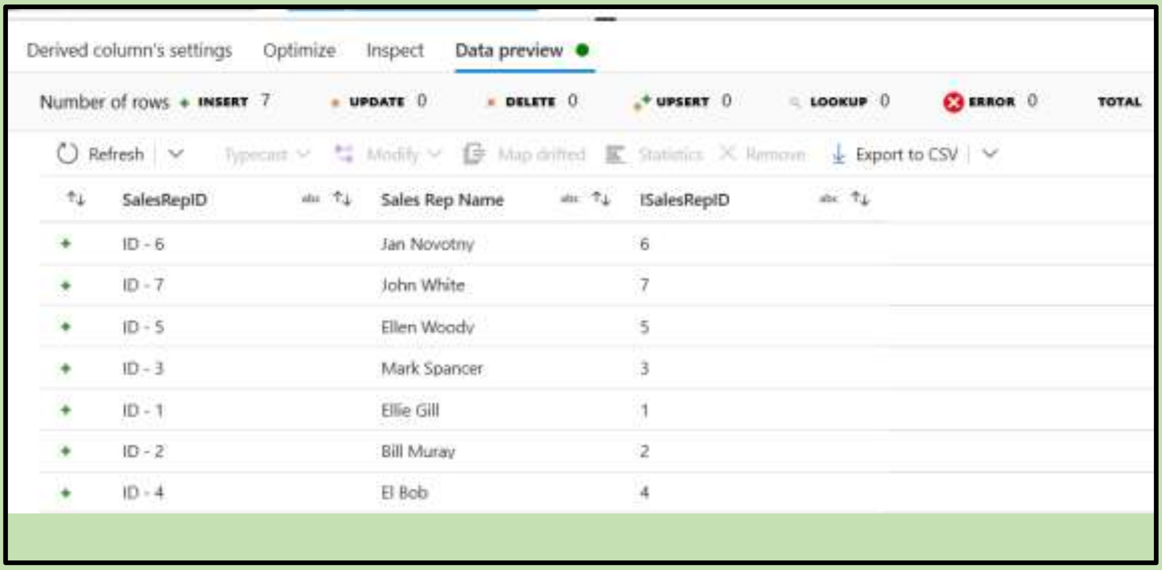
Data preview



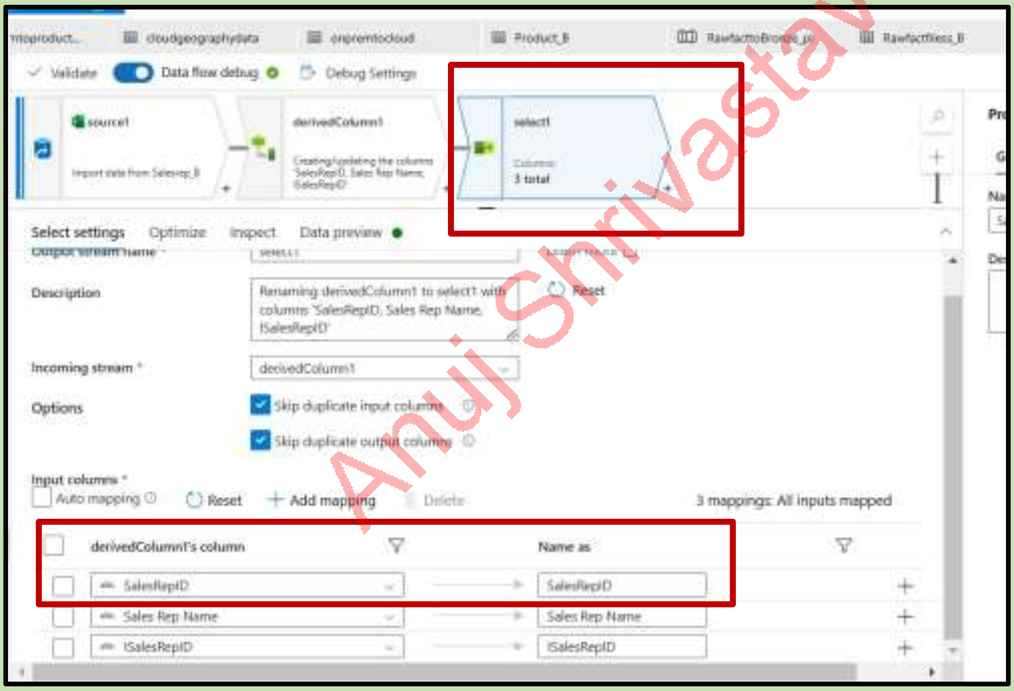
<u>Requirement</u>: remove 'ID

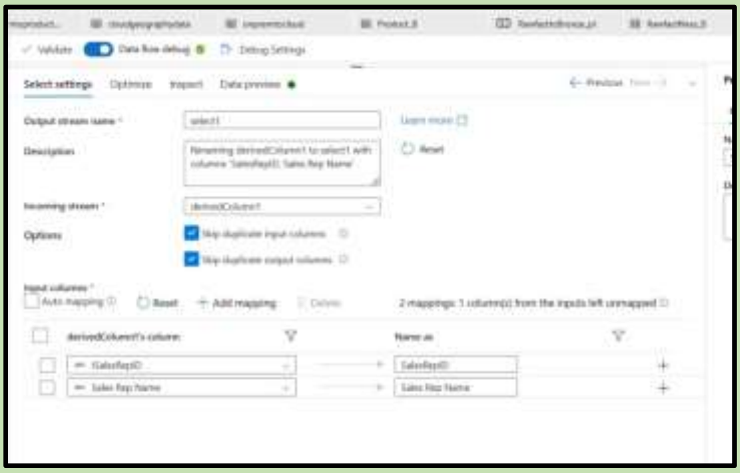
Derive column→ give expression



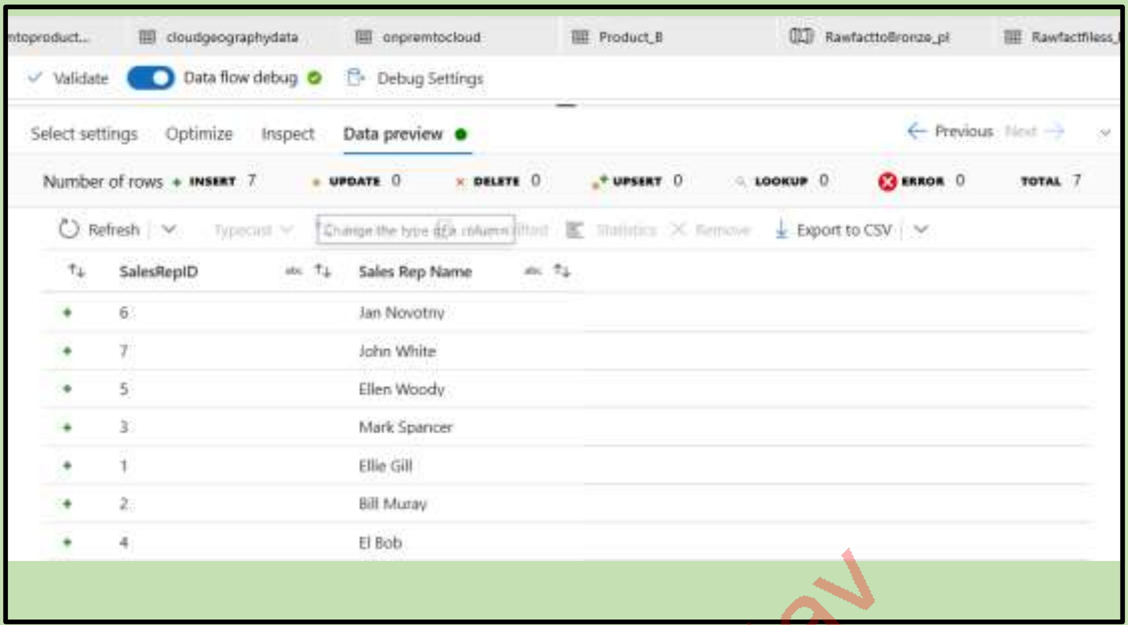


Now click on plus and add select and delete sales rep

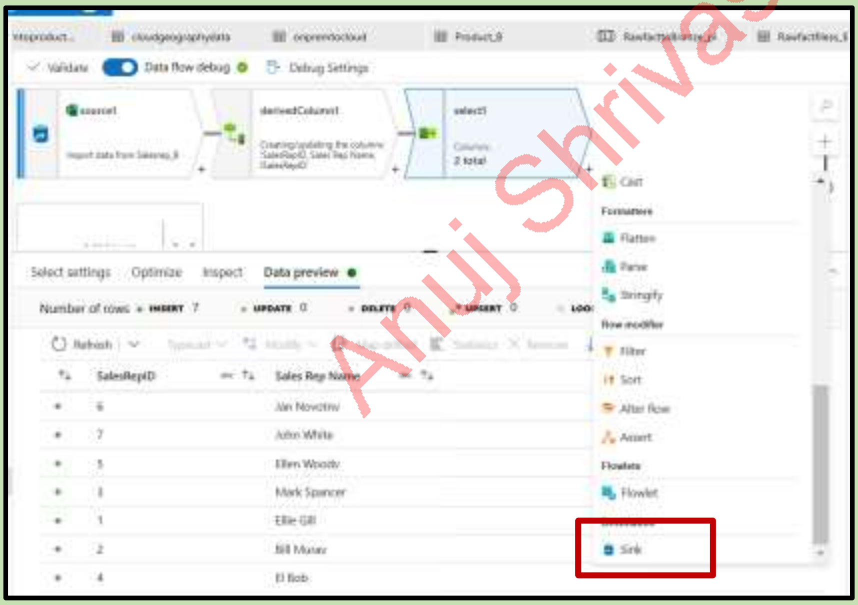


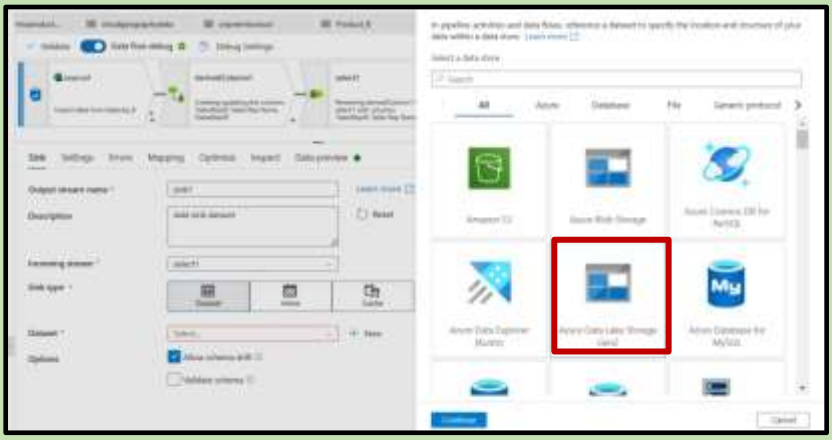


After clean→

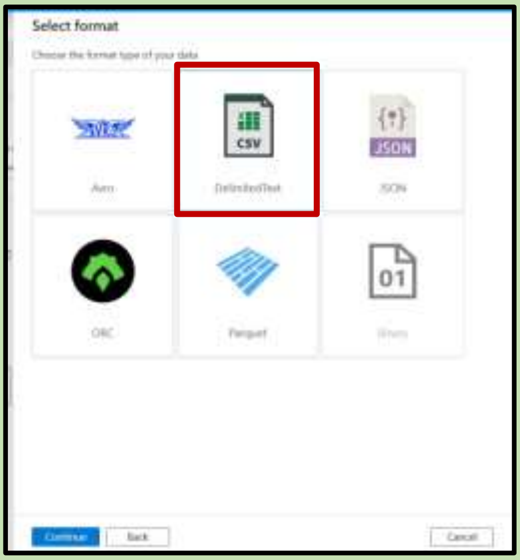


Select sink

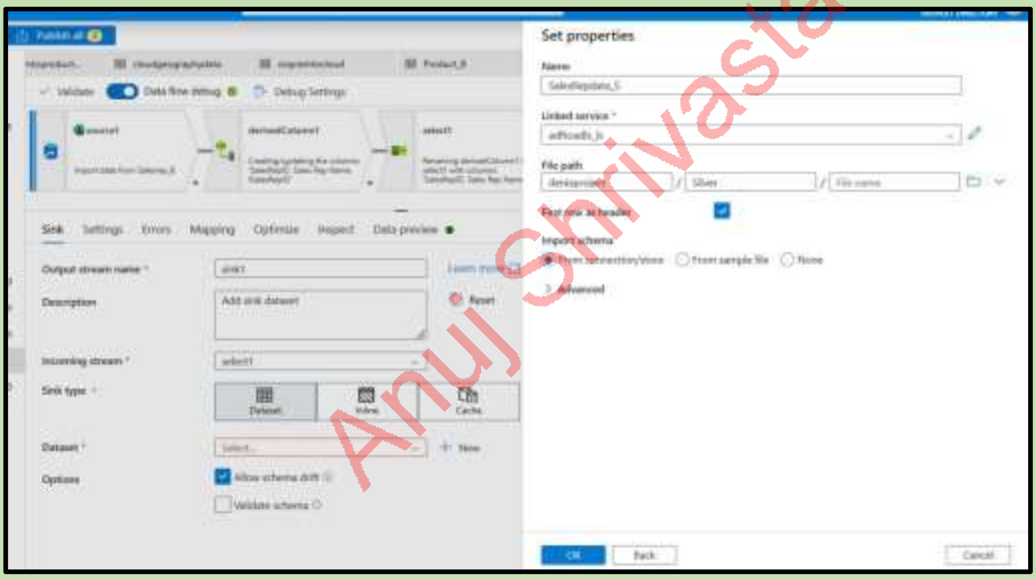




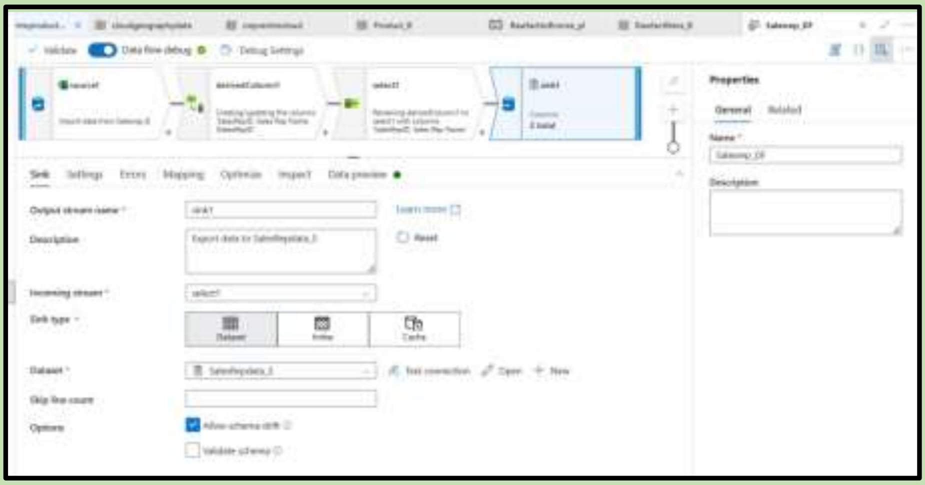
Select the format:



Set the propeties



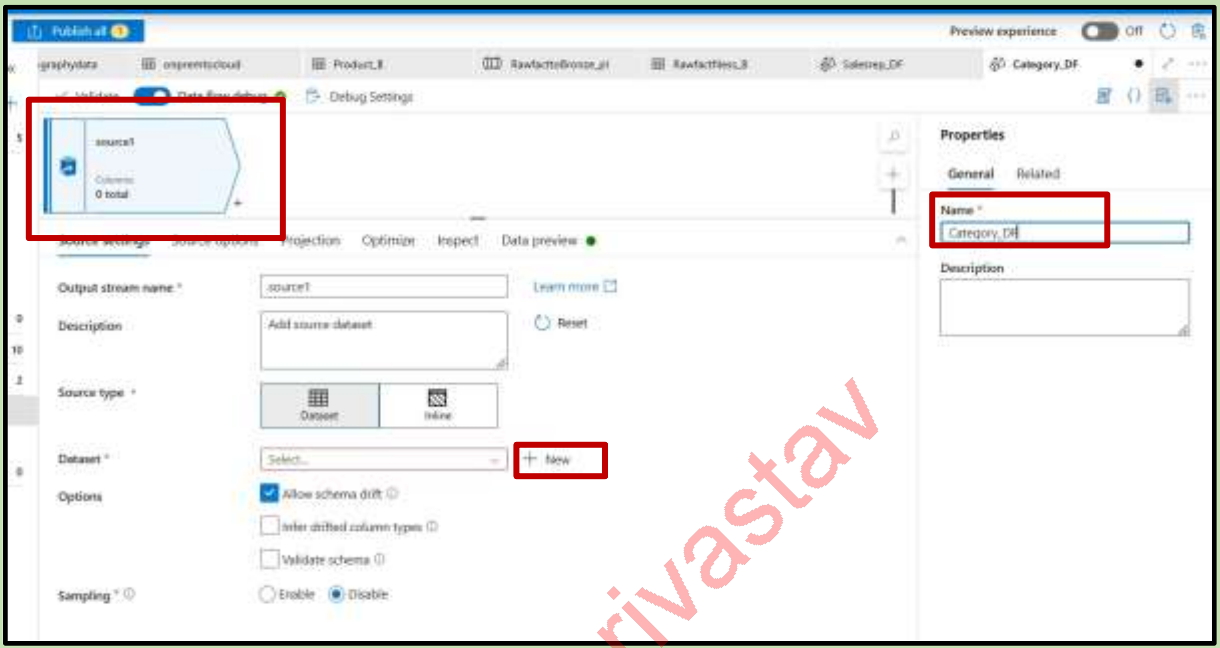
Data flow is created

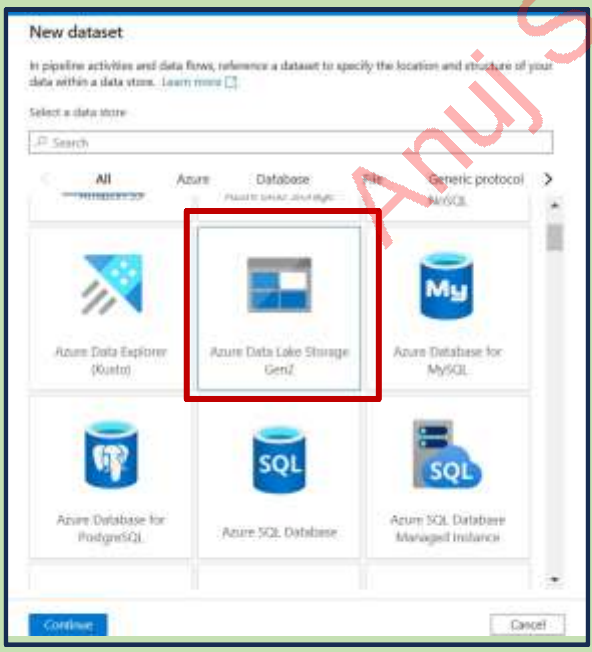


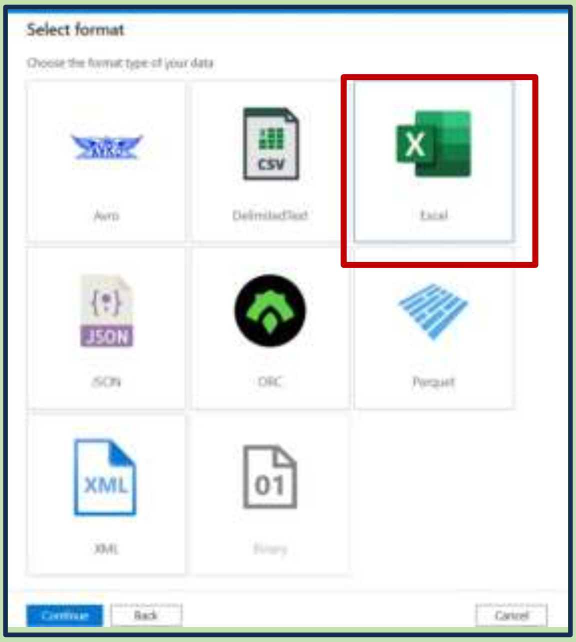
Data Flow2 :

- **Categorydataflow**

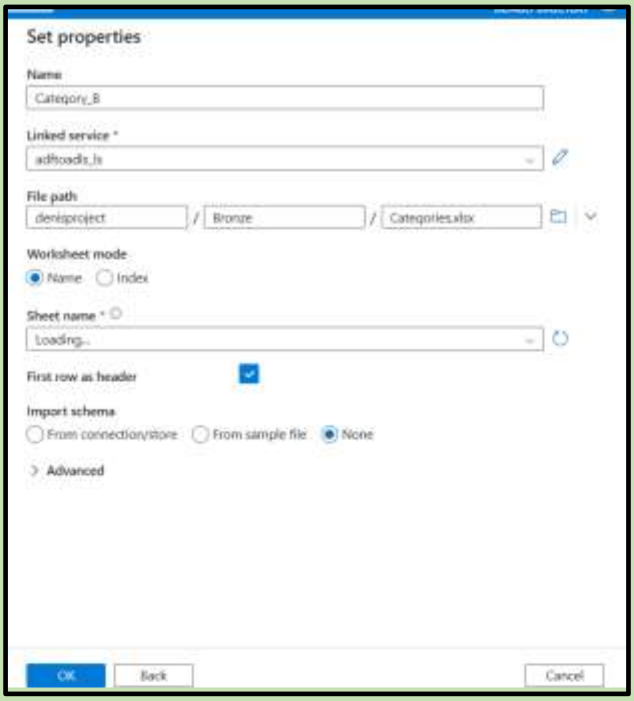
Create new data flow give the pipeline name and select new data set



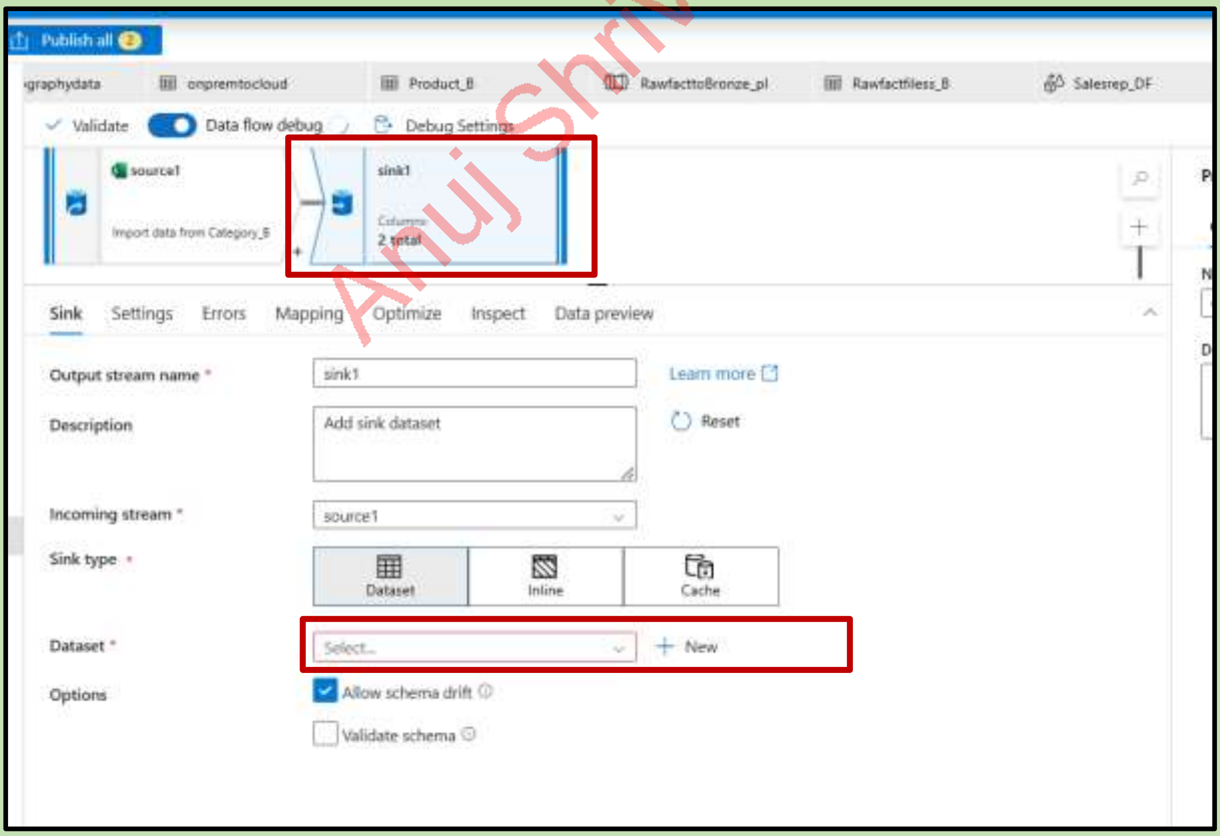


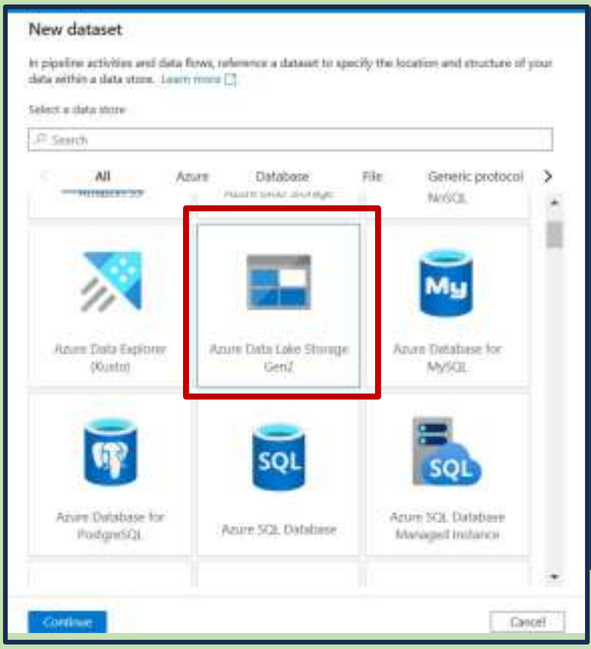


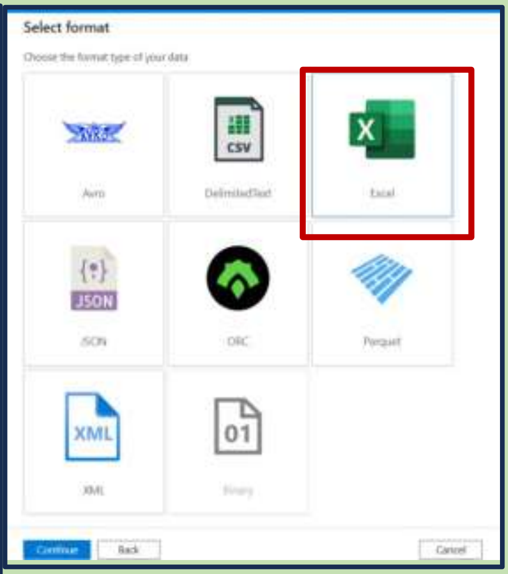
Select properties:



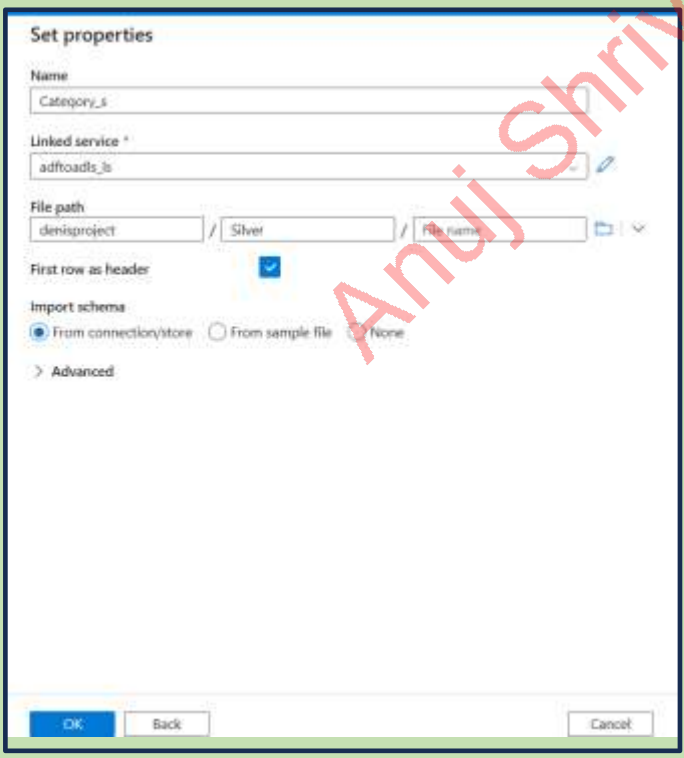
Now select sink because there is no need to clean data the data is good



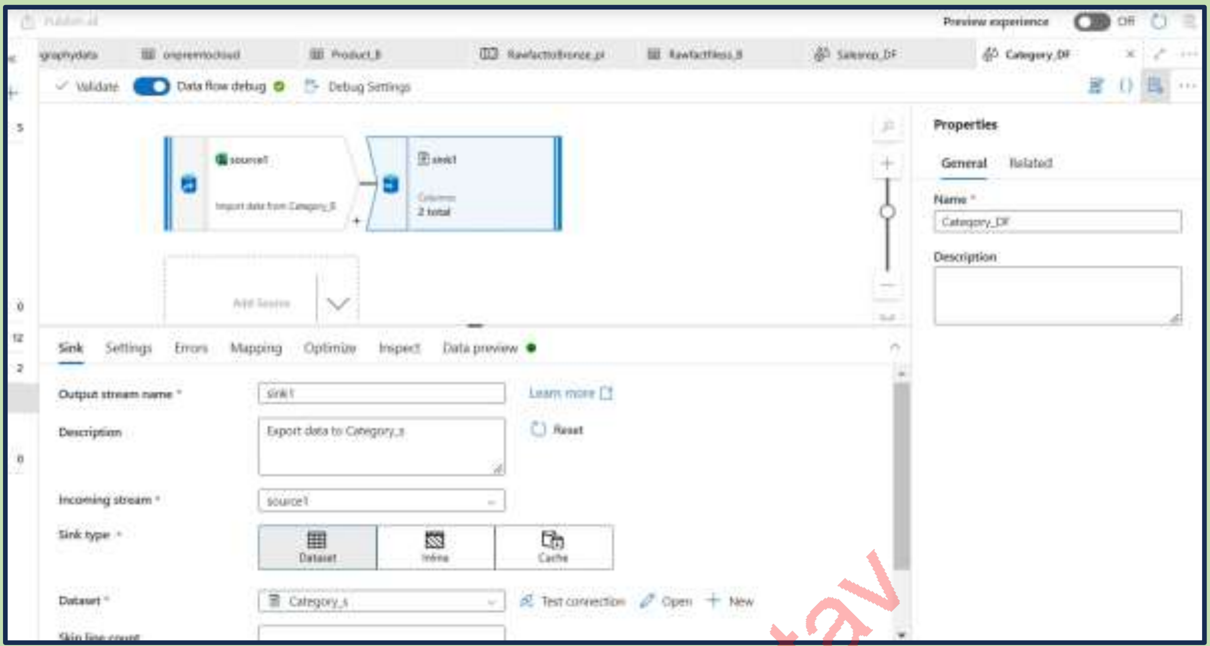




Set properties:



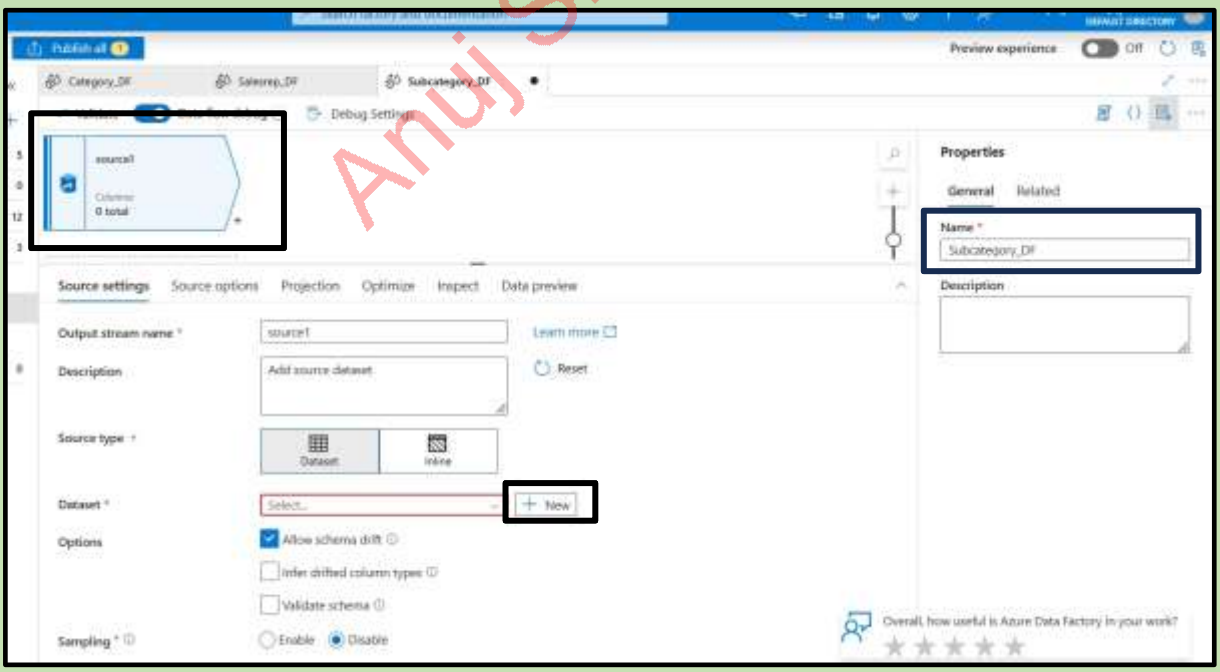
Dataflow created:

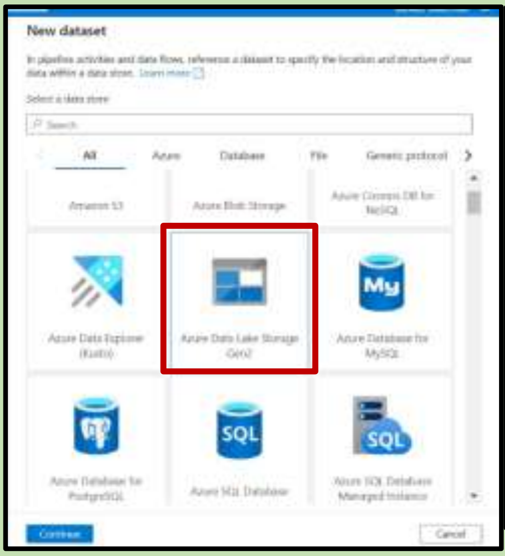


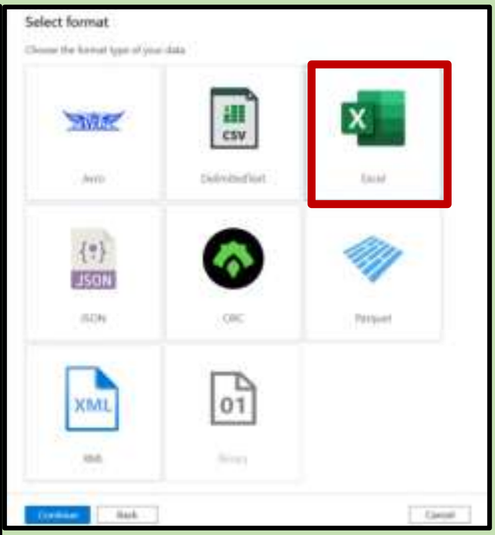
Data Flow3 :

- **Subcategorydataflow**

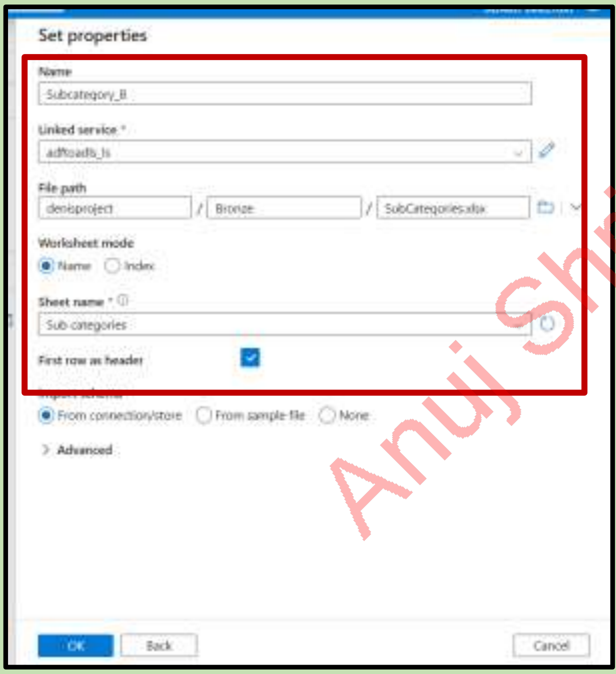
Create new data flow give the name of the data flow and select new in the data set



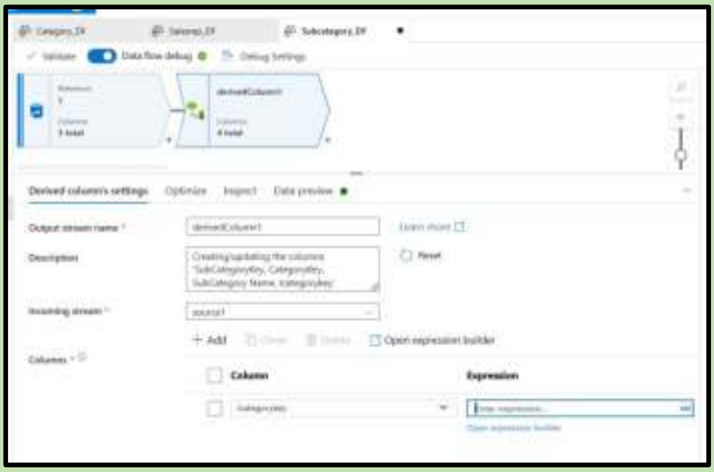




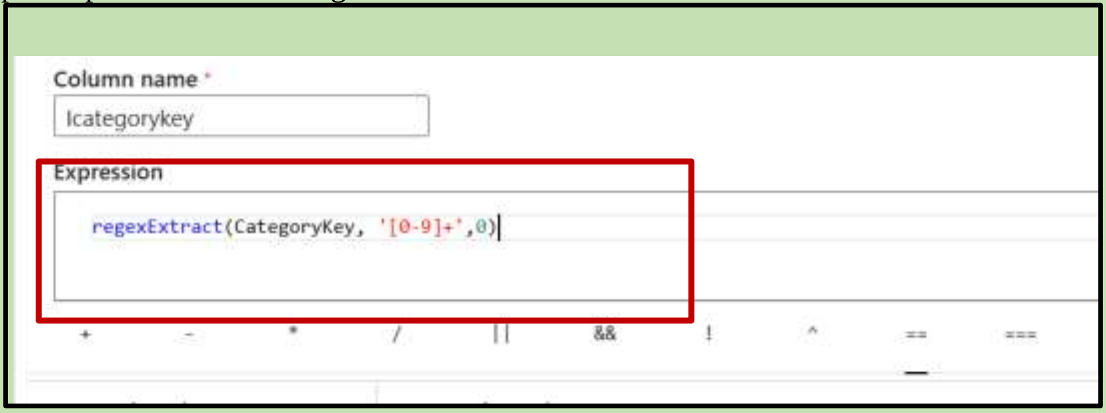
Set the properties



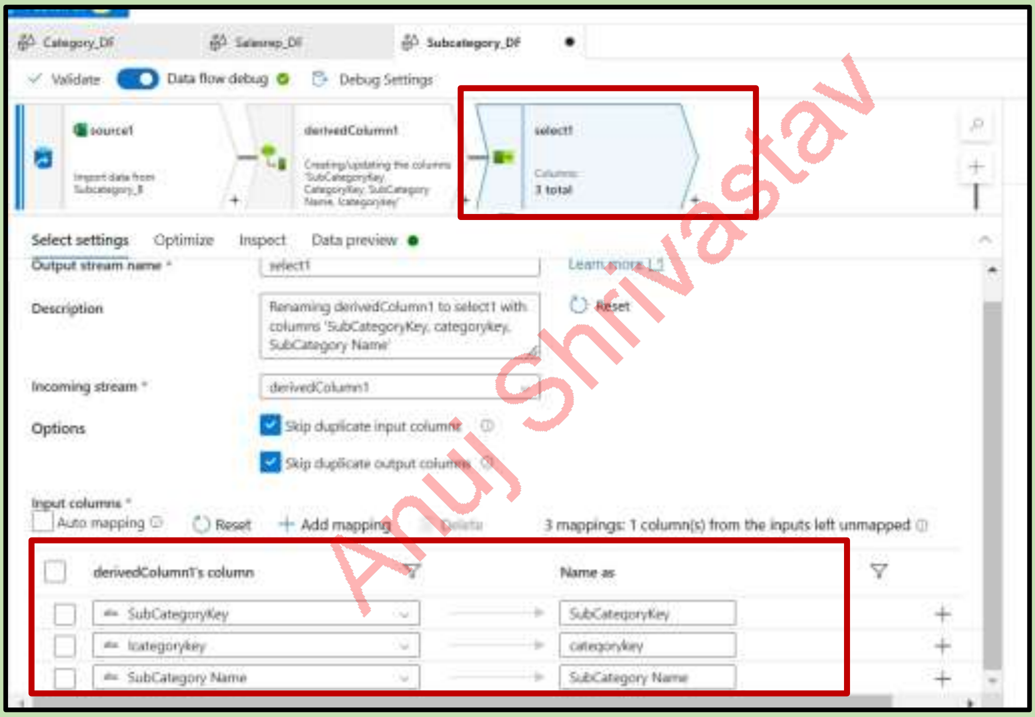
Now select the derived column and check the data preview



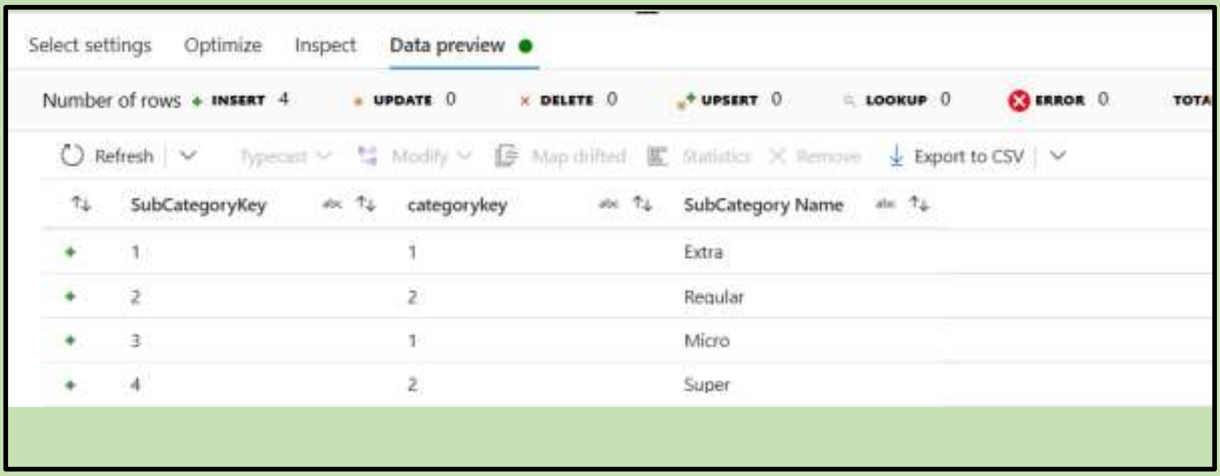
Open expression builder→ give command



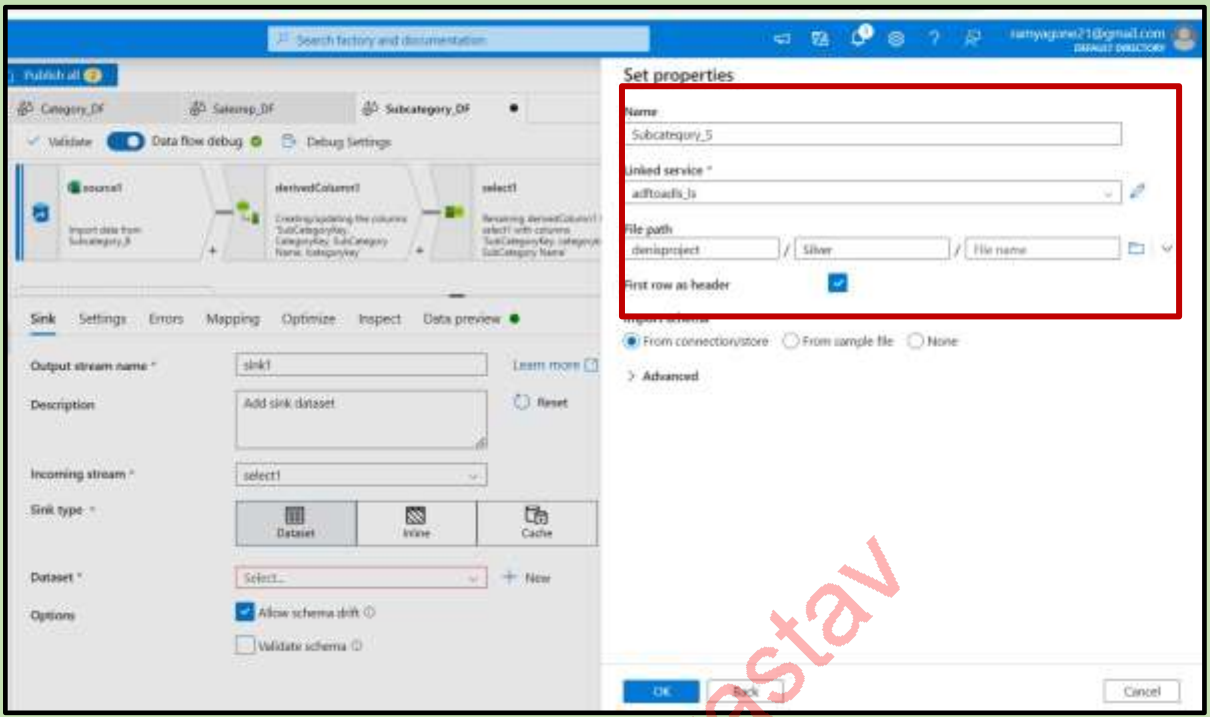
Now select one more column -> select and change the derived column name delete the previous derived column



We can see the data preview data is been sorted



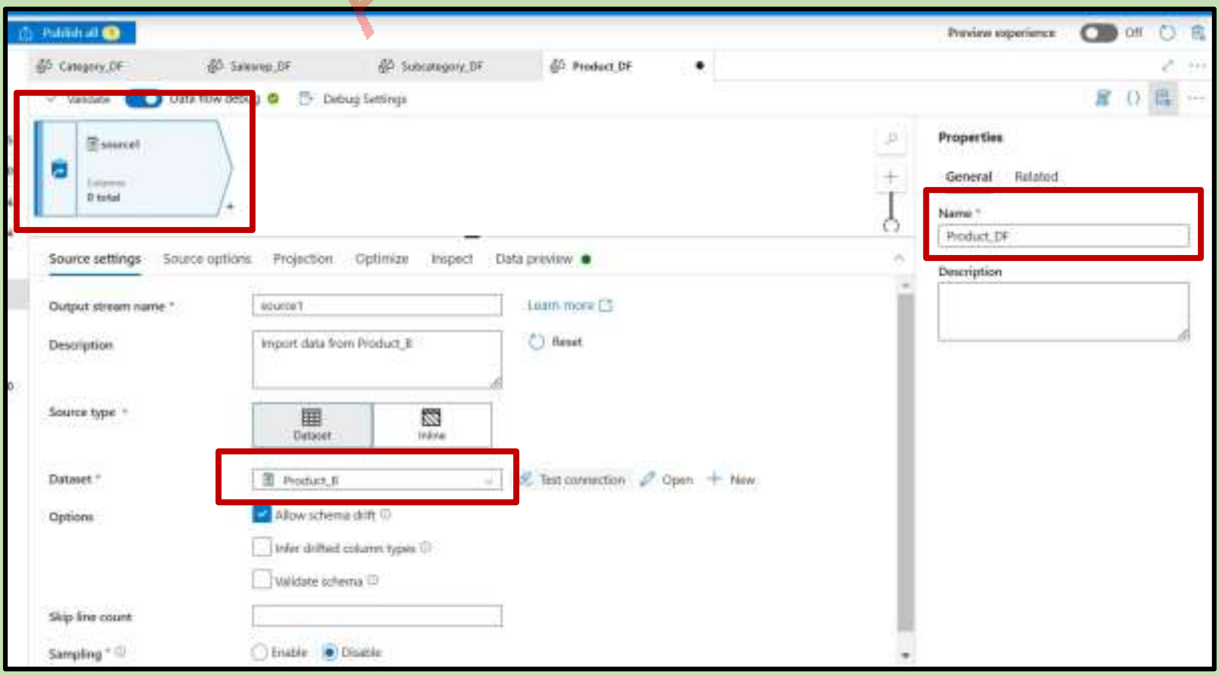
add more column to sink→set properties→publish



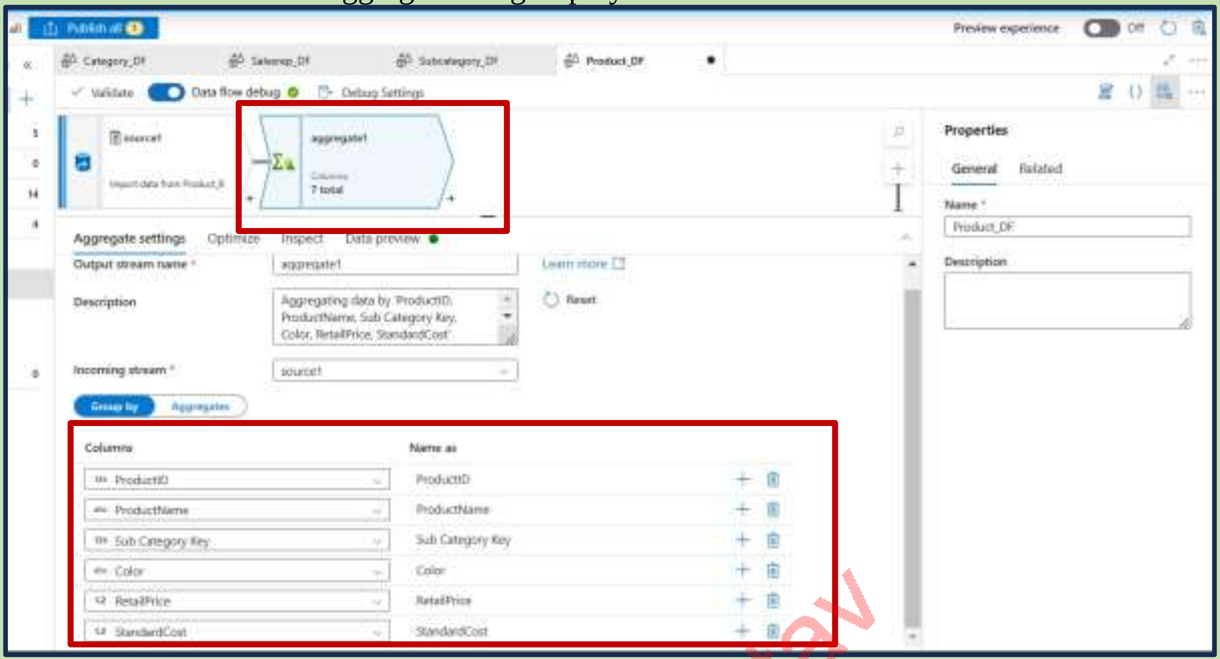
Data Flow4 :

- **Productdataflow**

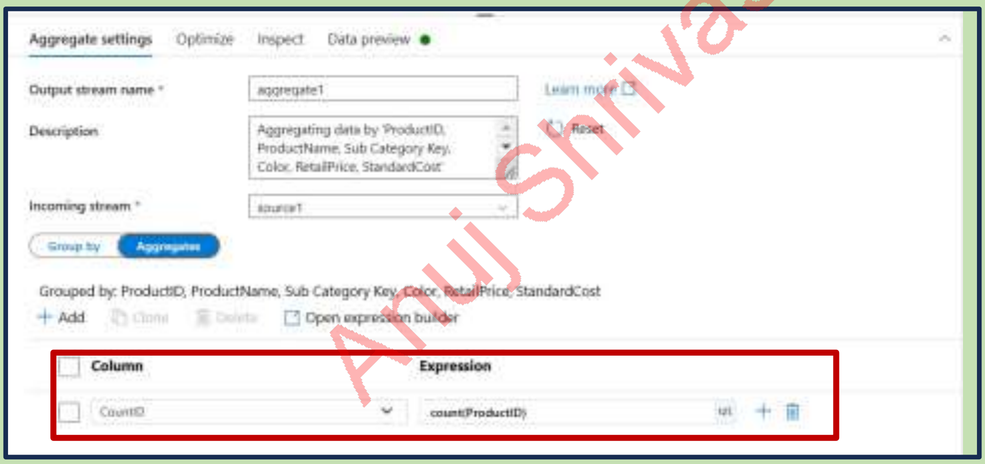
Add new data set name it as product_DF and add the source we can see that the new data set already there because we have brought the data from blob to adls bronze.



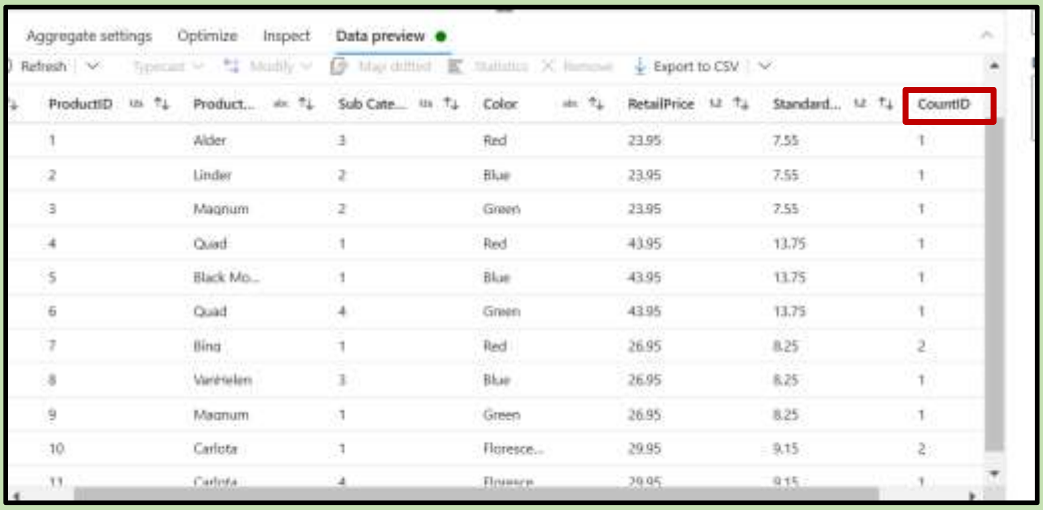
Select one more column →aggregation→group by all columns



Now we need to aggregate it

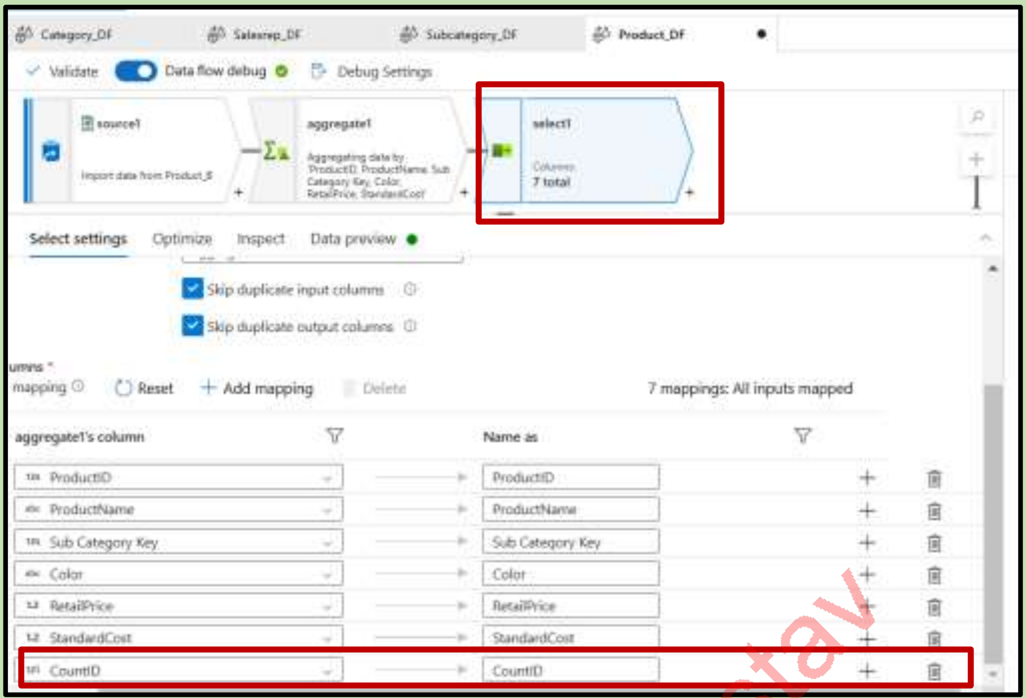


New column is added

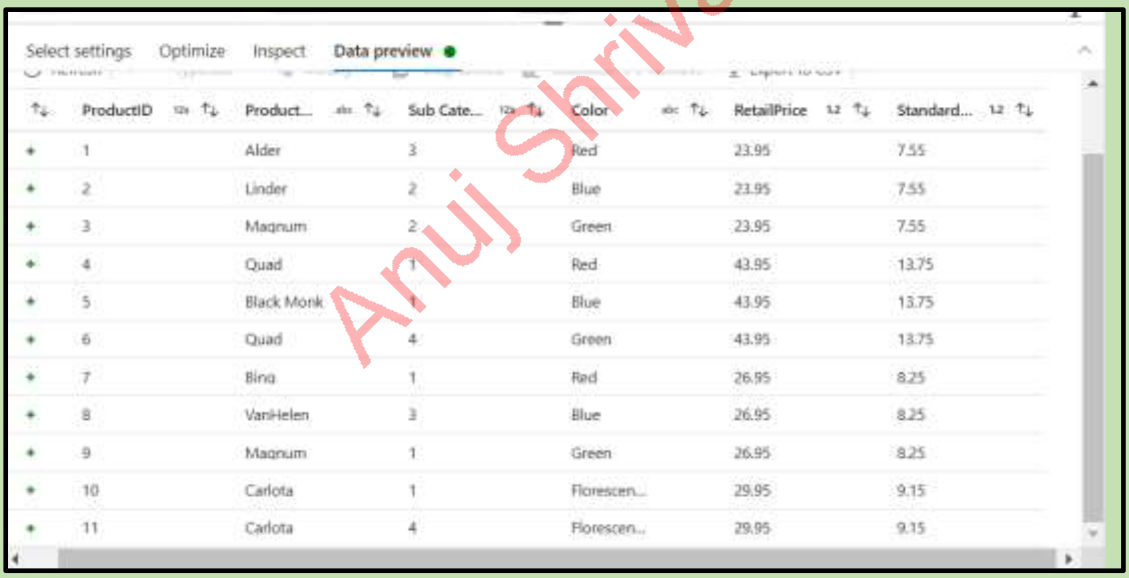


ProductID	Product...	Sub Cate...	Color	RetailPrice	Standard...	CountID
1	Alder	3	Red	23.95	7.55	1
2	Linder	2	Blue	23.95	7.55	1
3	Magnum	2	Green	23.95	7.55	1
4	Quad	1	Red	43.95	13.75	1
5	Black Mo...	1	Blue	43.95	13.75	1
6	Quad	4	Green	43.95	13.75	1
7	Bing	1	Red	26.95	8.25	2
8	VanHelen	3	Blue	26.95	8.25	1
9	Magnum	1	Green	26.95	8.25	1
10	Carlota	1	Floresce...	29.95	9.15	2
11	Carlota	4	Floresce	29.95	9.15	1

Add one more column→ select→delete countID

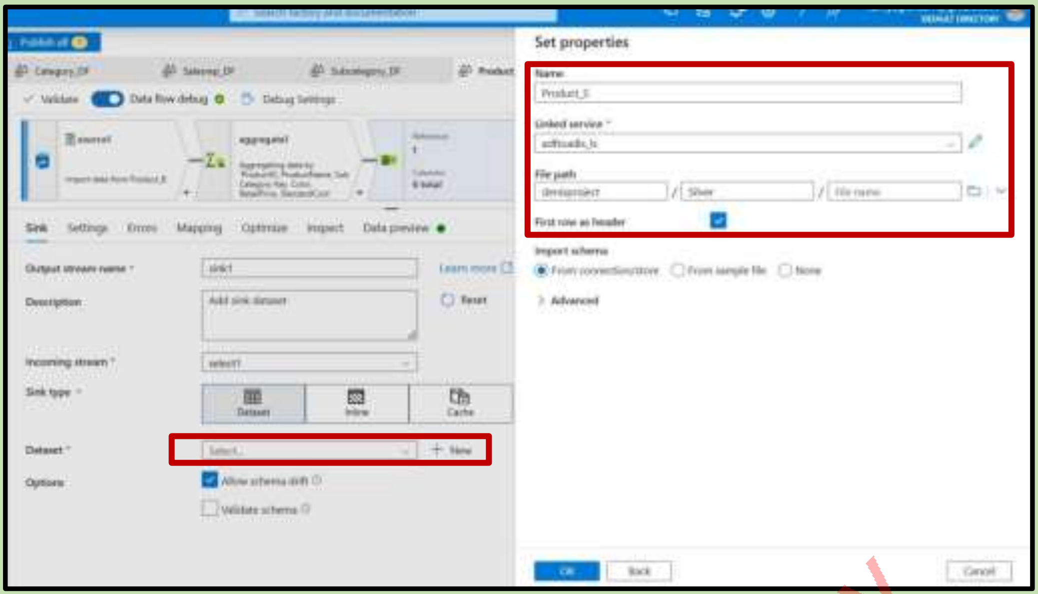


Successfully removed the duplicate values



ProductID	Product...	Sub Cate...	Color	RetailPrice	Standard...
1	Alder	3	Red	23.95	7.55
2	Linder	2	Blue	23.95	7.55
3	Magnum	2	Green	23.95	7.55
4	Quad	1	Red	43.95	13.75
5	Black Monk	1	Blue	43.95	13.75
6	Quad	4	Green	43.95	13.75
7	Bing	1	Red	26.95	8.25
8	VanHelen	3	Blue	26.95	8.25
9	Magnum	1	Green	26.95	8.25
10	Carlota	1	Florescen...	29.95	9.15
11	Carlota	4	Florescen...	29.95	9.15

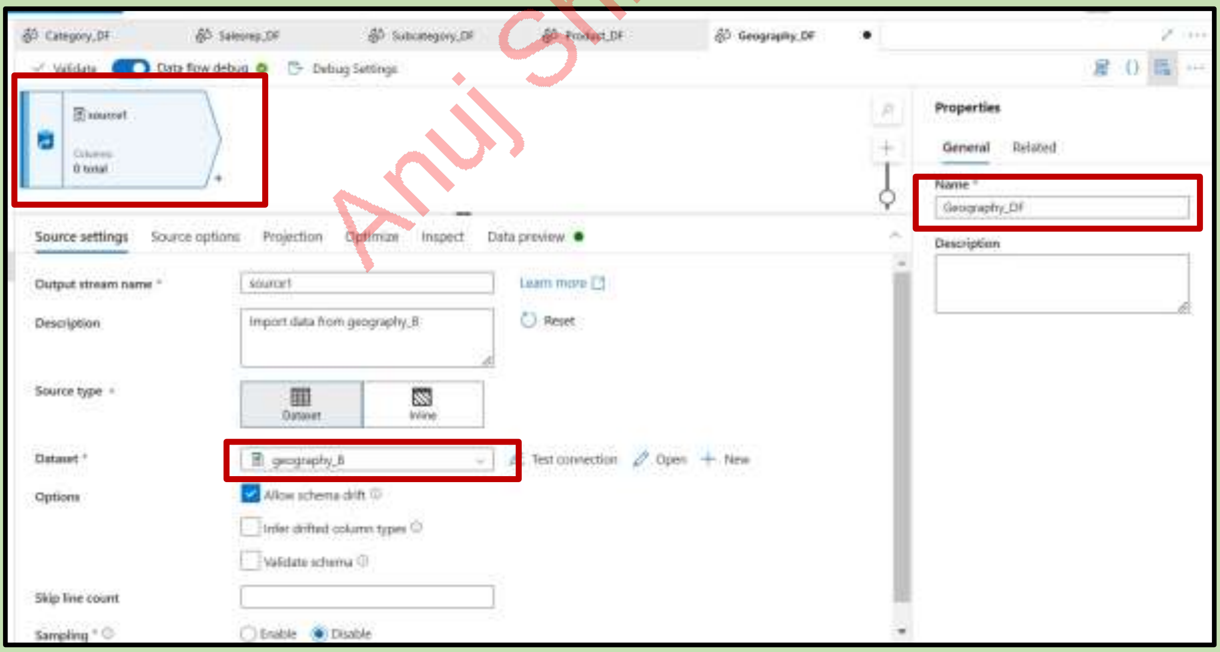
Add sink add data set give the set propertied details and publish



Data Flow5 :

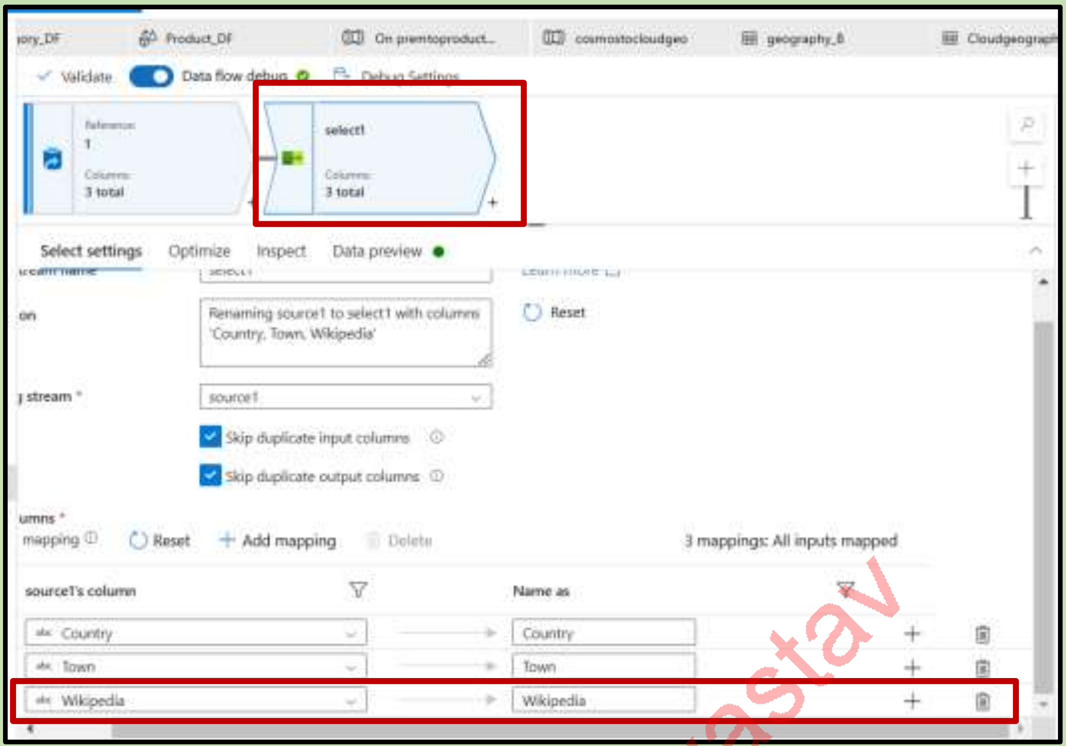
- **Geographydataflow**

Add source and select the data set

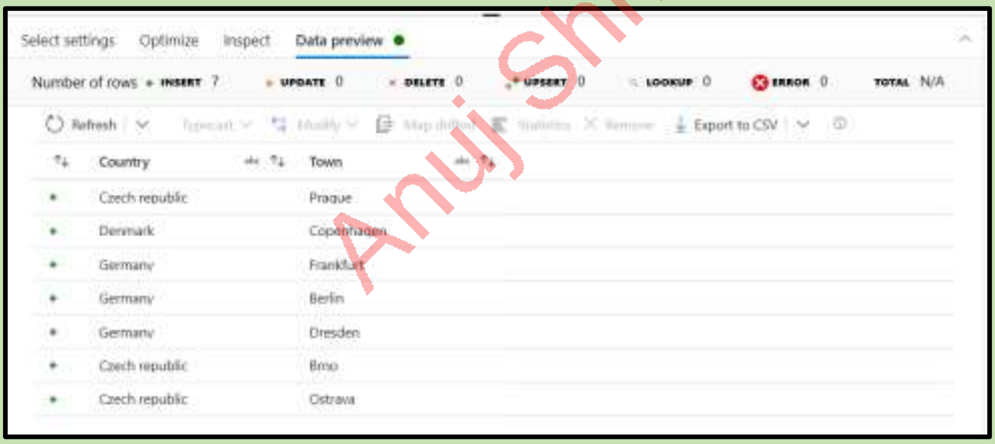


Requirement: Create unique key (Geo Key) and geography table

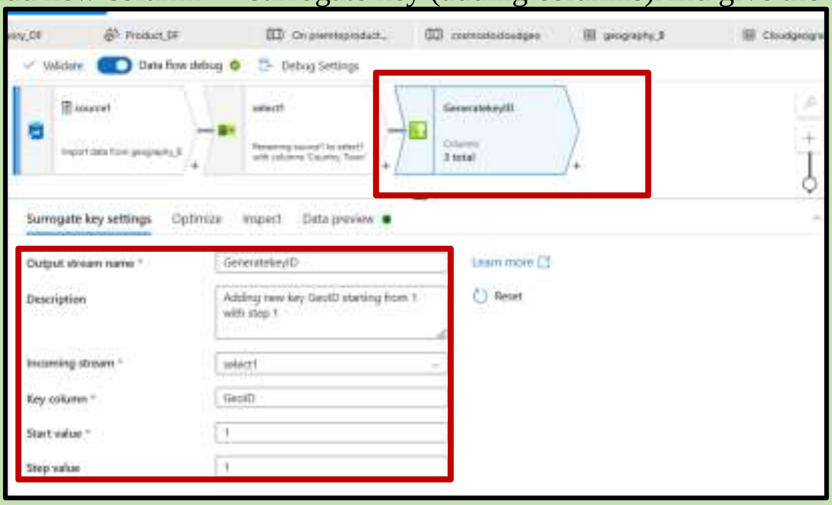
Add one more column --> select , delete the wikipeida column



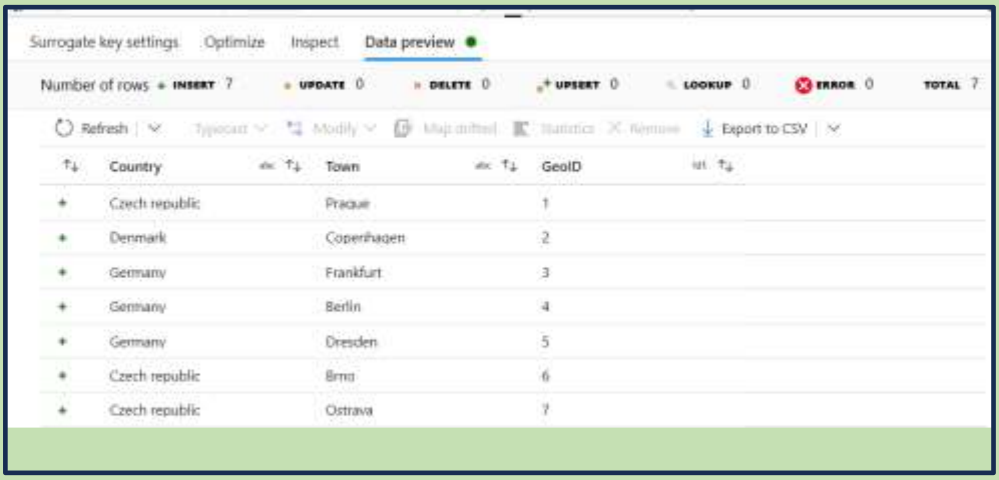
As we can see the column is deleted



Add new column--> surrogate key (adding columns)And give the details

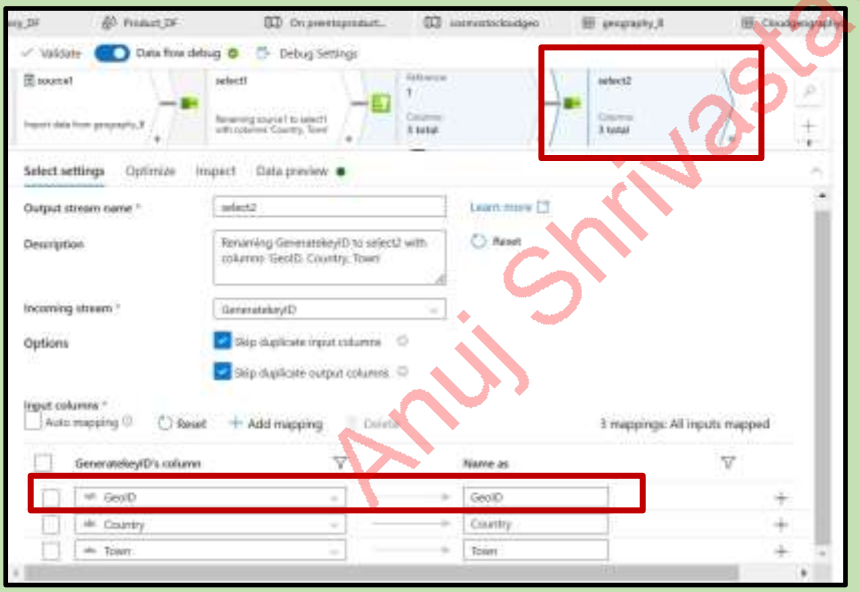


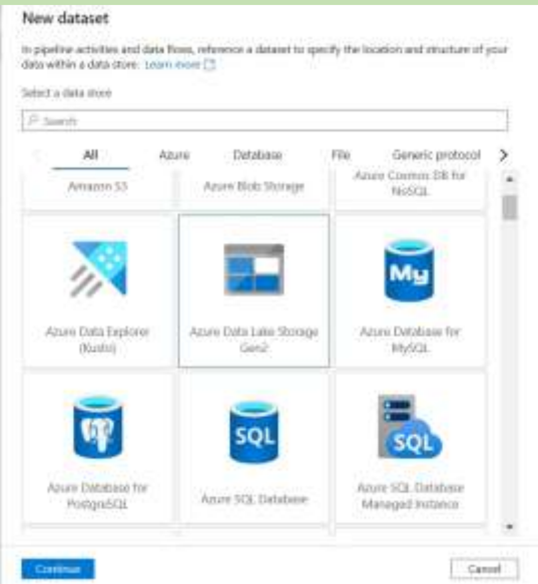
New column added

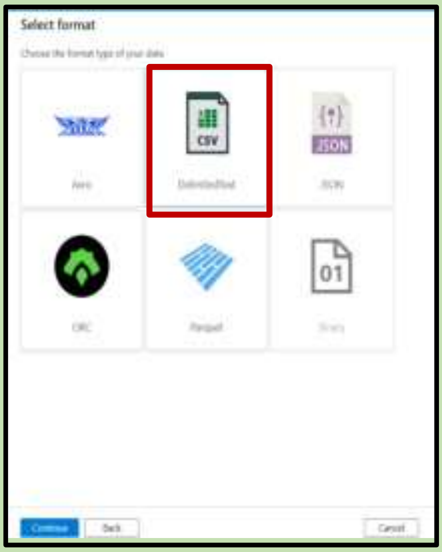


Changing order of columns

New column is added ---> Select and keep the columns in order

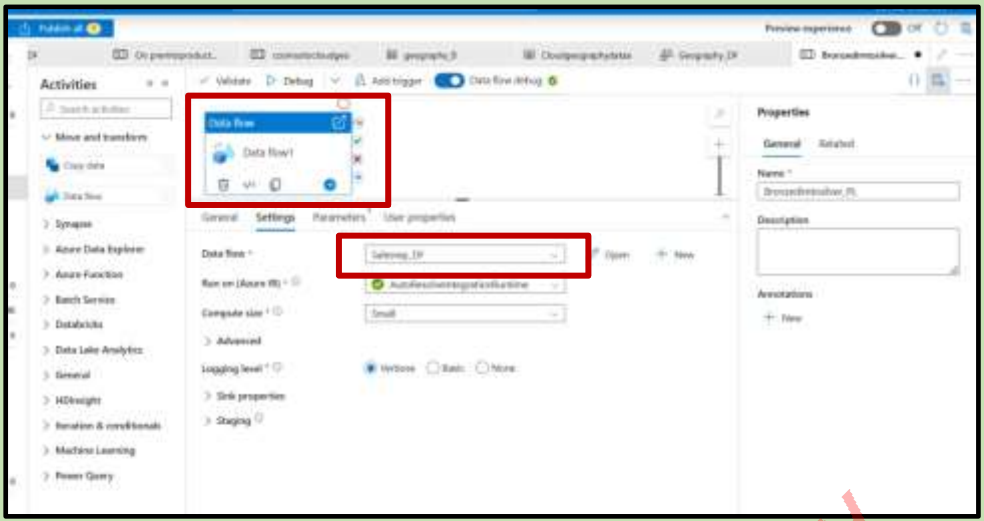


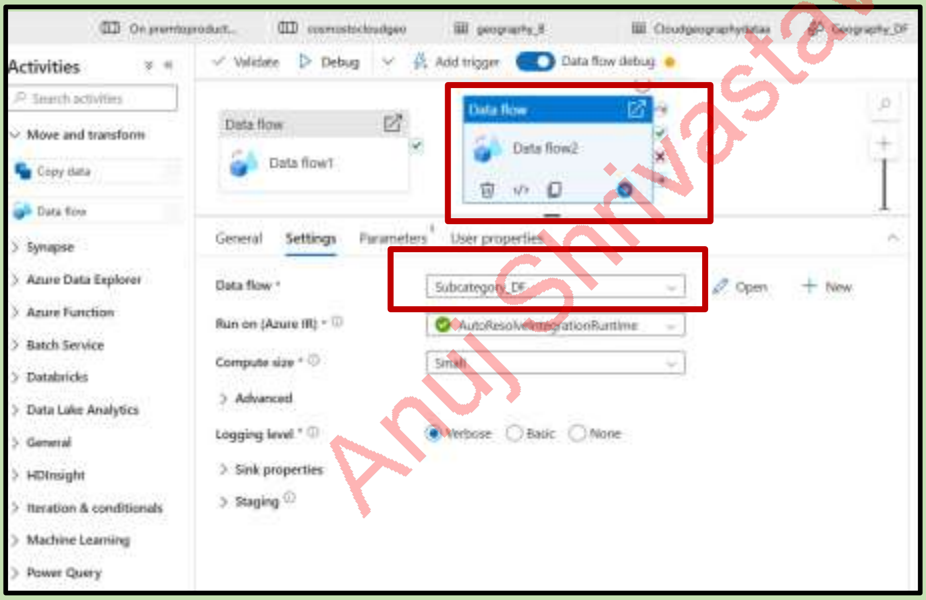


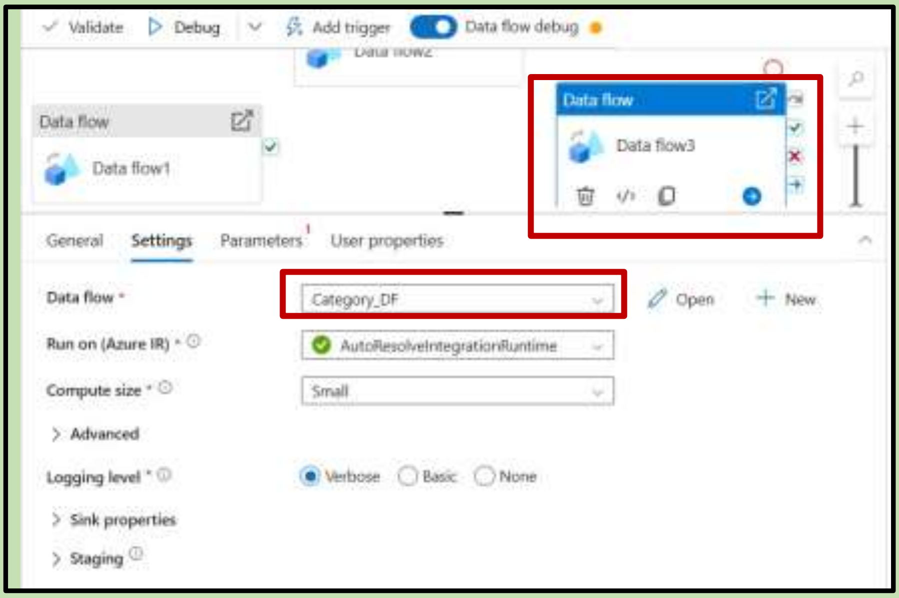


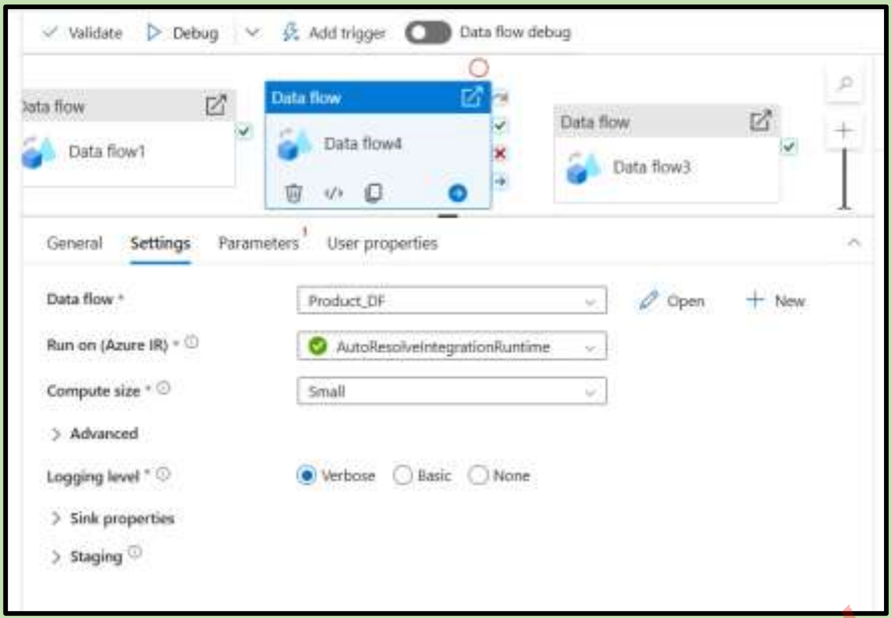
Pipeline:

Create new pipeline add all dataflow in the pipeline and give the pipeline name

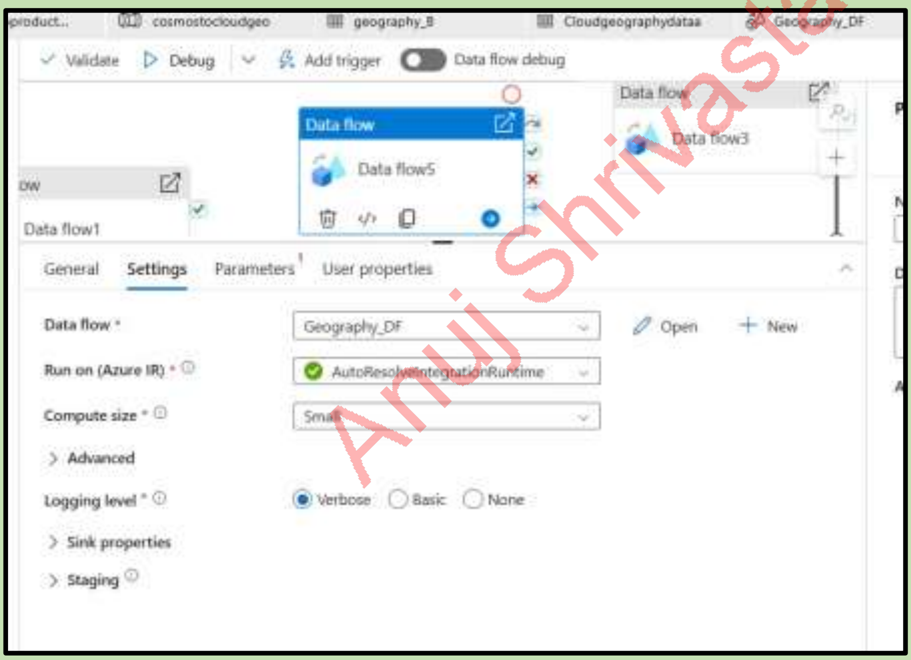






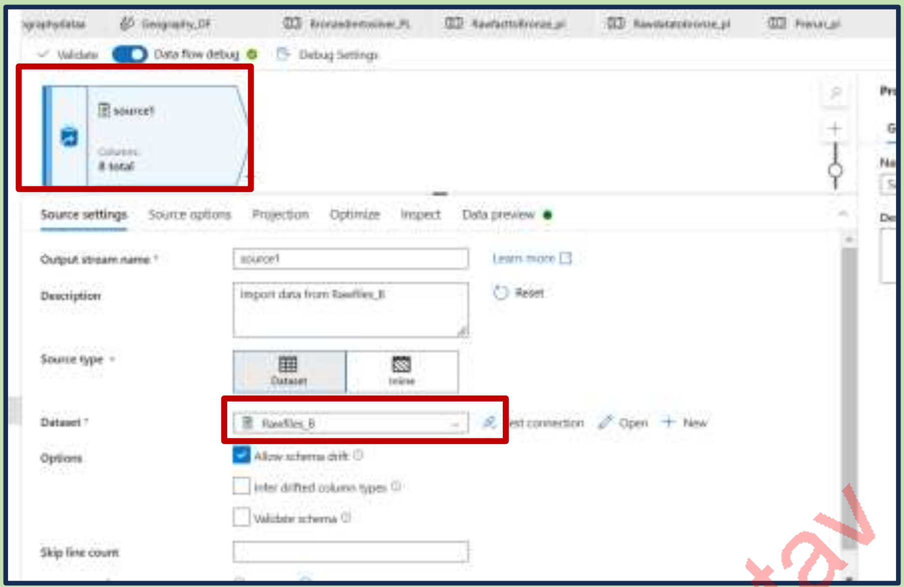


Five Data flows created and publish it

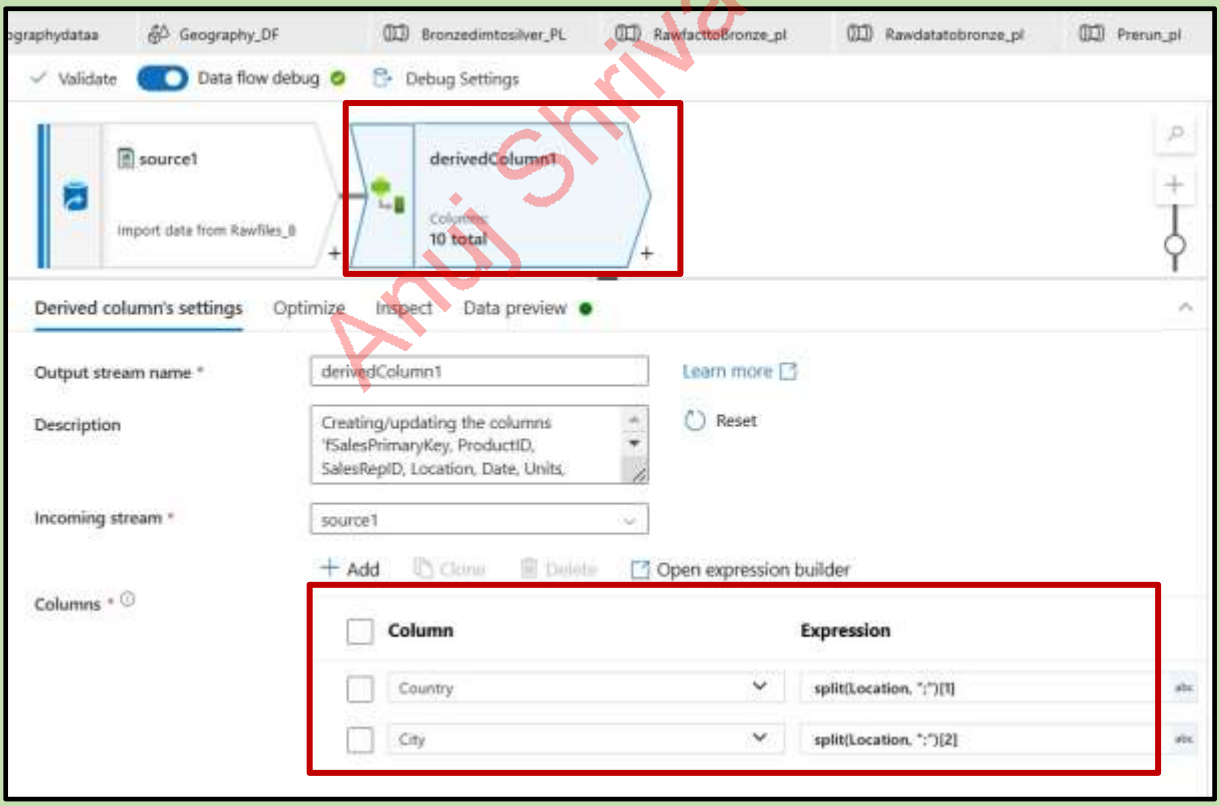


Data Flow6 :

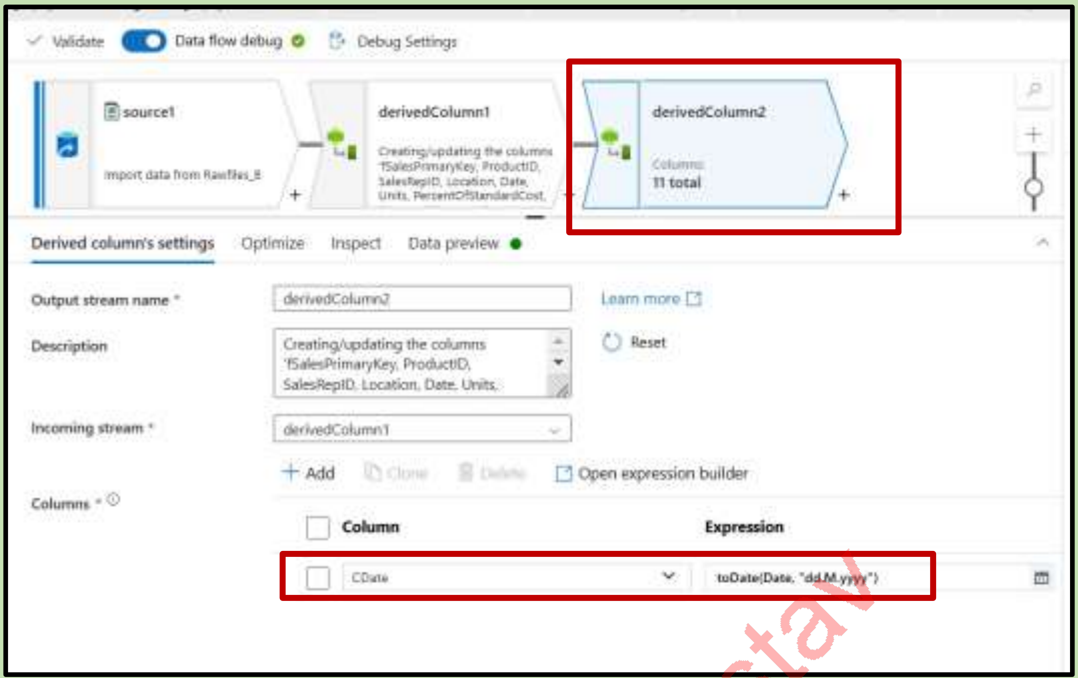
New data flow is created add source and give the data set



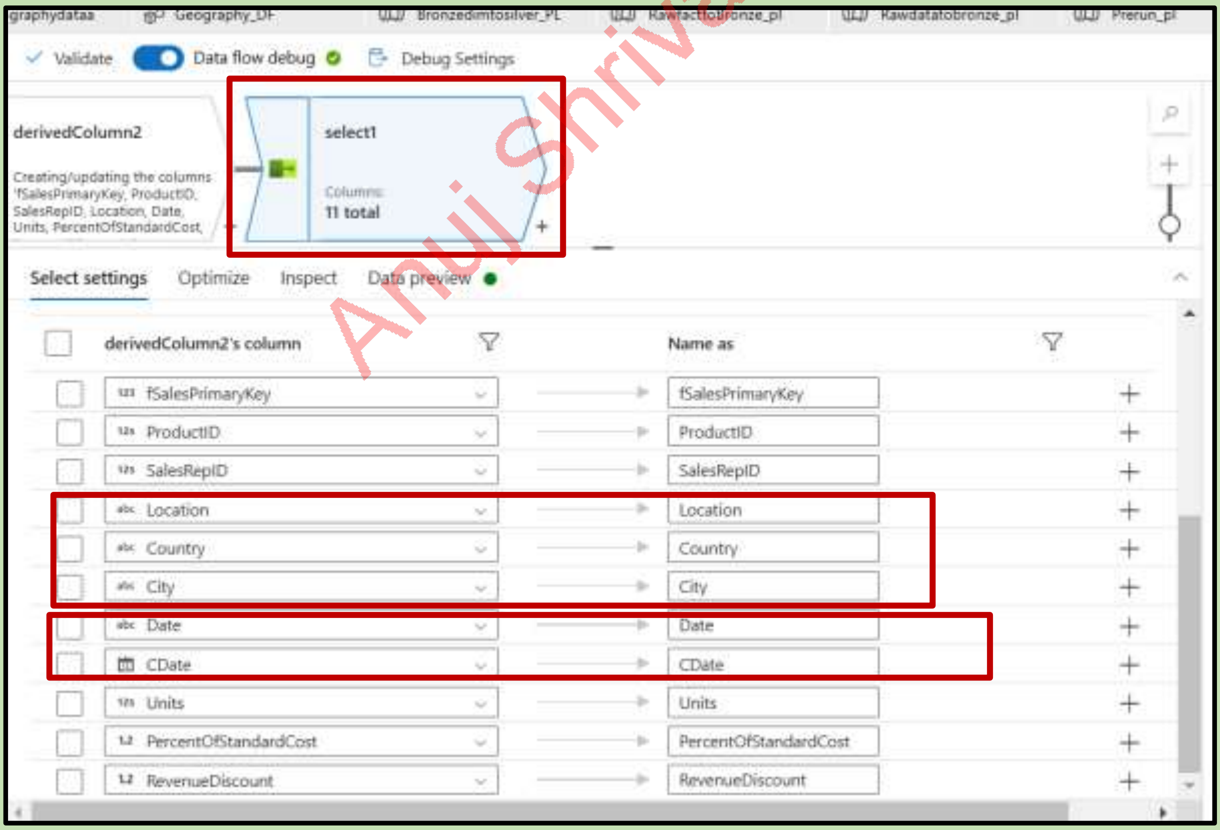
Add derived column

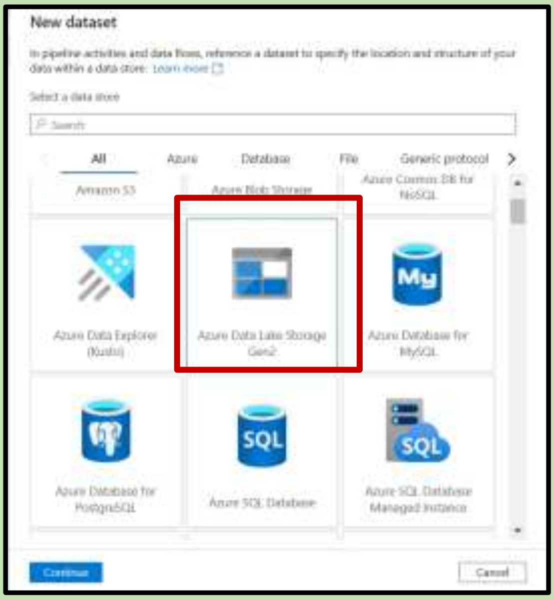


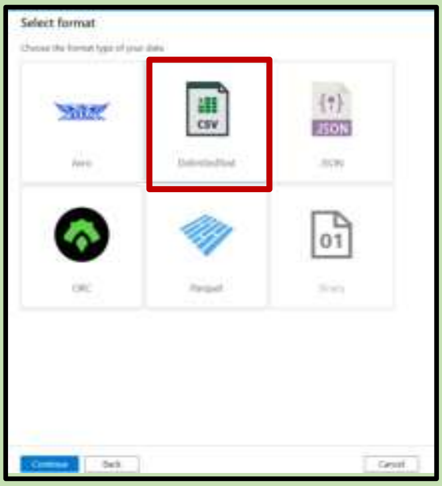
Add one more derived column



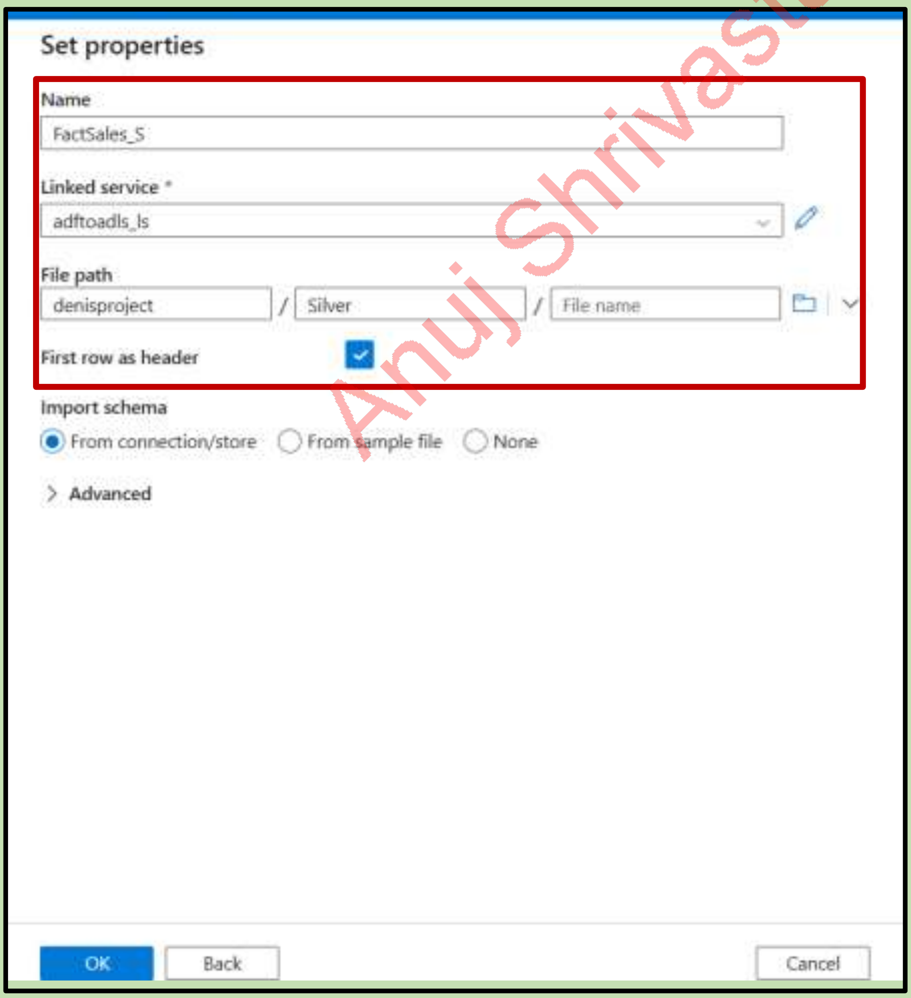
Add select column remove location and drag Country, City remove date and drag Cdate





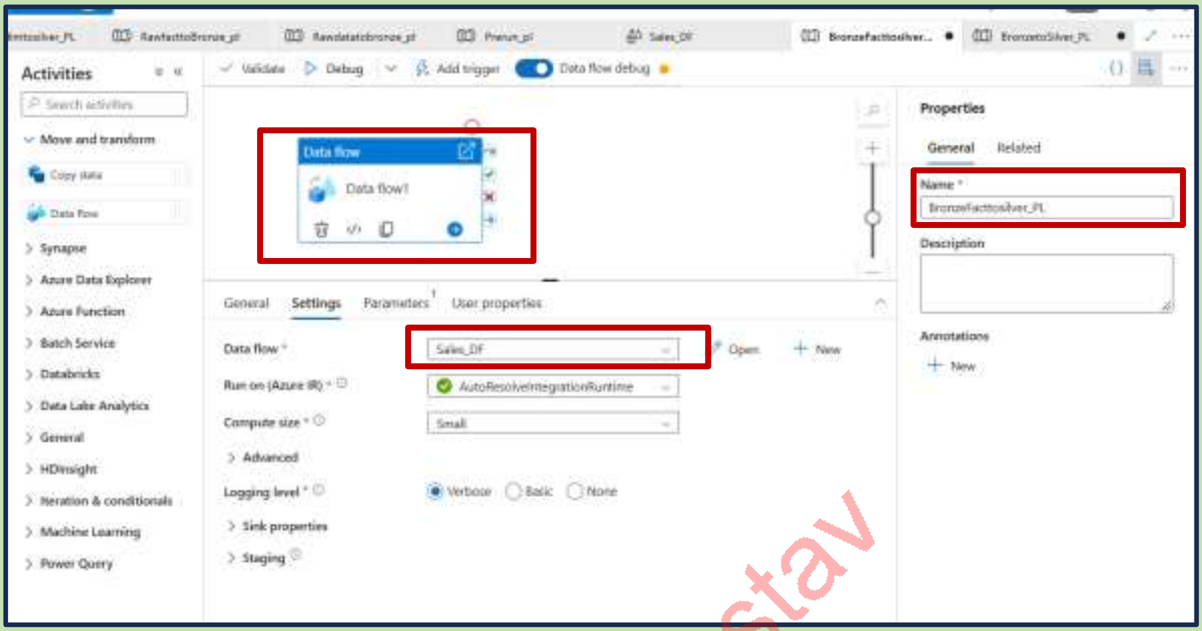


Set the properties and publish

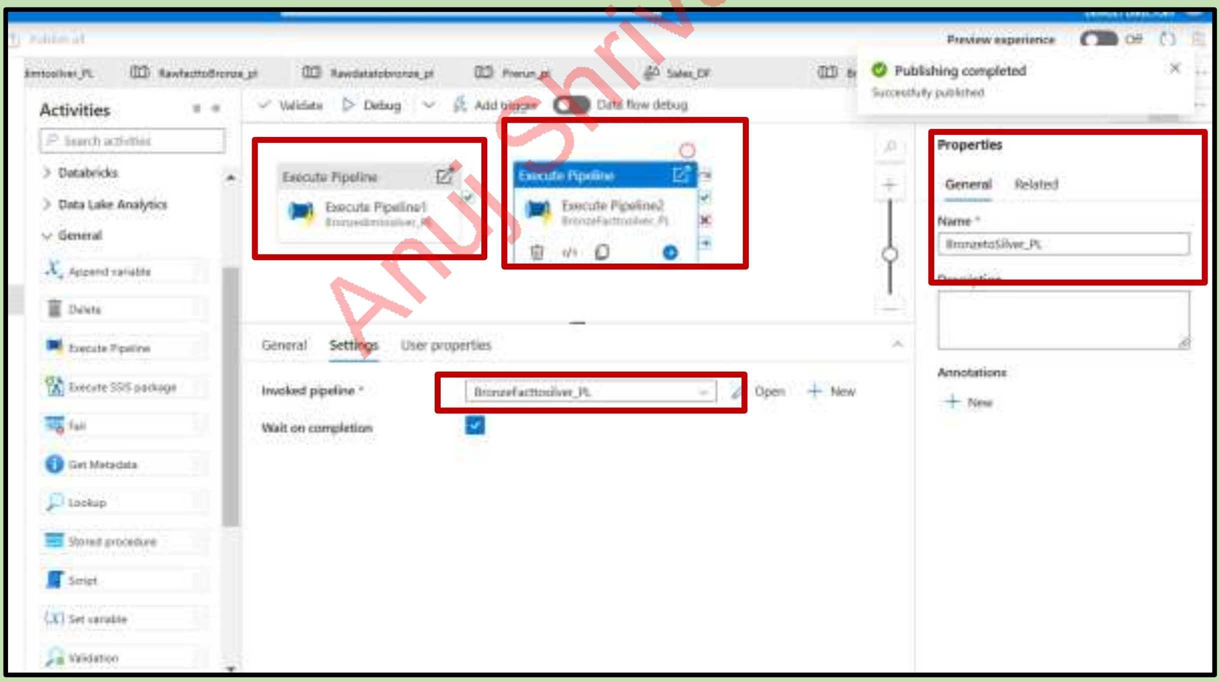


Pipeline:

Create new pipeline move the data flow add the name and data flow

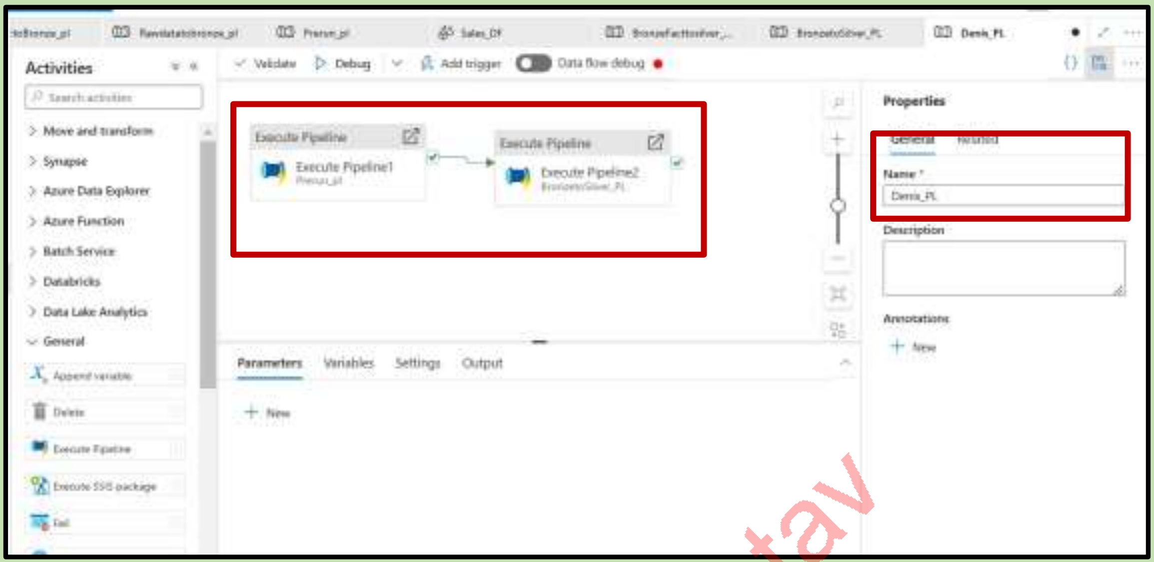


Now drag 2 execute pipeline and invoke two pipelines from bronze to Silver and publish



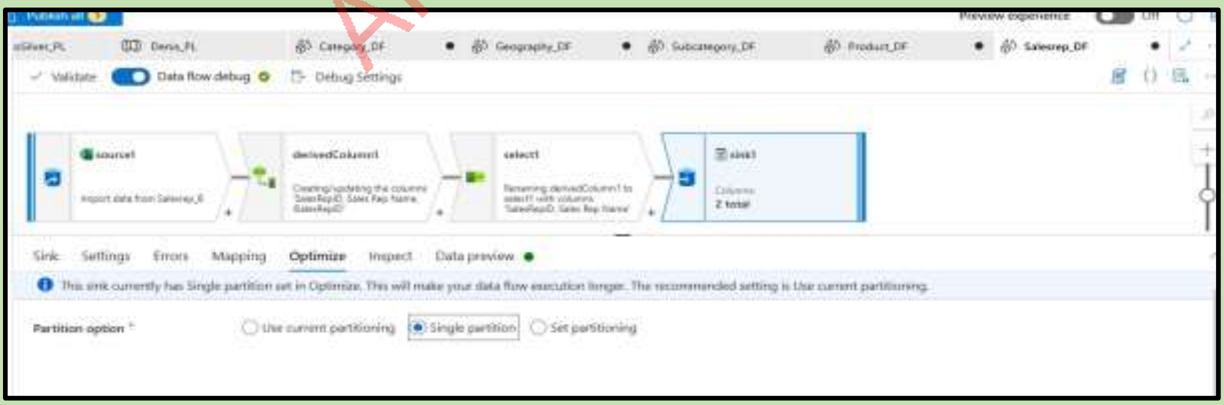
Final Pipeline:

Now we need to add two execute pipelines --->Prerun , Bronze to silver and publish it and run it



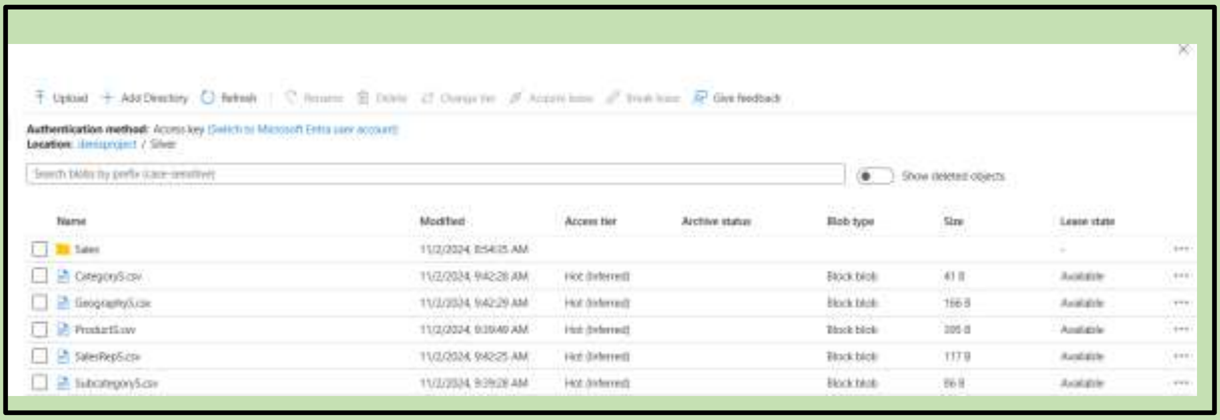
After successful pipeline run if we get the files in this format

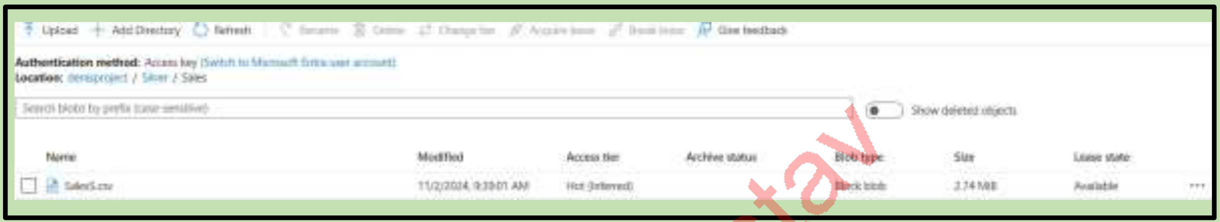
Go to the data flows and give it as single partition in the sink side



Pipeline run successfully

Now we can check in adls→ silver



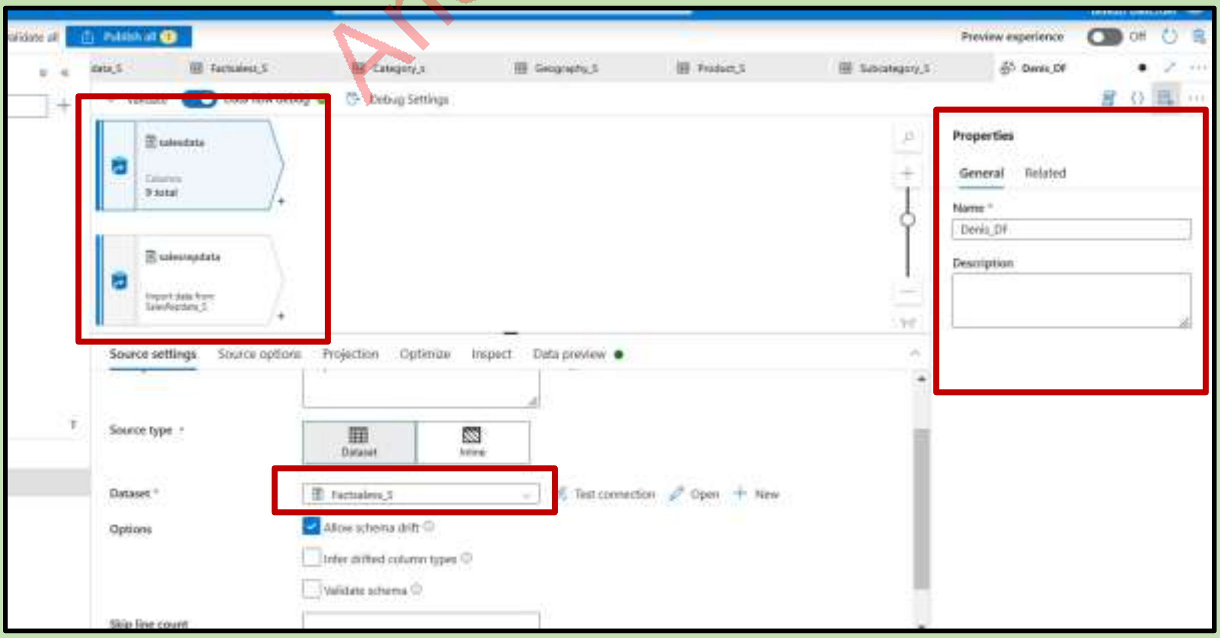


Final Data Flow :

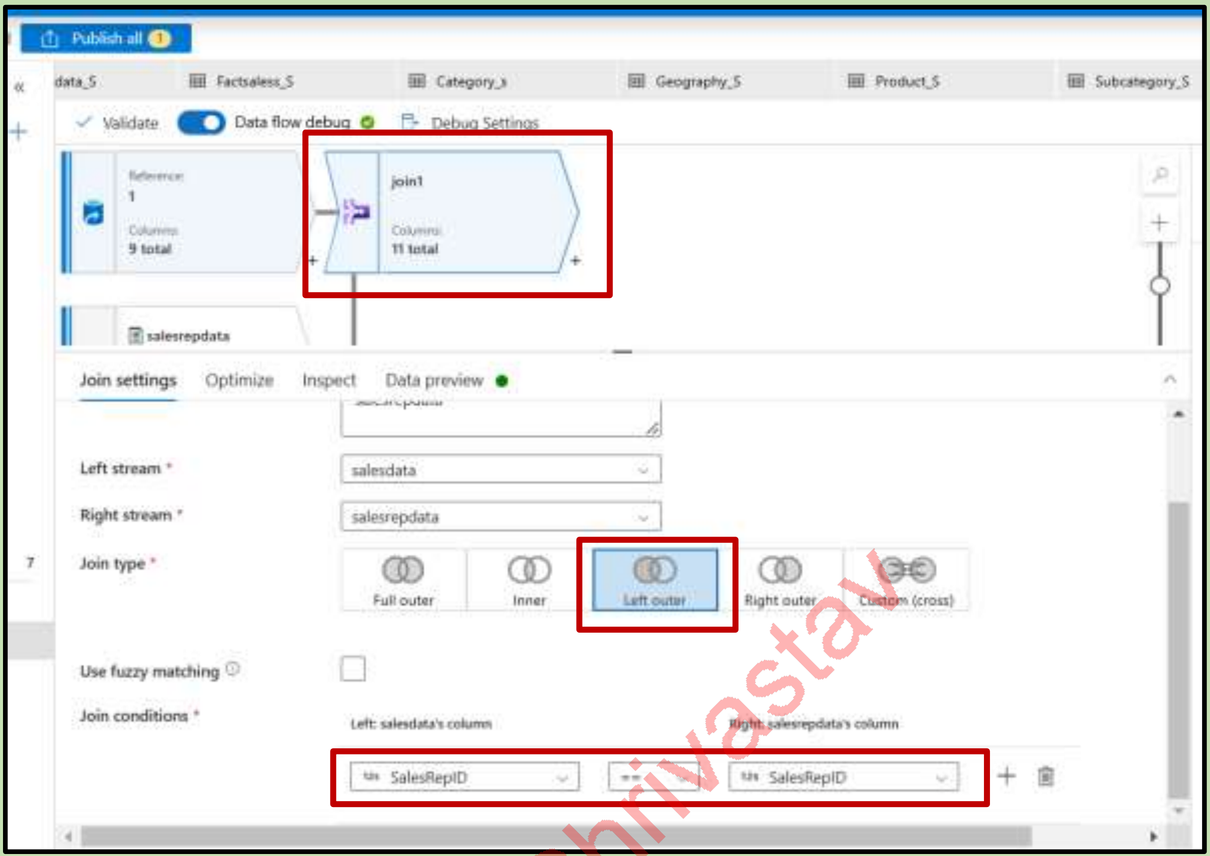
Name the data flow as Denis_DF

Add the source and add the data set salesrep

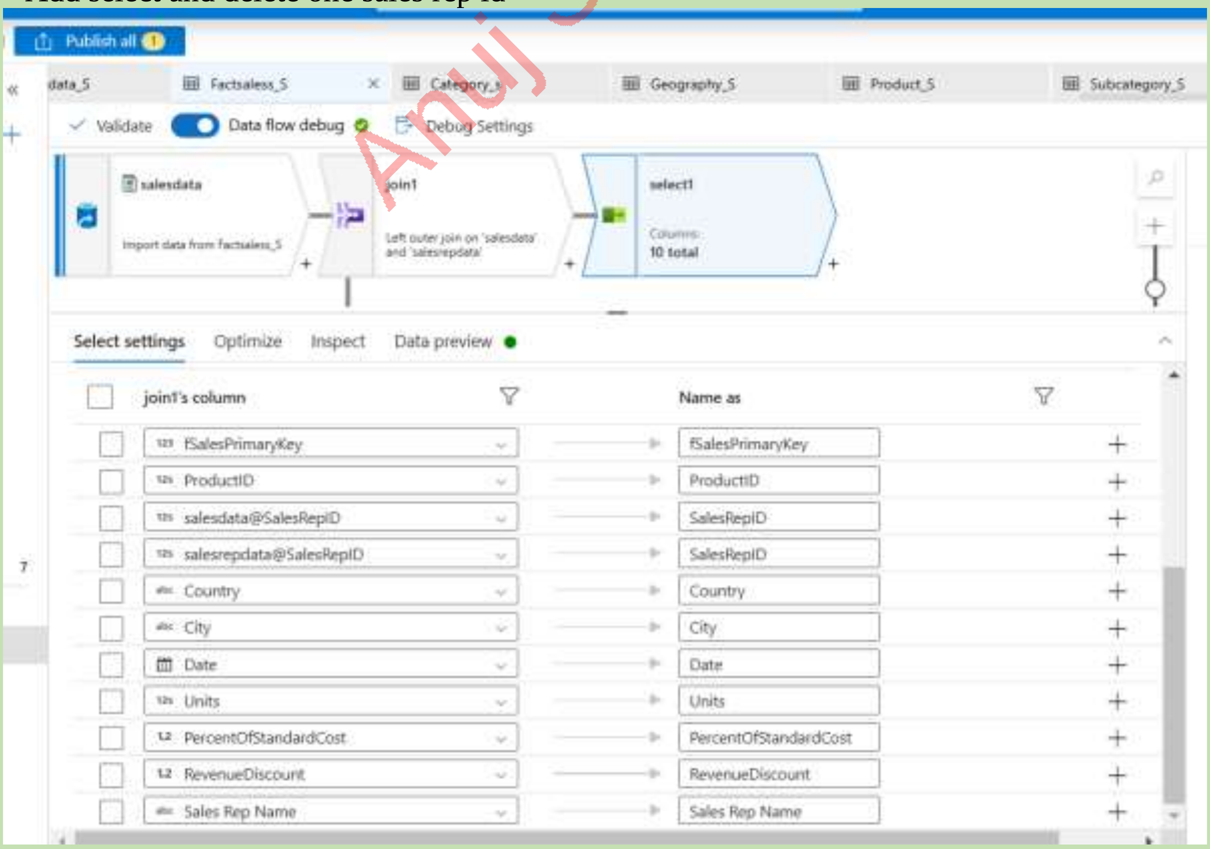
Add one more source add the salresrep dataset-filename-salesrep



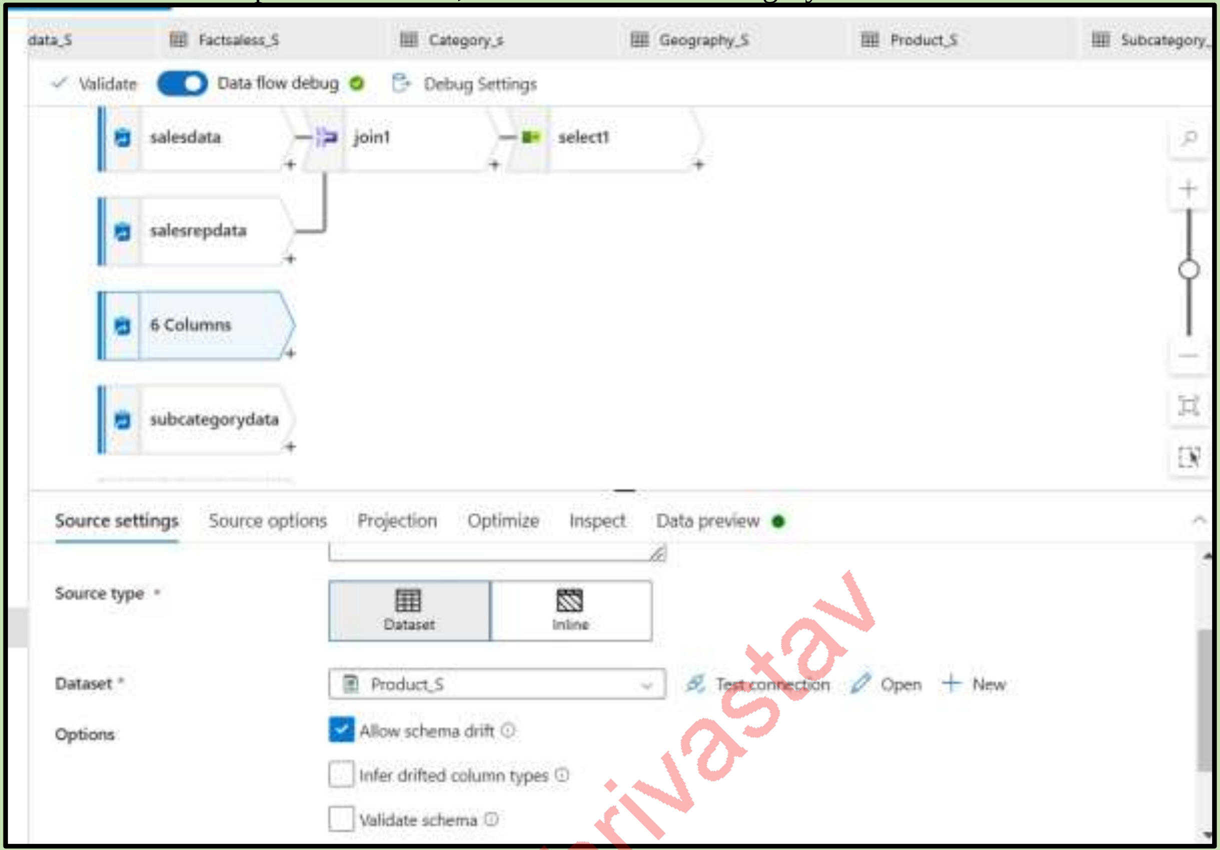
Add join and give details



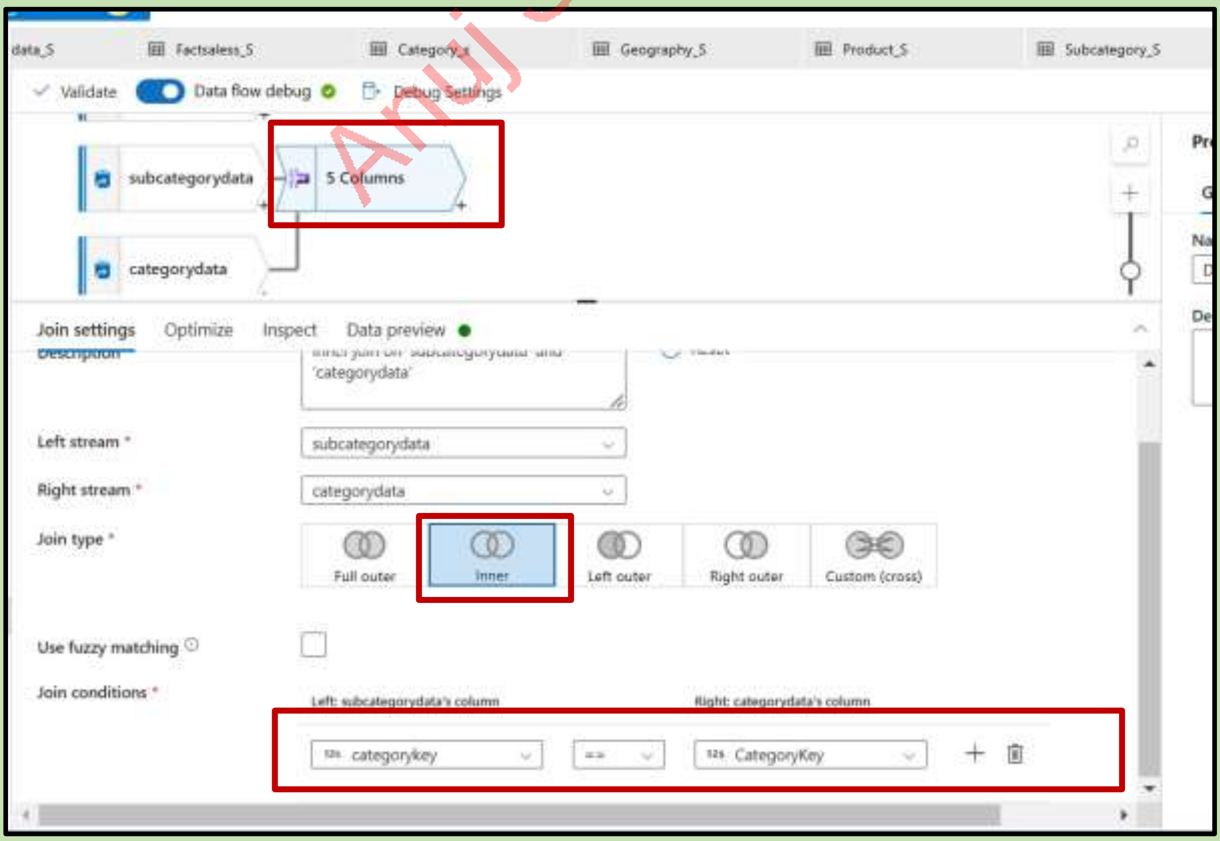
Add select and delete one sales rep id



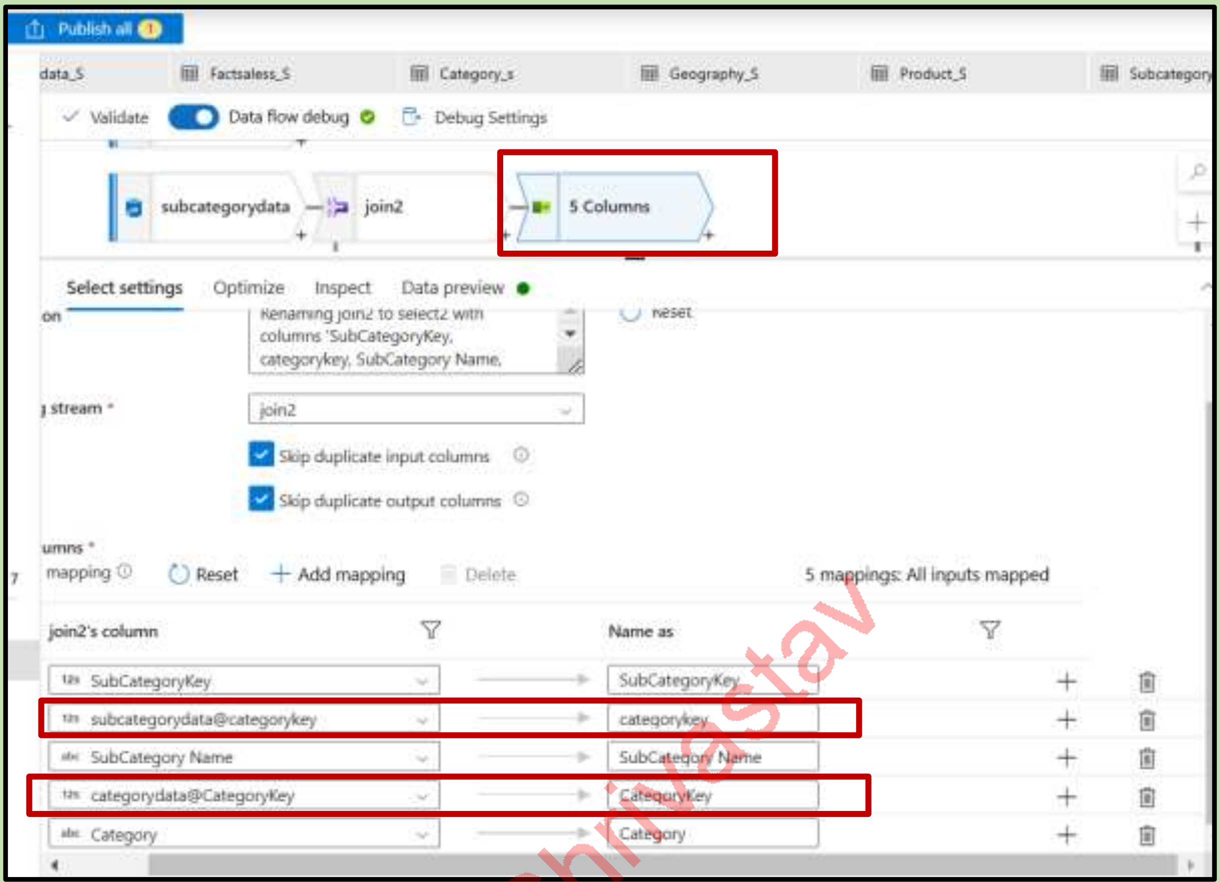
Now add source ---> product dataset , add source --> subcategory data



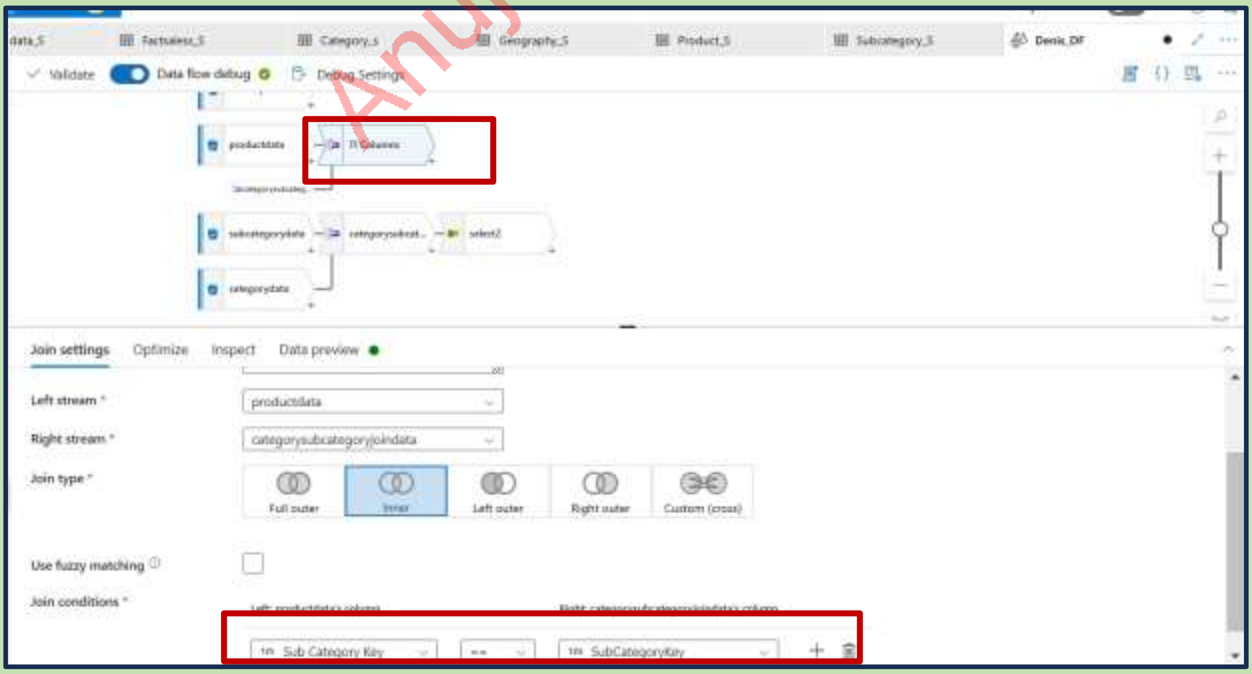
Add source -->Category data --> add join for subcategory data and join with category data and add the name as categorysubcategoryjoindata



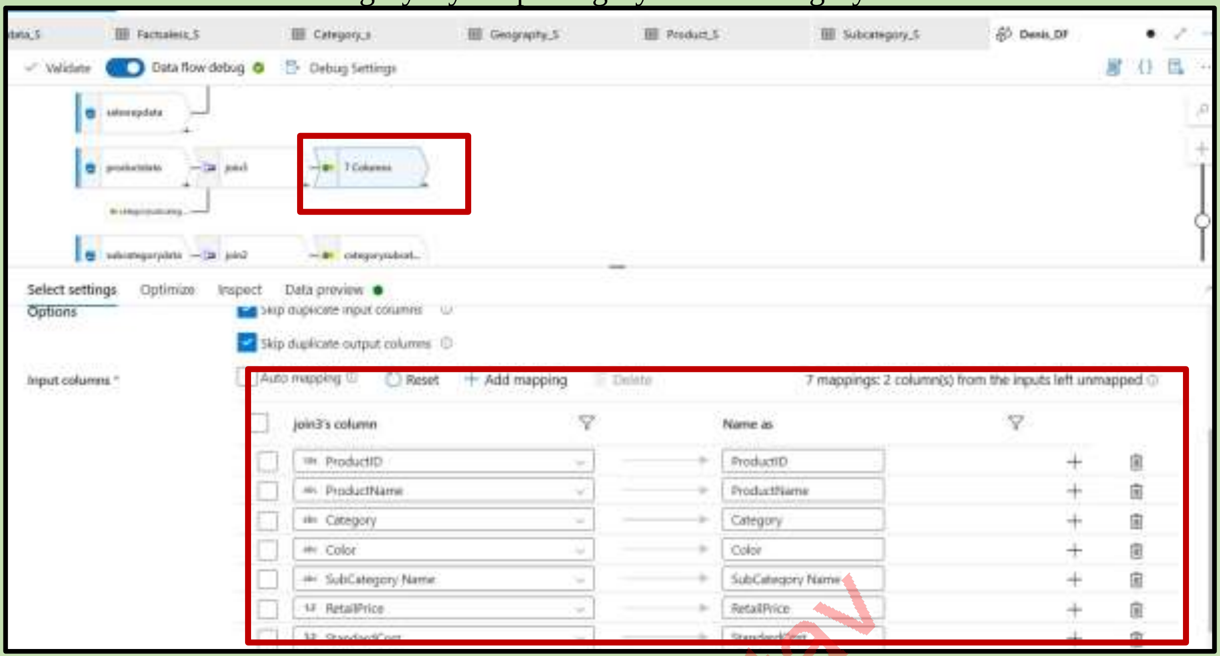
Add select column and delete similar columns



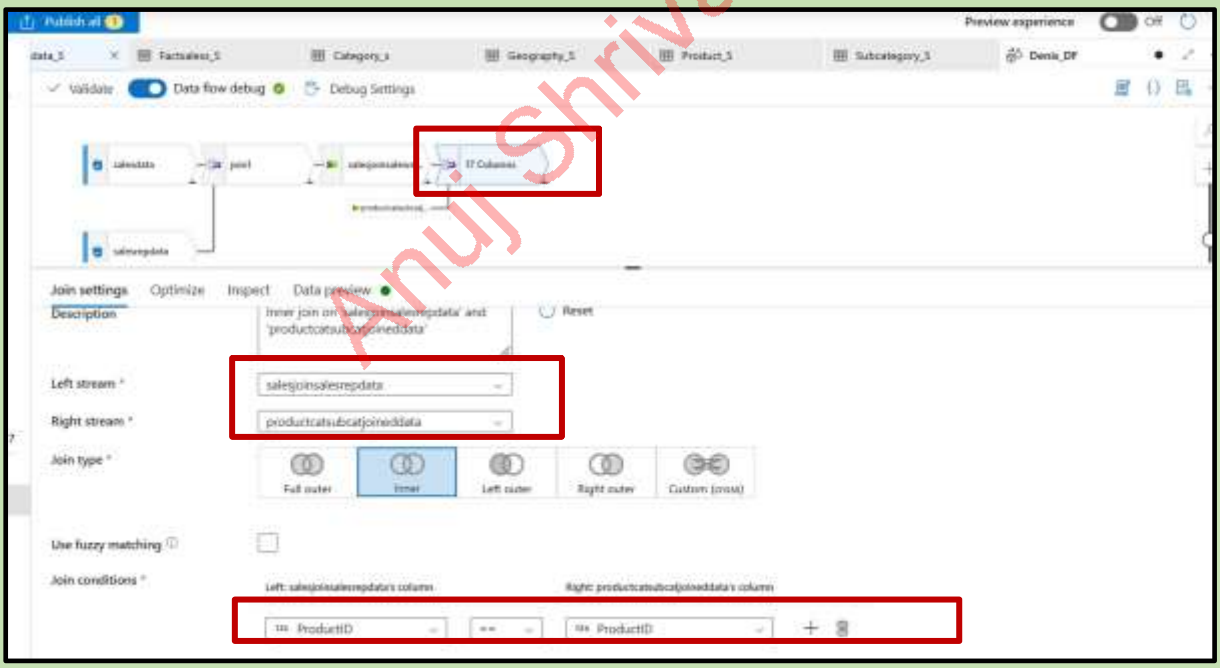
Now add join to product data join it with categorysubcategoryjoindata and give the conditions check the data preview



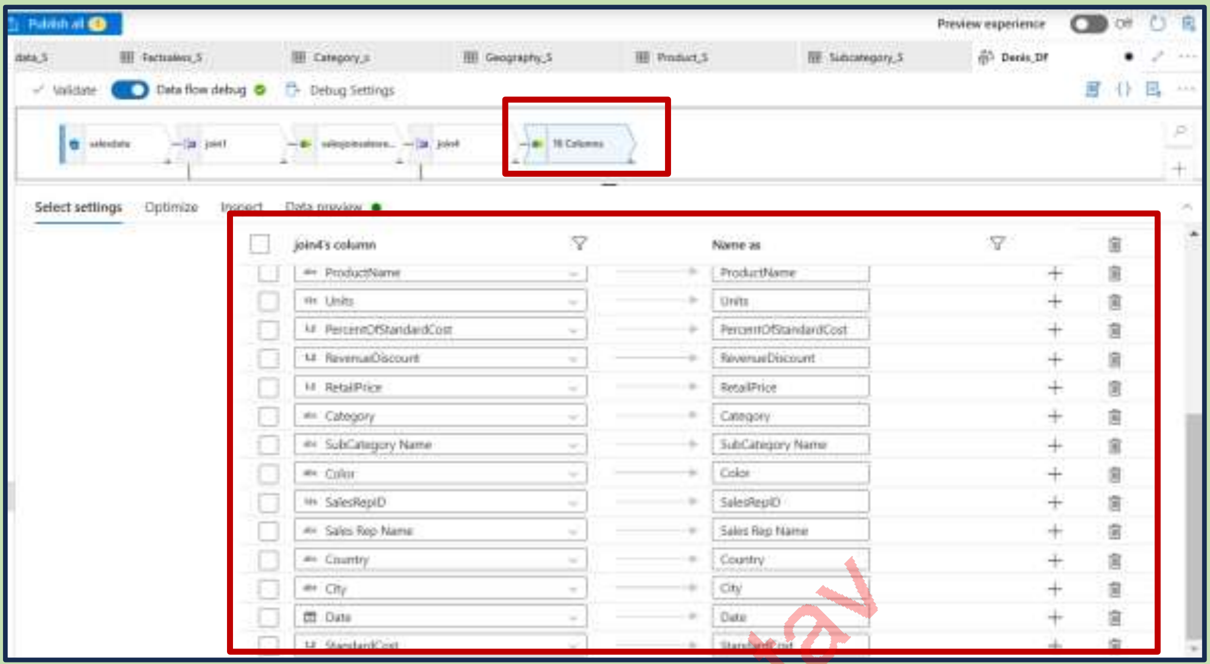
Add select and delete subcategorykey keep category and subcategory name in order



Now salesjoinsalesrepdata add join with productcatsubcatjoindata Give the condition and preview the data



Add one select column to join 4 and delete similar columns product ID arrange it in serial order

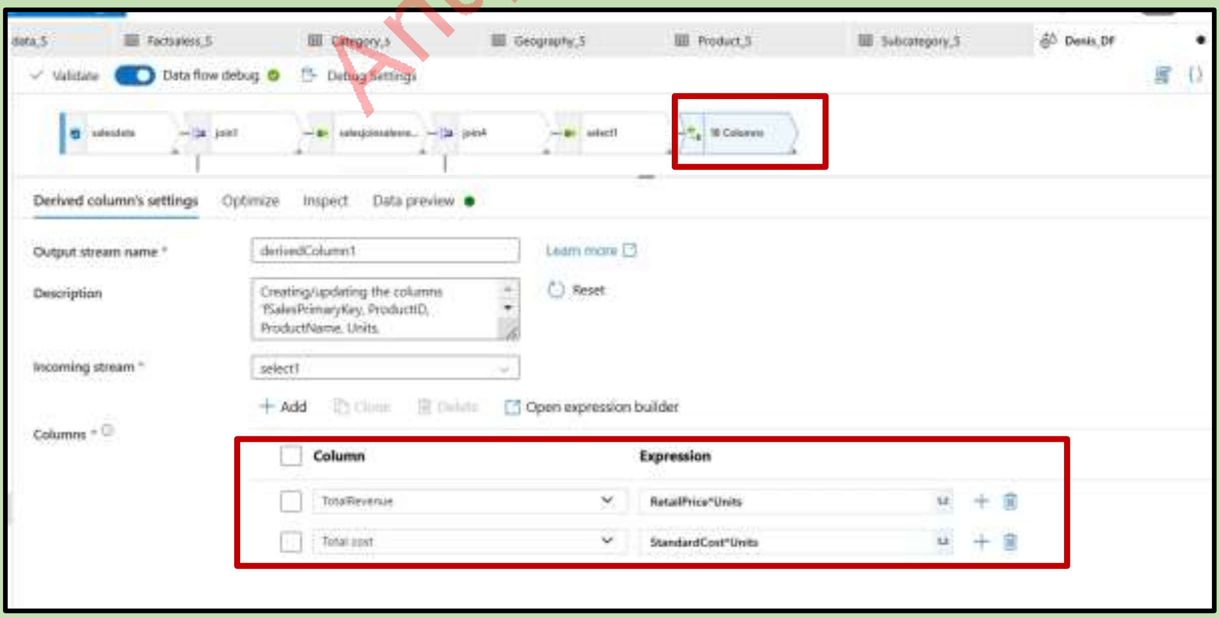


BUISINESS REQUIREMENT:

Calculate **Total Revenue** in sales table using product's Retail Price and multiplying it by the units

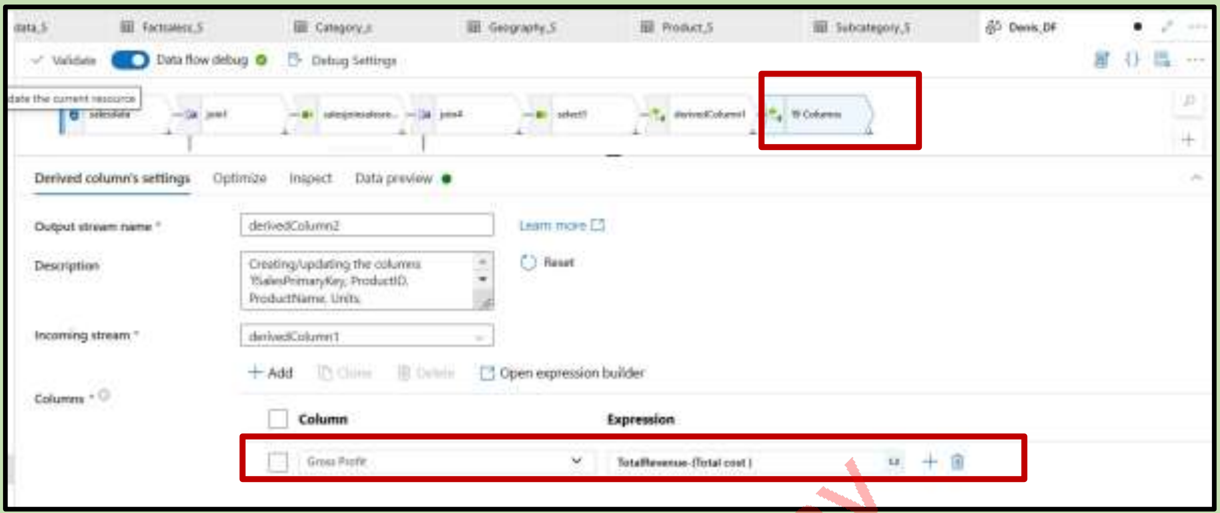
Calculate **Total Cost** in Sales Table, using Product's standard cost, and multiplying it by the units

Add the derived column and give the conditions



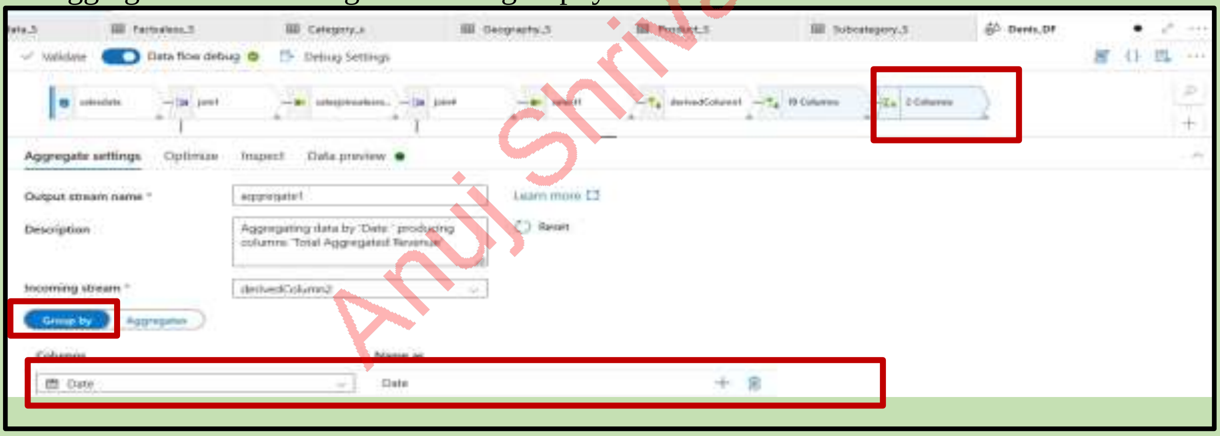
Calculate **Gross Profit** in Sales: Total Revenue – Total cost

Add one more derived column and give the condition

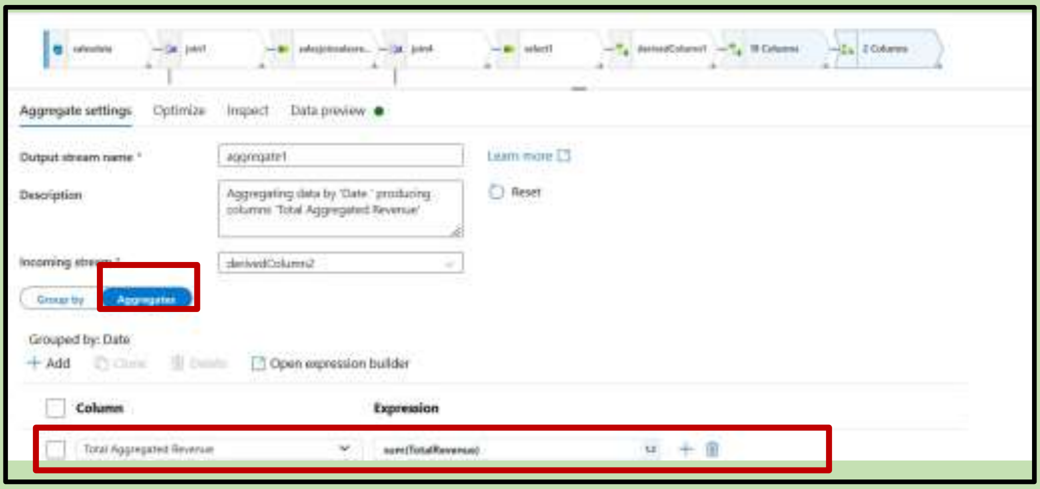


Calculate a measure for a **AVG Sales per day** – This is average sum of Total Revenue per day based on the date of actual sales

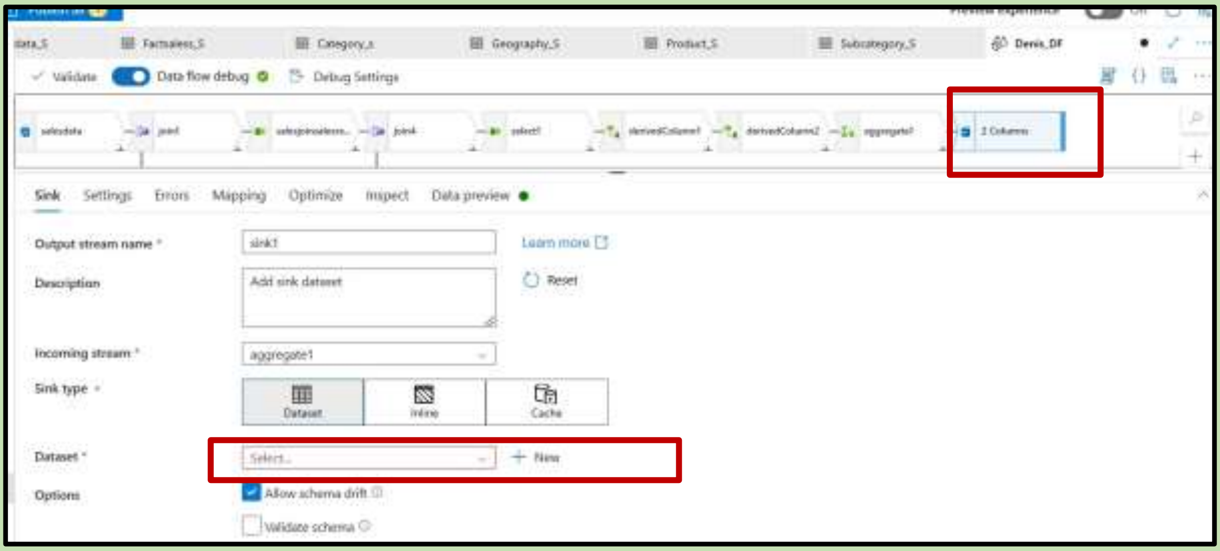
Add Aggregate column and give date in groupby

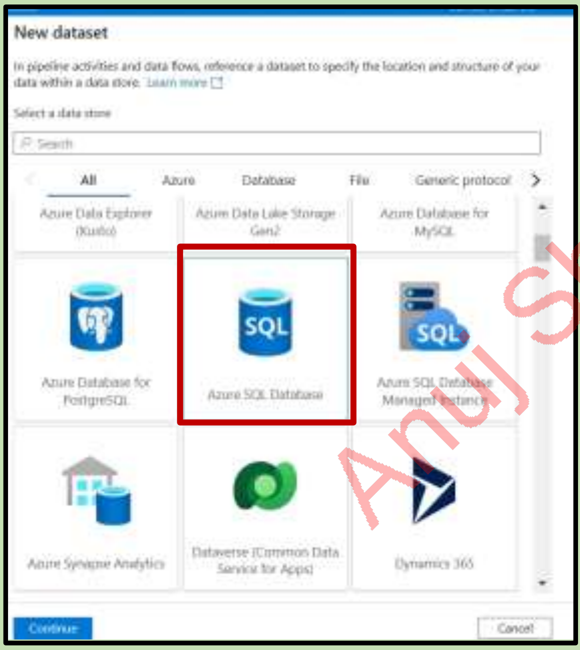


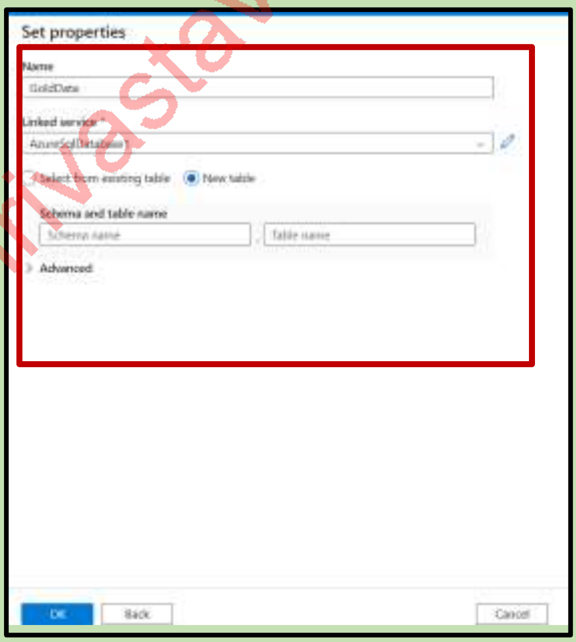
Give Total Aggregated Revenue sum(TotalRevenue) in Aggregation

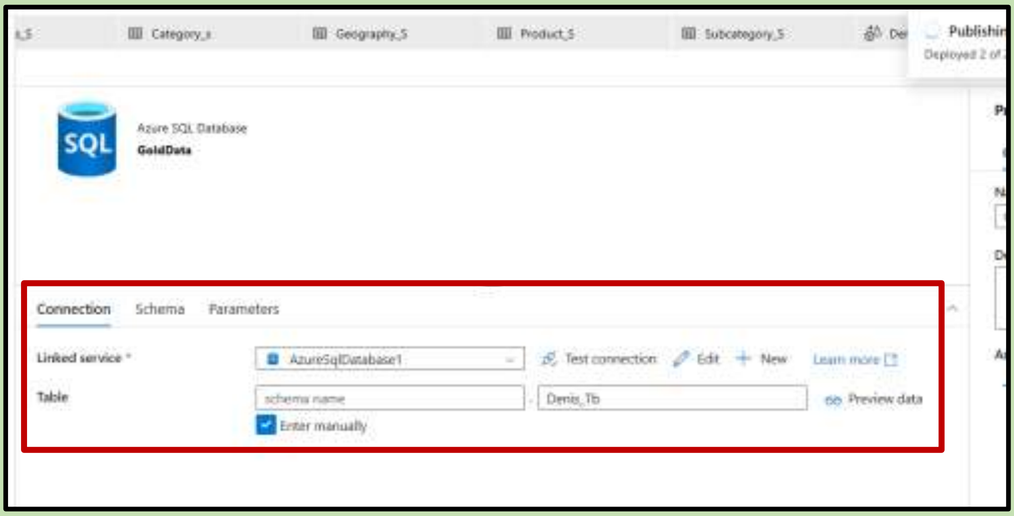


Now add the sink and add the data set



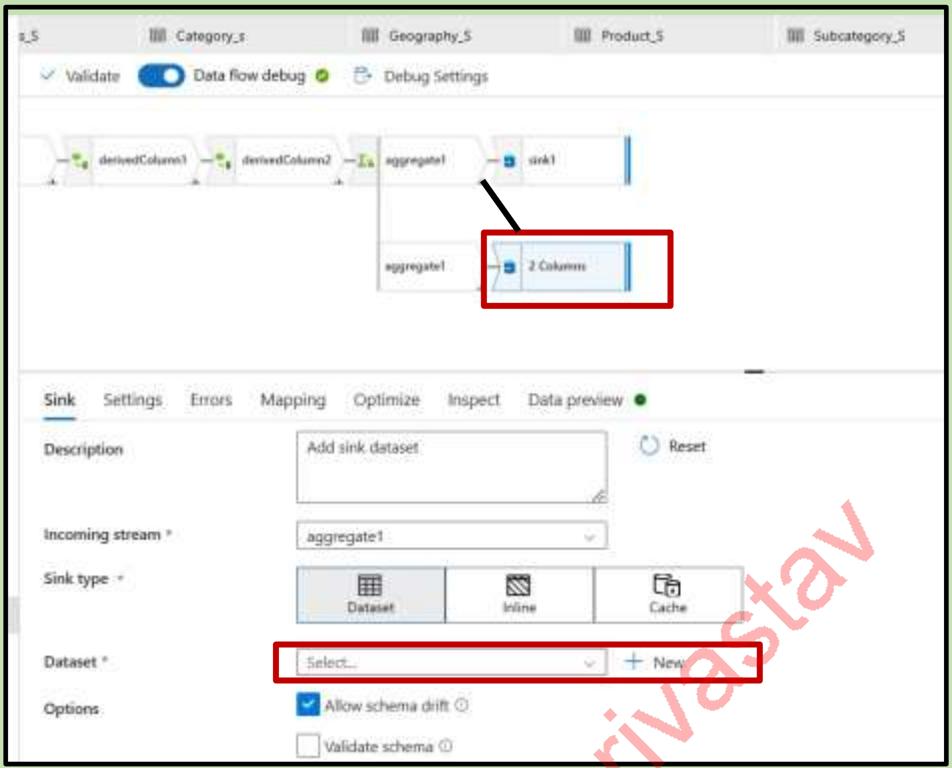


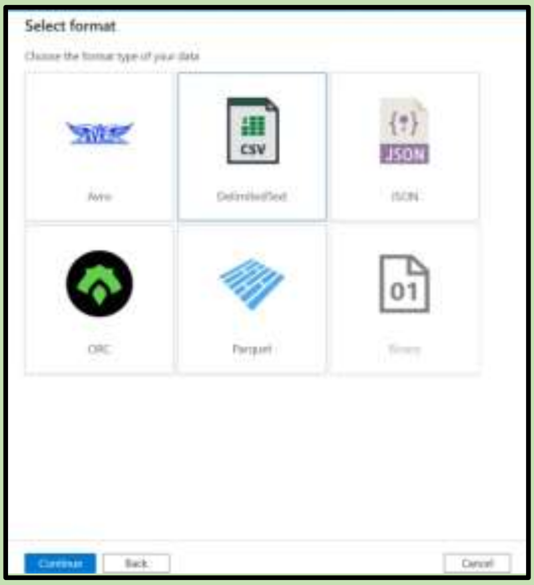




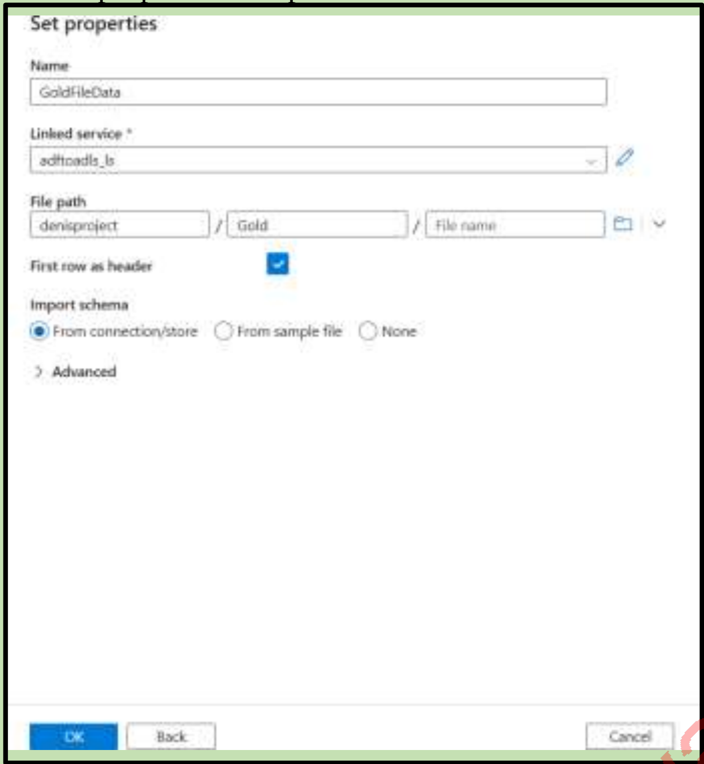
We can also store one copy in gold

Add one more sink

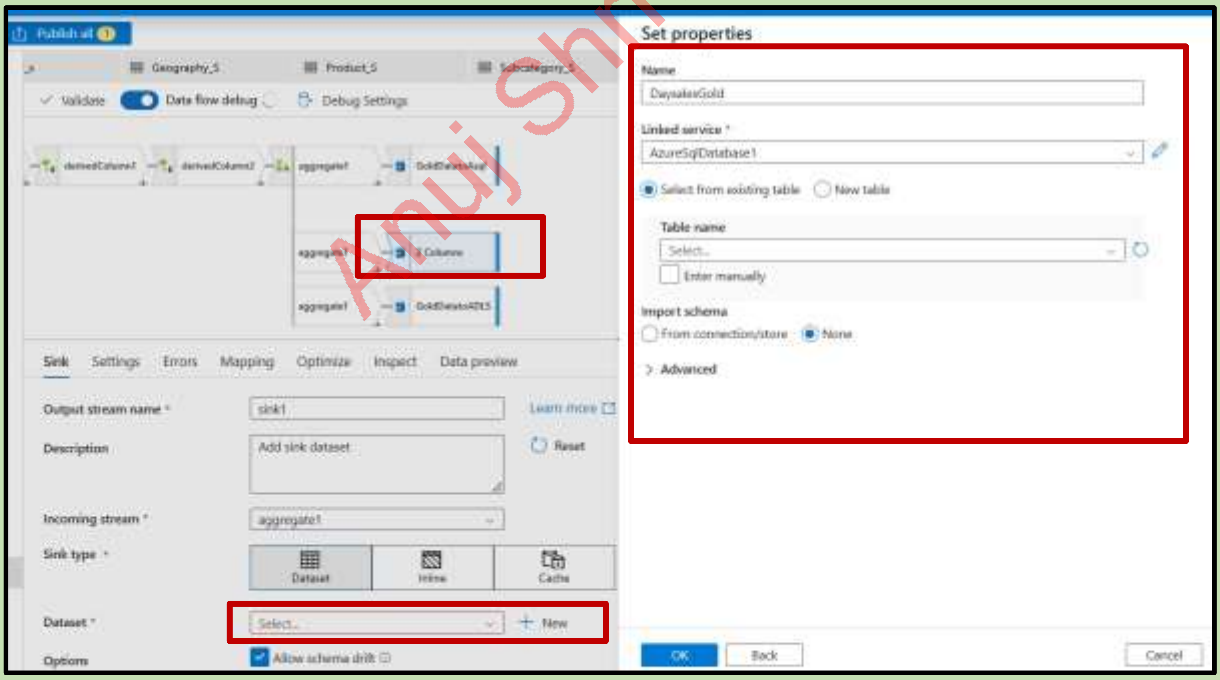




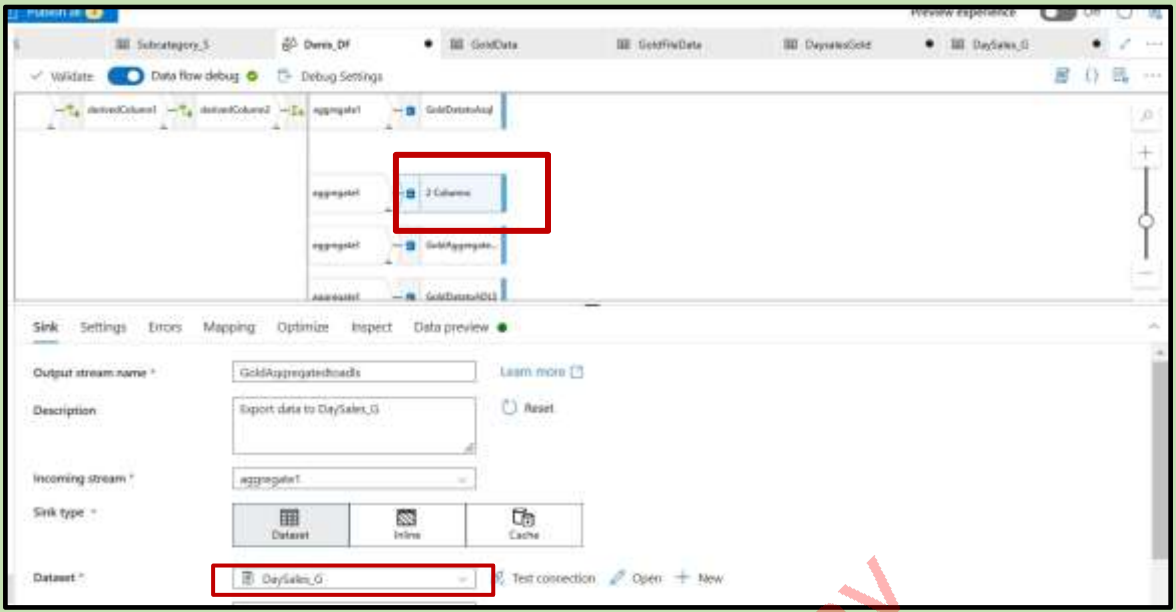
Give the properties and provide linked service



Add one more sink after aggregation to sql db

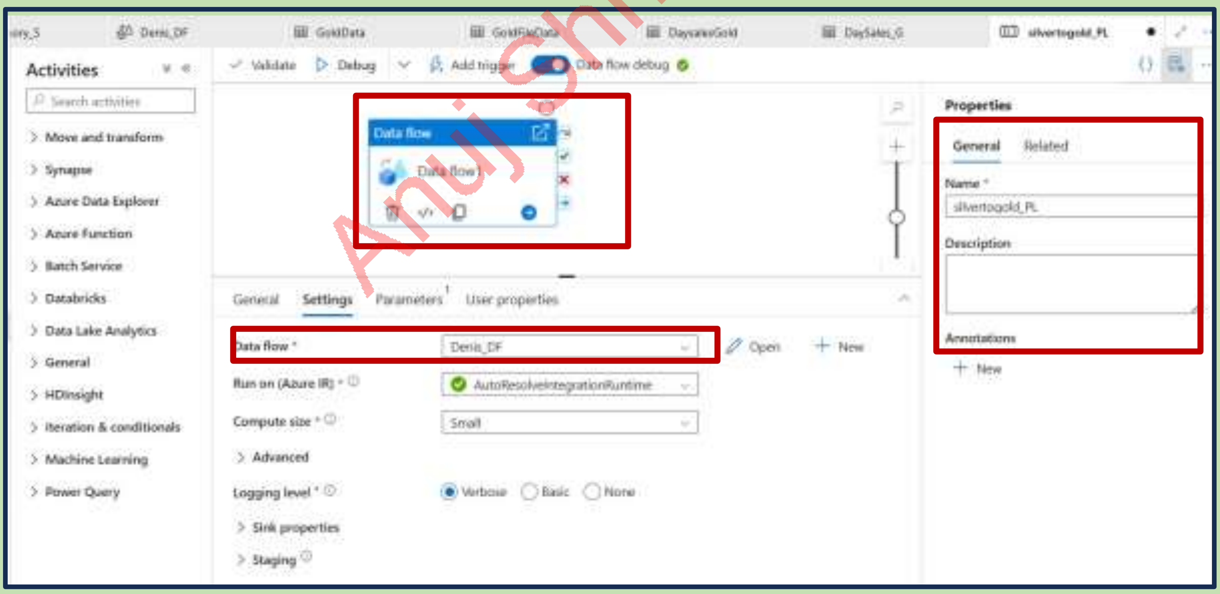


Add one more sink for GoldAggregatedtoadls

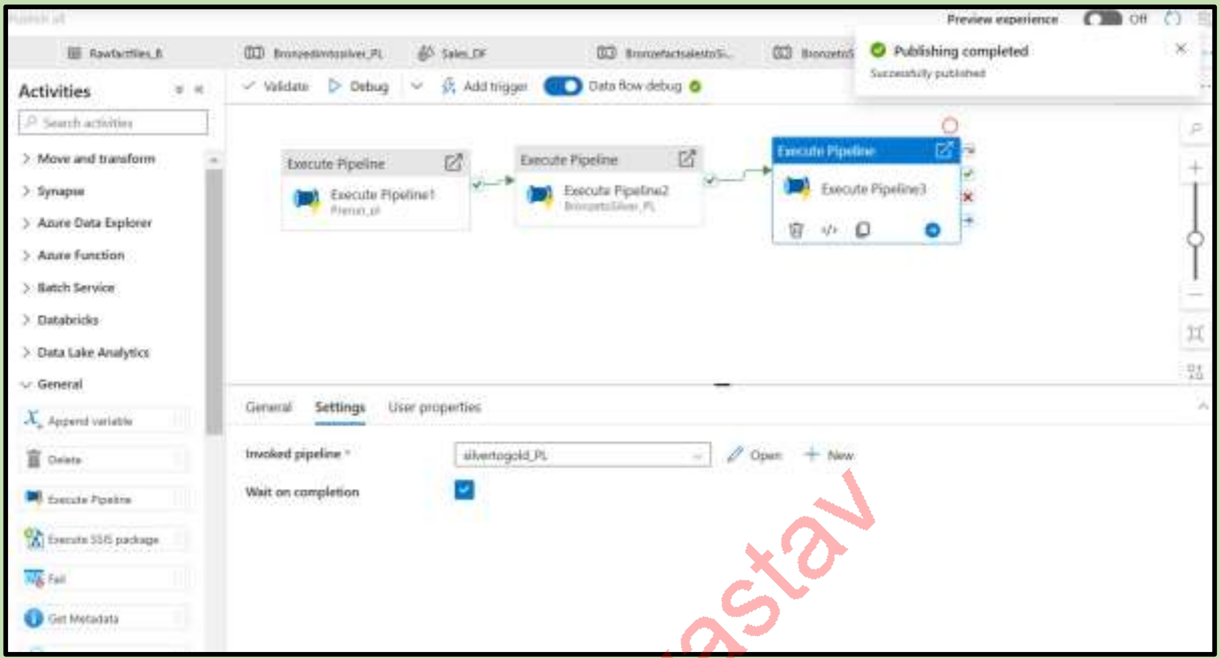


Pipeline:

Create new pipeline – silver to gold drag data flow and add the denies df

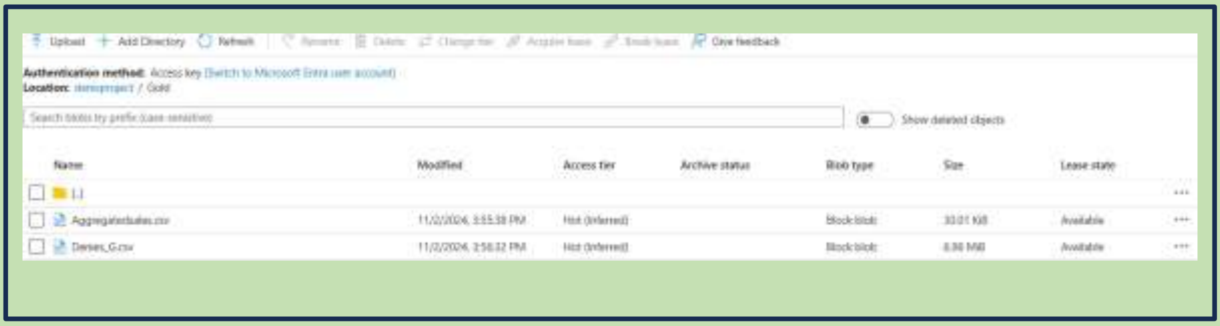


Now open the final pipeline denies pipeline and drag execute pipeline and invoke the silvertogold pipeline and run the pipeline

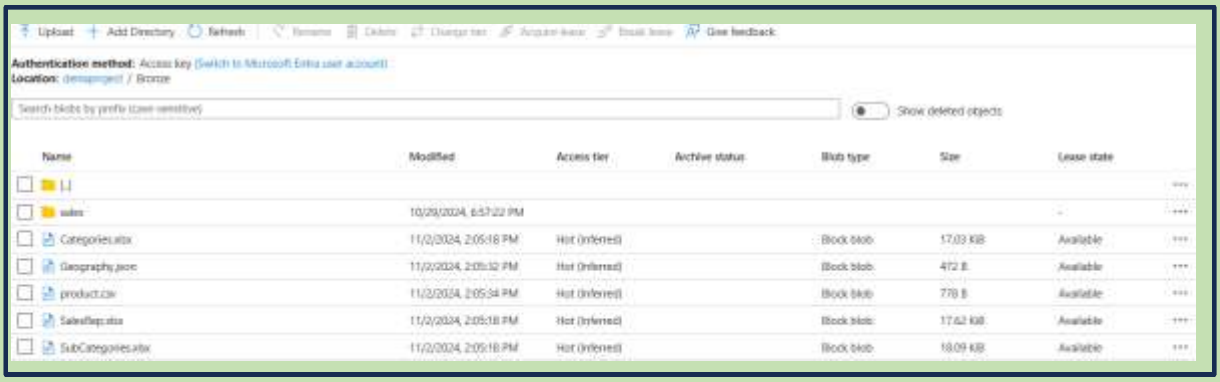


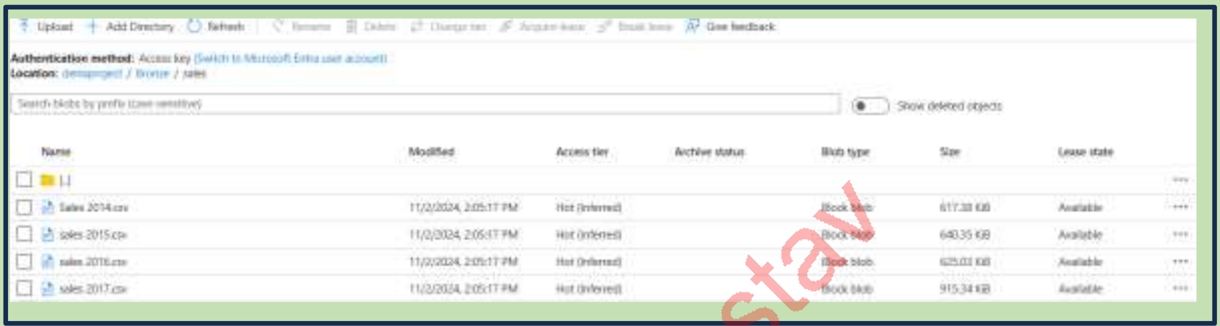
Pipelines got run successful

We can see the output

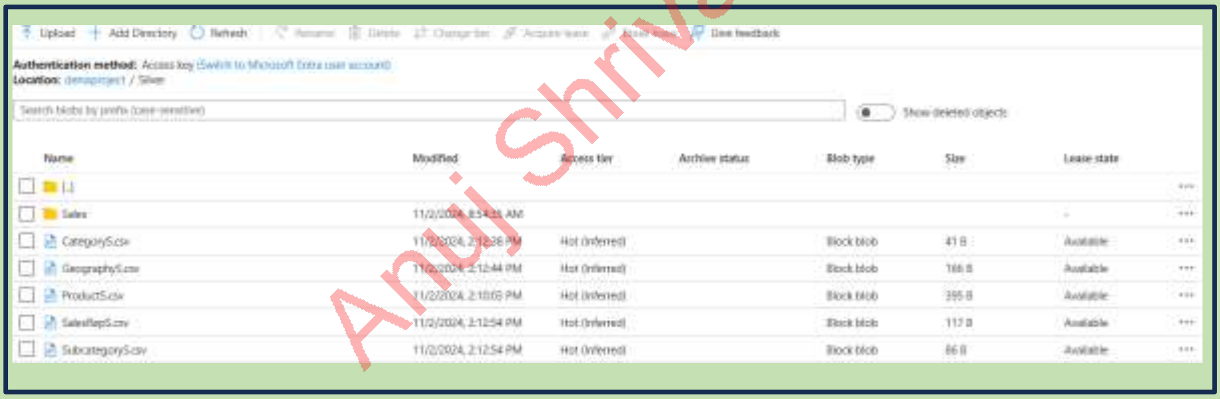


Bronze

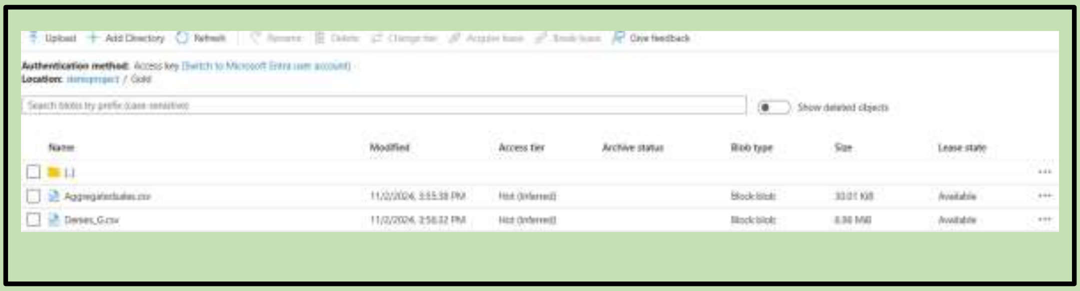




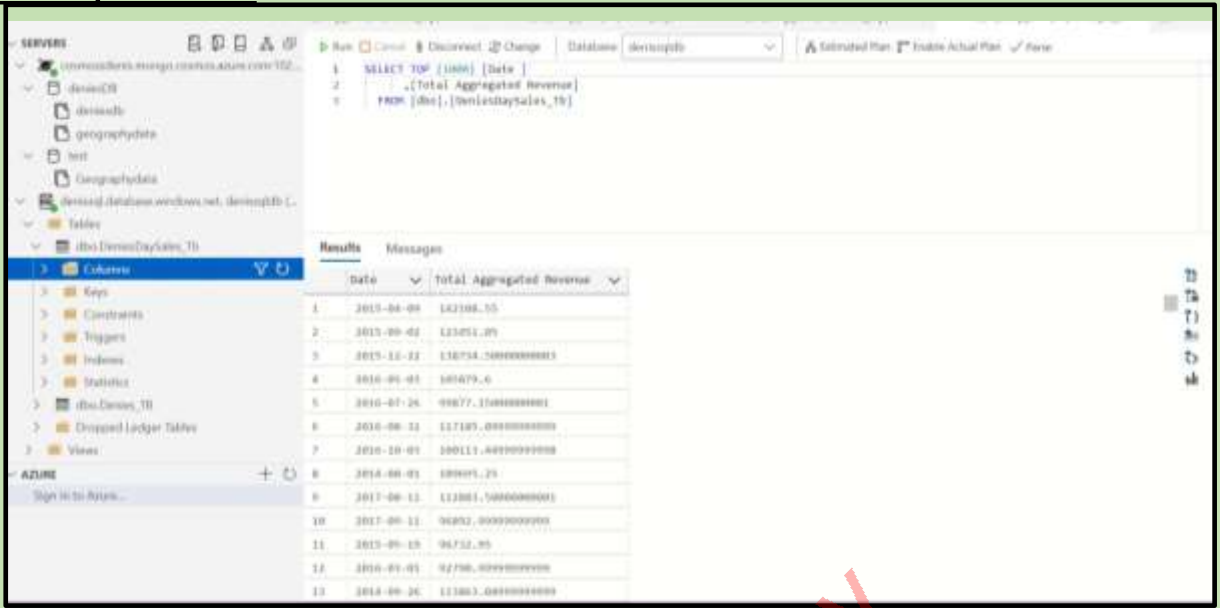
SILVER

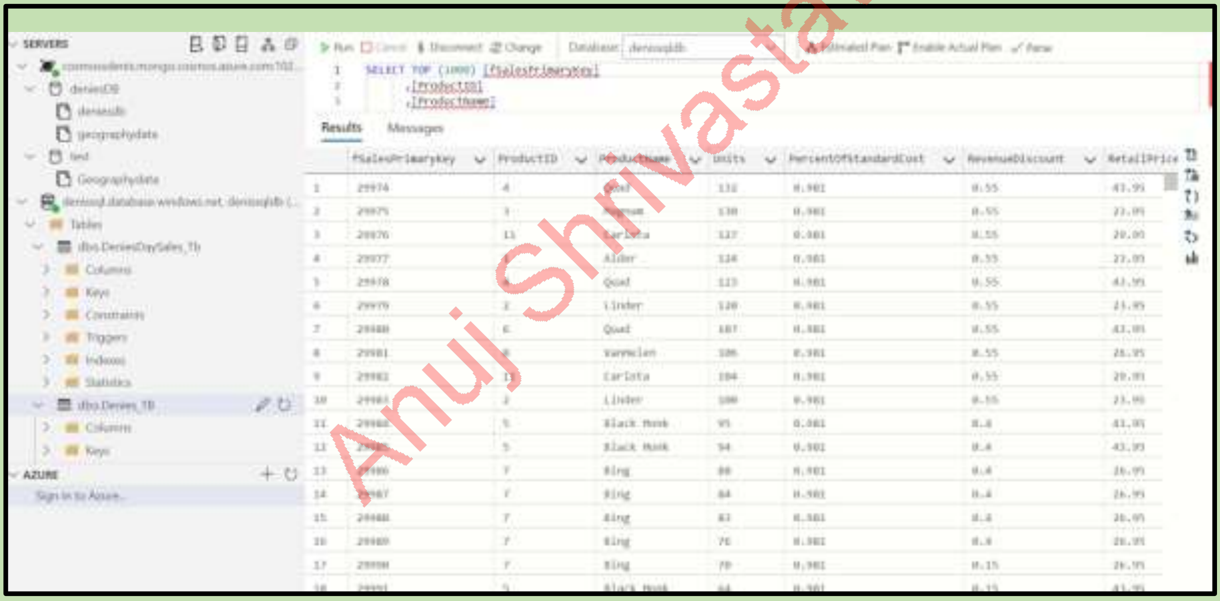


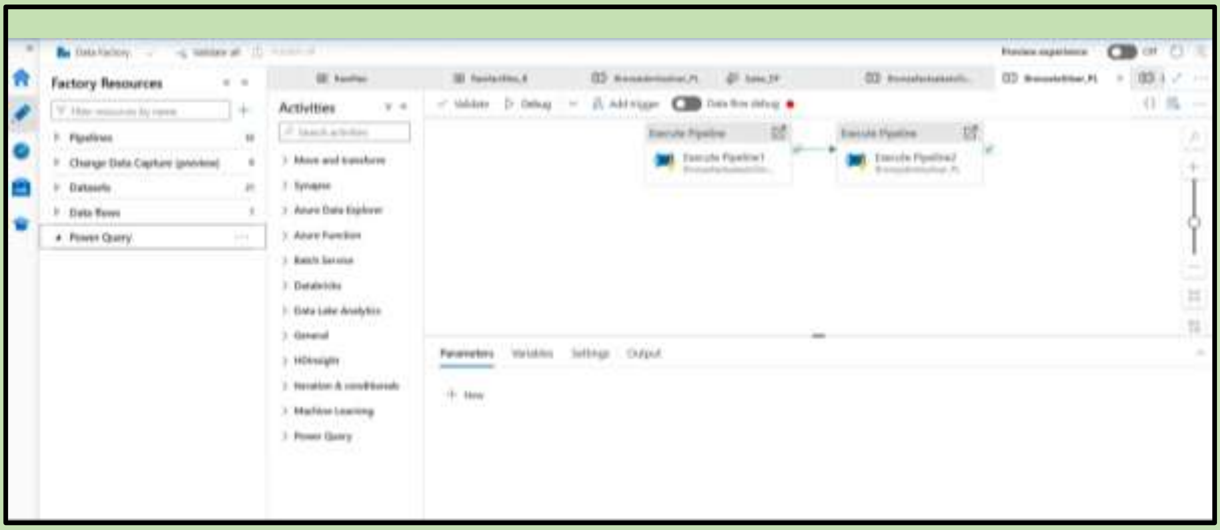
GOLD



Azure sql Database:



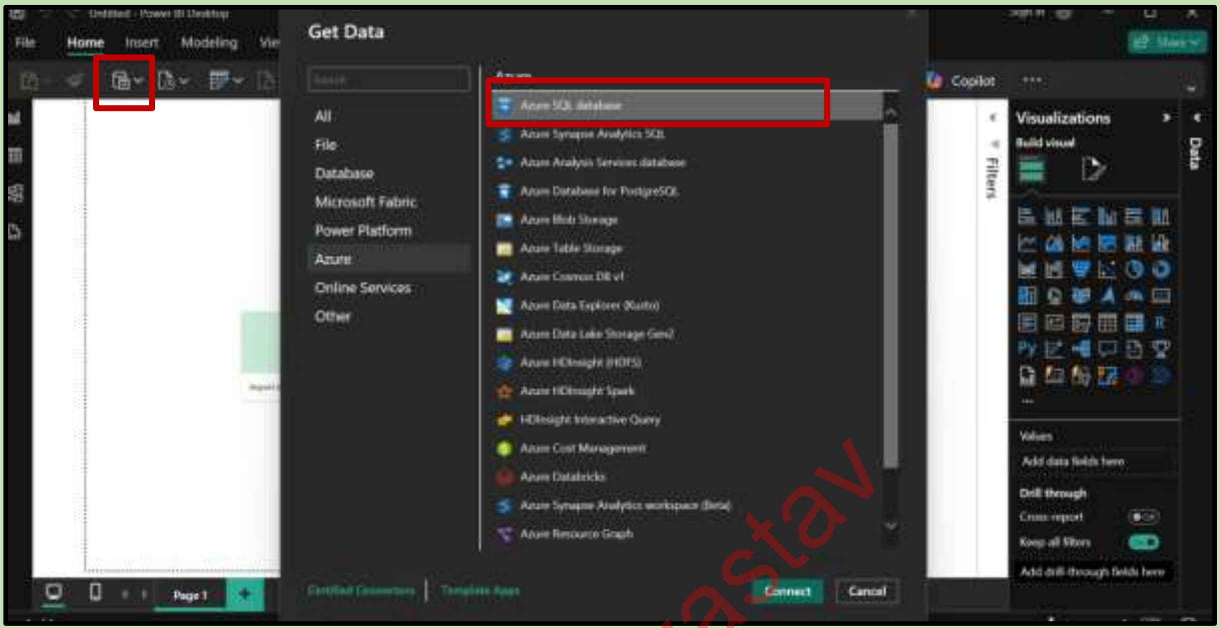




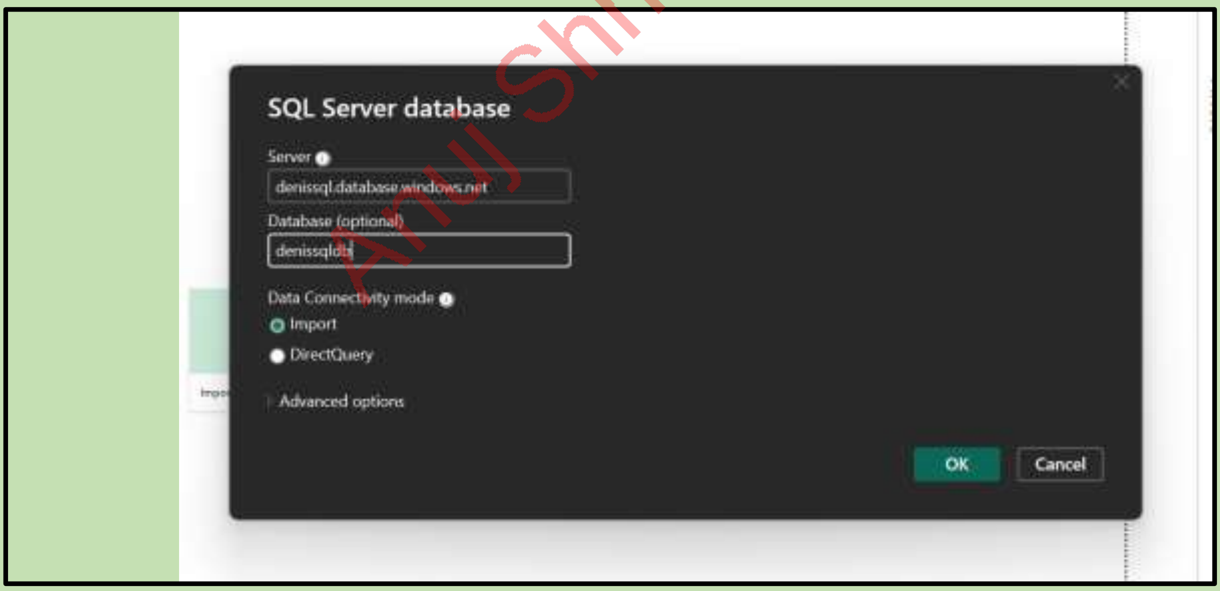
FINAL STEP:

Loading the data in power bi desktop

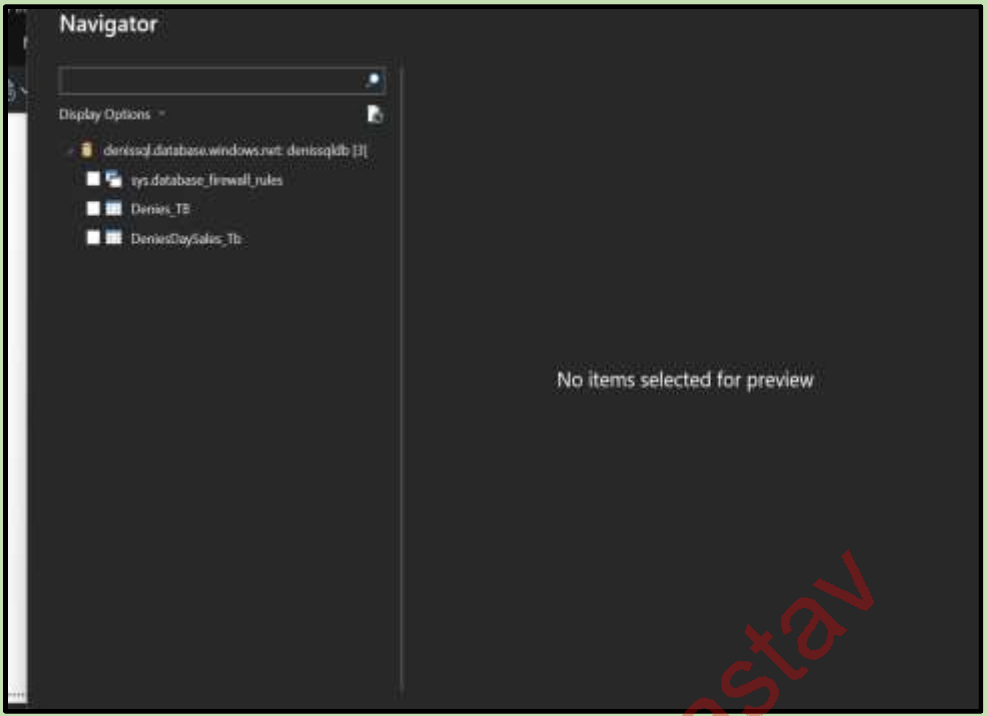
Open power Bi App → open azure sql database



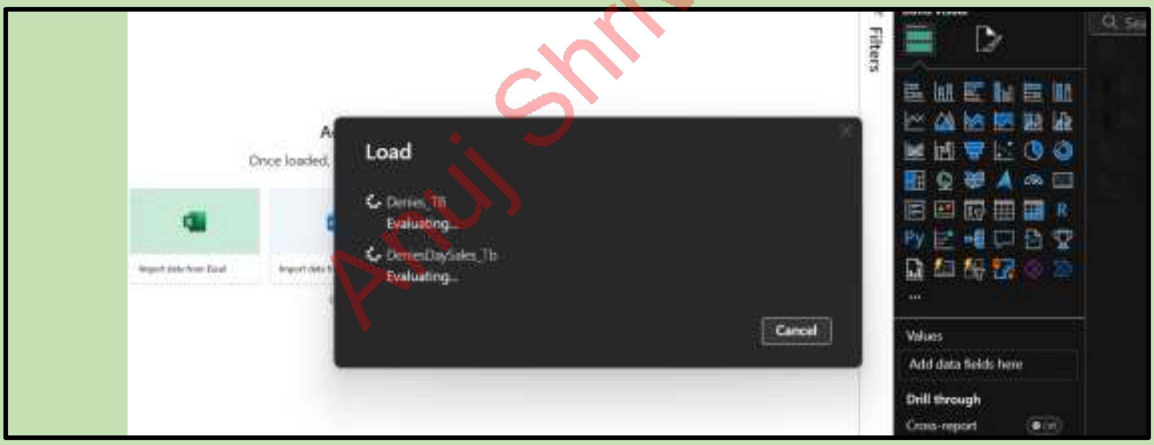
Give the server details from the azure sql data base and give the sql name and password



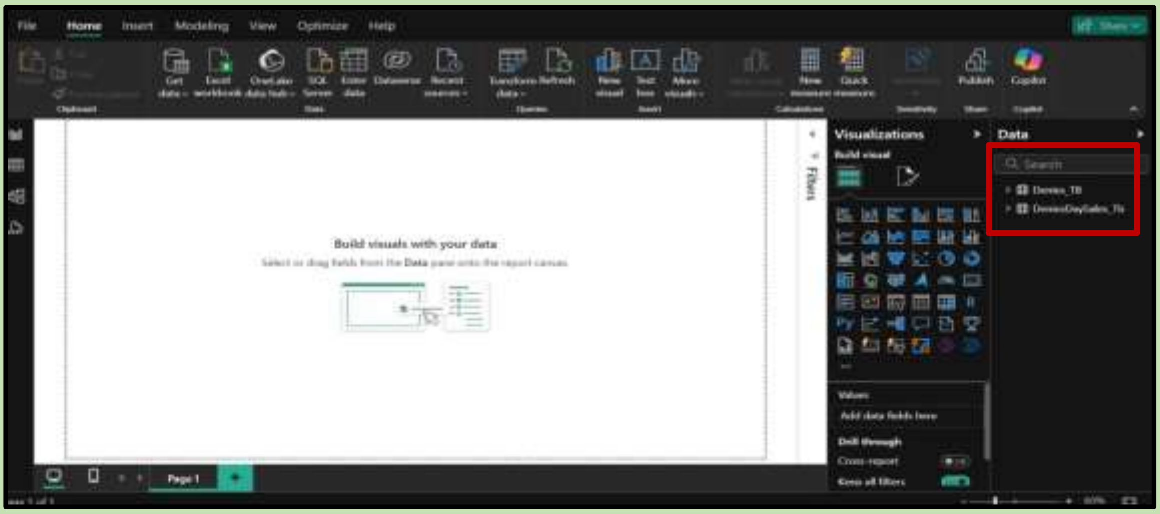
SQL TABLES:



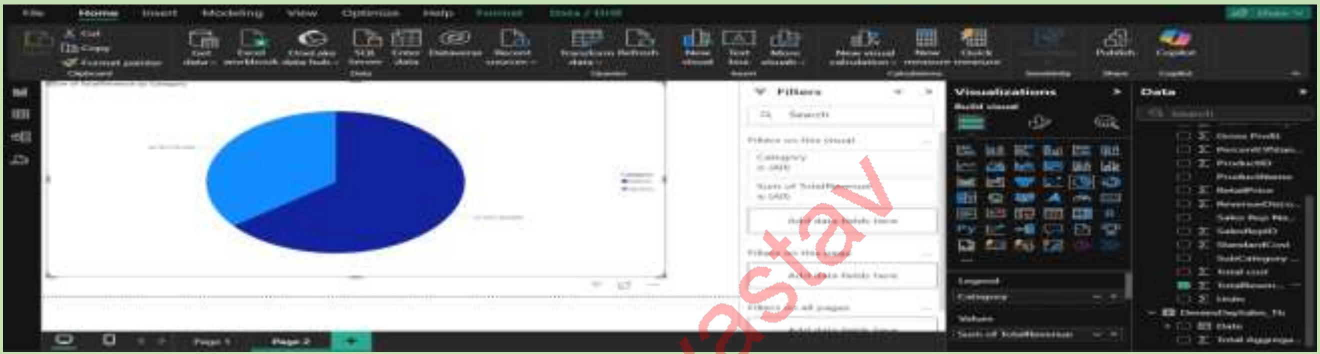
As we can see data is loading



As we can see data is loaded in power Bi




File
Home
Insert
Modeling
View
Optimize
Help
Format
Data / Drill
Filters
Visualizations
Data
Add data fields here
Page 1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View
Optimize
Help
Format
Data / Drill
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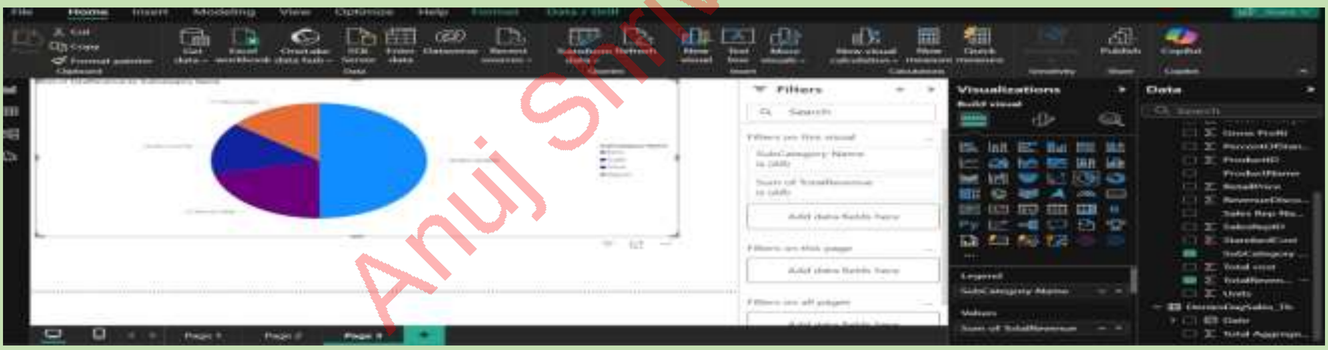
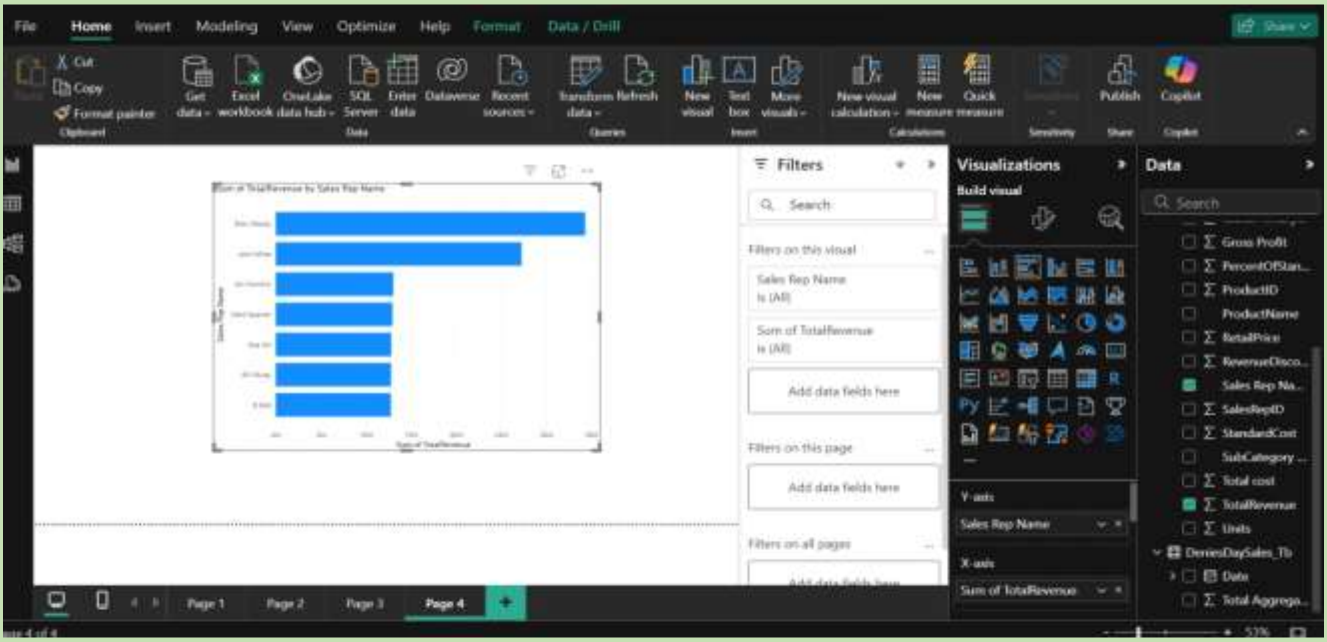
Visualizations
Data
Page 1
Page 2


Filters
Visualizations
Data
Page 1
Page 2
Page 3


File
Home
Insert
Modeling
View
Optimize
Help
Format
Data / Drill
Share
Cut
Copy
Format painter
Clipboard
Get data
Excel workbook
OneLake data hub
SQL Server
Enter data
Dataverse
Recent sources
Data
Transform data
Refresh
Queries
New visual
Text box
More visuals
Insert
New visual calculation
New measure
Quick measure
Calculations
Sensitivity
Publish
Share
Copilot
Sum of TotalRevenue by Sales Rep Name
Filters
Search
Filters on this visual
Sales Rep Name
is (All)
Sum of TotalRevenue
is (All)
Add data fields here
Filters on this page
Filters on all pages
Visualizations
Build visual
Y-axis
Sales Rep Name
X-axis
Sum of TotalRevenue
Data
Gross Profit
ProductID
ProductName
RetailPrice
Sales Rep Na...
SalesRepID
StandardCost
SubCategory ...
Total cost
TotalRevenue
Units
DeriesDaySales_Tb
Date
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-------------------END---------------------

Anuj Shrivastav

Lead Data Engineer Pepsico